My Learning Draft

Labix

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Contents

Part I

Combinatorial Commutative Algebra

1 Monomial Ideals

1.1 Basic Definitions

Definition 1.1.1 (Monomial Notation)
Let R be a commutative ring. Let $u = (u_1, \dots, u_n) \in \mathbb{N}^n$. Define

$$x^u = x_1^{u_1} \cdots x_n^{u_n} \in R[x_1, \dots, x_n]$$

Definition 1.1.2 (Monomial Ideals)
Let R be a commutative ring. Let $I \subseteq R[x_1, \dots, x_n]$ be an ideal. We say that I is a monomial ideal if I is generated by monomials.

Proposition 1.1.3
Let R be a commutative ring. Let $I \subseteq R[x_1, \dots, x_n]$ be an ideal. Then the following are equivalent.
- I is a monomial ideal.
- Suppose that $f = \sum_{\alpha \in \mathbb{N}^n} c_\alpha x^\alpha \in I$. If $c_\alpha \neq 0$, then $x^\alpha \in I$.
- Suppose that $(c_1, \dots, c_n) \in (R^*)^n$. Then $(c_1, \dots, c_n) \cdot I = I$.

Proposition 1.1.4
Let R be a commutative ring. Let $m_1, \dots, m_k \in R[x_1, \dots, x_n]$ be monomials. Let $f = \sum_{\alpha \in \mathbb{N}^n} c_\alpha x^\alpha \in R[x_1, \dots, x_n]$. Then $f \in (m_1, \dots, m_k)$ if and only if $c_\alpha x^\alpha$ is a multiple of one of $m_1, \dots, m_k$.

Proposition 1.1.5
Let R be a commutative ring. Let $I \subseteq R[x_1, \dots, x_n]$ be a monomial ideal. Then there exists unique $u_1, \dots, u_m \in \mathbb{N}^n$ such that

$$I = (x^{u_1}, \dots, x^{u_m})$$

Proposition 1.1.6
Let R be a commutative ring. Then the following are true.
- The sum of monomial ideals is a monomial ideal.
- The product of monomial ideals is a monomial ideal.
- The intersection of monomial ideals is a monomial ideal.

Definition 1.1.7 (Squarefree Monomials)
Let R be a commutative ring. Let $u = (u_1, \dots, u_n) \in \mathbb{N}^n$. We say that x^u is squarefree if $u_1, \dots, u_n \leq 1$.

Definition 1.1.8 (Squarefree Monomial Ideals)
Let R be a commutative ring. Let $I \subseteq R[x_1, \dots, x_n]$ be a monomial ideal. We say that I is squarefree if it is generated by squarefree monomial.

Proposition 1.1.9
Let R be a commutative ring. Let $u_1, \dots, u_m \in \mathbb{N}^n$. Then we have

$$\sqrt{(x^{u_1}, \dots, x^{u_m})} = \left(x^{\min\{u_1, 1\}}, \dots, x^{\min\{u_m, 1\}}\right) \in R[x_1, \dots, x_n]$$

Definition 1.1.10 (Quadratic Monomials)
Let R be a commutative ring. Let $u = (u_1, \dots, u_n) \in \mathbb{N}^n$. We say that x^u is squarefree if $\sum_{i=1}^n u_i = 2$.

Definition 1.1.11 (Quadratic Monomial Ideals)
Let R be a commutative ring. Let $I \subseteq R[x_1, \dots, x_n]$ be a monomial ideal. We say that I is quadratic if it is generated by quadratic monomial.

Lemma 1.1.12
Let R be a commutative ring. Let $I \subseteq R[x_1, \dots, x_n]$ be an ideal. Then I is a quadratic monomial ideal if and only if $\frac{R[x_1,\dots,x_n]}{I}$ is a quadratic algebra.

1.2 Monomial Matrices

Definition 1.2.1 (Partial Order on $\mathbb{N}^n$)
Let $x = (x_1, \dots, x_n), y = (y_1, \dots, y_n) \in \mathbb{N}^n$. We say that $x \succeq y$ if $x_i \geq y_i$ for all i.

Definition 1.2.2 (Monomial Matrices)
Let R be a commutative ring. Let $a_1, \dots, a_p, b_1, \dots, b_q \in \mathbb{N}^n$. Let $\phi : \bigoplus_{i=1}^p R[x_1, \dots, x_n](-a_i) \to \bigoplus_{j=1}^q R[x_1, \dots, x_n](-b_j)$ be a $R[x_1, \dots, x_n]$-module homomorphism. Let $A = (a_{ij}) \in M_{q\times p}(R[x_1, \dots, x_n])$ be the matrix representing representing ϕ. We say that A is a monomial matrix if A is of the form
$$a_{ij} = \begin{cases} \lambda_{ij} x^{a_i - b_j} & \text{if } a_i \succeq b_j \\ 0 & \text{otherwise} \end{cases}$$
for some $\lambda_{ij} \in R$.

Definition 1.2.3 (Minimal Monomial Matrices)
Let R be a commutative ring. Let $a_1, \dots, a_p, b_1, \dots, b_q \in \mathbb{N}^n$. Let $\phi : \bigoplus_{i=1}^p R[x_1, \dots, x_n](-a_i) \to \bigoplus_{j=1}^q R[x_1, \dots, x_n](-b_j)$ be a $R[x_1, \dots, x_n]$-module homomorphism. Let $A = (a_{ij}) \in M_{q\times p}(R[x_1, \dots, x_n])$ be the matrix representing representing ϕ. We say that A is a minimal monomial matrix if A is of the form
$$a_{ij} = \begin{cases} \lambda_{ij} x^{a_i - b_j} & \text{if } a_i \succeq b_j \text{ and } a_i \neq b_j \\ 0 & \text{otherwise} \end{cases}$$

Proposition 1.2.4
Let R be a commutative ring. Let $I \subseteq R[x_1, \dots, x_n]$ be a monomial ideal. Let $(F_\bullet, d)$ be a graded free resolution of I. Then $F_\bullet$ is a minimal graded free resolution of I if and only if for all $k \in \mathbb{N}$, the maps $d_k : F_k \to F_{k-1}$ are represented by minimal monomial matrices.

2 Cellular Resolutions

3 The Stanley-Reisner Ring

3.1 Basic Definitions

Definition 3.1.1 (The Stanley-Reisner Ideal)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Define the Stanley-Reisner ideal of $\mathcal{K}$ to be

$$I_{\mathcal{K}} = \left\langle \prod_{j\in J} x_j \;\middle|\; J \in \mathcal{P}(\{1,\dots,n\}) \setminus \mathcal{K} \right\rangle \subseteq R[x_1,\dots,x_n]$$

Proposition 3.1.2
Let R be a commutative ring. Let $n \in \mathbb{N}$. Let $\mathcal{K}$ be an abstract simplicial complex on $\{1,\dots,n\}$. Then we have

$$I_{\mathcal{K}} = \bigcap_{J\in\mathcal{K}} \langle x_i \mid i \notin J\rangle$$

Proposition 3.1.3
Let R be a commutative ring. Let $n \in \mathbb{N}$. Then there is a one-to-one bijection

$$\left\{ \mathcal{K} \subseteq \mathcal{P}(\{1,\dots,n\}) \;\middle|\; \mathcal{K} \text{ is a simplicial complex} \right\} \overset{1:1}{\longleftrightarrow} \left\{ I \subseteq R[x_1,\dots,x_n] \;\middle|\; \begin{array}{c} I \text{ is a squarefree} \\ \text{monomial ideal} \end{array} \right\}$$

given by $\mathcal{K} \mapsto I_{\mathcal{K}}$.

Definition 3.1.4 (The Stanley-Reisner Ring)
Let R be a commutative ring. Let $n \in \mathbb{N}$. Let $\mathcal{K}$ be an abstract simplicial complex on $\{1,\dots,n\}$. Define the Stanley-Reisner ring of $\mathcal{K}$ to be

$$R[\mathcal{K}] = \frac{R[x_1,\dots,x_n]}{I_{\mathcal{K}}}$$

Example 3.1.5
Let R be a commutative ring. The Stanley-Reisner ring of the standard n-simplex Δ^n is given by

$$R[\Delta^n] = k[x_1,\dots,x_n]$$

Example 3.1.6
Let R be a commutative ring. The Stanley-Reisner ring of the abstract simplicial complex $S = \mathcal{P}(\{1,2,3\}) \cup \mathcal{P}(\{1,3,4\}) \cup \mathcal{P}(\{2,4\})$ over $\{1,2,3,4\}$ is given by

$$R[S] = \frac{k[x_1,x_2,x_3,x_4]}{(x_1x_2x_4, x_2x_3x_4)}$$

Proposition 3.1.7
Let R be a commutative ring. Let $\mathcal{K}_1, \mathcal{K}_2$ be simplicial complexes. Then we have

$$R[\mathcal{K}_1 * \mathcal{K}_2] \cong R[\mathcal{K}_1] \otimes_R R[\mathcal{K}_2]$$

Definition 3.1.8 (Induced Map of Stanley-Reisner Rings)
Let R be a commutative ring. Let $\mathcal{K}$ and $\mathcal{L}$ be simplicial complexes on $\{1,\dots,n\}$ and $\{1,\dots,m\}$ respectively. Let $f : \mathcal{K} \to \mathcal{L}$ be a simplicial map. Define the induced map of Stanley-Reisner rings $f^* : R[\mathcal{L}] \to R[\mathcal{K}]$ by

$$f^*(y_j) = \sum_{i\in f^{-1}(j)} x_i$$

Theorem 3.1.9 (Bruns-Gubeladze)
Let k be a field. Let $\mathcal{K}$ and $\mathcal{L}$ be simplicial complexes on $\{1, \dots, n\}$ and $\{1, \dots, m\}$ respectively. If $k[\mathcal{K}]$ and $k[\mathcal{L}]$ are isomorphic, then there exists a bijection $f : \{1, \dots, n\} \to \{1, \dots, m\}$ such that the induced map $f : \mathcal{K} \to \mathcal{L}$ is an isomorphism.

Proposition 3.1.10
Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex. Then we have

$$\dim(R[\mathcal{K}]) = \dim(\mathcal{K}) + 1$$

Proposition 3.1.11
Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are equivalent.
- $R[\mathcal{K}]$ is a quadratic algebra.
- $R[\mathcal{K}]$ is a Koszul algebra.
- $I_{\mathcal{K}}$ is a quadratic monomial ideal.
- $\mathcal{K}$ is a flag complex.

Theorem 3.1.12 (Stanley)
Let k be a field. Let $n \in \mathbb{N}^+$. Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $(f_{-1}, \dots, f_{d-1})$ be the face vector of $\mathcal{K}$. Let $(h_0, \dots, h_d)$ be the h-vector of $\mathcal{K}$. Then the following are true.
- The Hilbert function of $k[\mathcal{K}]$ is given by

$$HF_{k[\mathcal{K}]}(n) = \begin{cases} \sum_{i=-1}^{d-1} f_i \binom{n-1}{i} & \text{if } n \geq 1 \\ 0 & \text{if } n = 0 \end{cases}$$

- The Hilbert series of $k[\mathcal{K}]$ is given by

$$HS_{k[\mathcal{K}]}(t) = \sum_{i=-1}^{d-1} f_i \frac{t^{i+1}}{(1-t)^{i+1}} = \frac{1}{(1-t)^d} \sum_{i=0}^{d} h_i t^i$$

- The Hilbert polynomial of $k[\mathcal{K}]$ is given by

$$HP_{k[\mathcal{K}]}(n) = \sum_{i=-1}^{d-1} f_i \binom{n-1}{i}$$

Example 3.1.13
Let Δ^n be the standard n-simplex. Then the following are true.
- The Hilbert series for the Stanley-Reisner ring of Δ^n is given by

$$HS_{k[\Delta^n]}(t) = \frac{1}{(1-t^2)^{n+1}}$$

- The Hilbert series for the Stanley-Reisner ring of $\partial\Delta^n$ is given by

$$HS_{k[\partial\Delta^n]}(t) = \frac{1}{(1-t^2)^{n+1}} \sum_{i=0}^{n+1} t^{2i}$$

We take $k[x_1, \dots, x_n]$ to be graded by $\deg(x_i) = 2$ so that $k[x_1, \dots, x_n]$ is graded-commutative as a graded k-algebra. It follows that $\text{Tor}_p^{k[x_1, \dots, x_n]}(k[\mathcal{K}], k)$ also acquires a grading (in which the odd degree components are 0).

Proposition 3.1.14
Let k be a field or $\mathbb{Z}$. Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.

- There is an graded isomorphism k-vector spaces

$$\text{Tor}_p^{k[x_1,\dots,x_n]}(k[\mathcal{K}],k)\cong H_p(K_\bullet(x_1,\dots,x_n;k[\mathcal{K}]))$$

- There is an isomorphism of graded k-algebras

$$\text{Tor}_*^{k[x_1,\dots,x_n]}(k[\mathcal{K}],k)\cong H_*(K_\bullet(x_1,\dots,x_n;k[\mathcal{K}]))$$

Proposition 3.1.15
Let k be a field or $\mathbb{Z}$. Let $\mathcal{K}$ and $\mathcal{L}$ be simplicial complexes on $\{1,\dots,n\}$ and $\{1,\dots,m\}$ respectively. Let $f:\mathcal{K}\to\mathcal{L}$ be a simplicial map. Then f induces a graded k-algebra homomorphism

$$f^*:\text{Tor}_*^{k[y_1,\dots,y_m]}(k[\mathcal{L}],k)\to\text{Tor}_*^{k[x_1,\dots,n]}(k[\mathcal{K}],k)$$

Theorem 3.1.16 (Hochster I)
Let k be a field or $\mathbb{Z}$. Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following are true.

- There is an isomorphism of k-vector spaces

$$\text{Tor}_p^{k[x_1,\dots,n]}(k[\mathcal{K}],k)_{2j}\cong\bigoplus_{\substack{J\subseteq\{1,\dots,n\}\\|J|=j}}\widetilde{H}^{j-i-1}(\mathcal{K}_J;k)$$

- There is an graded isomorphism of k-vector spaces

$$\text{Tor}_p^{k[x_1,\dots,x_n]}(k[\mathcal{K}],k)\cong\bigoplus_{J\subseteq\{1,\dots,n\}}\widetilde{H}^{p-|J|-1}(\mathcal{K}_J,k)$$

We are missing a description of the cup product in terms of Hochster's formula. It is given by the following definition.

Definition 3.1.17 (Ring Structure in Hochster's Formula)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Define a ring structure in $\bigoplus_{J\subseteq\{1,\dots,n\}}\widetilde{H}^*(\mathcal{K}_\mathcal{J};\mathbb{Z})$ as follows. Let α_L be the cochain basis element in $C^p(\mathcal{K}_I;\mathbb{Z})$ corresponding to the simplex $L\subseteq I$. The ring structure is induced by the map

$$C^p(\mathcal{K}_I;\mathbb{Z})\otimes_\mathbb{Z}C^q(\mathcal{K}_\mathcal{J};\mathbb{Z})\to C^{p+q+1}(\mathcal{K}_{I\cup J})$$

defined by

$$\alpha_L\otimes\alpha_M\mapsto\begin{cases}\varepsilon(L,I)\varepsilon(M,J)\varepsilon(L\cup M,I\cup J)\alpha_{L\amalg M} & \text{if } I\cap J=\emptyset\\ 0 & \text{otherwise}\end{cases}$$

where $\varepsilon(L,I)=\prod_{j\in L}\varepsilon(j,J)$ and $\varepsilon(j,J)=(-1)^{r-1}$ if j is in the rth position in J.

Prove: the above defines a ring structure on $\bigoplus_{J\subseteq\{1,\dots,n\}}\widetilde{H}^*(\mathcal{K}_J,k)$.

Theorem 3.1.18 (Hochster II)
Let k be a field or $\mathbb{Z}$. Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then there is a natural isomorphism of graded k-algebras

$$\text{Tor}_*^{k[x_1,\dots,x_n]}(k[\mathcal{K}],k)\cong\bigoplus_{J\subseteq\{1,\dots,n\}}\widetilde{H}^*(\mathcal{K}_J,k)$$

3.2 Cohen Macaulay Complexes

Definition 3.2.1 (Cohen Macaulay Complexes)
Let $\mathcal{K}$ be a simplicial complex. Let R be a commutative ring. We say that $\mathcal{K}$ is Cohen Macaulay over R if $R[\mathcal{K}]$ is a Cohen-Macaulay ring.

See 1.3.6

Proposition 3.2.2
Let k be a field. Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then $k[\mathcal{K}]$ is Cohen-Macaulay if and only if the following are true.

- $\dim_k \left(\text{Tor}_p^{k[x_1,\dots,x_n]}(k[\mathcal{K}], k)\right) = 0$ for all $p \geq n - \dim(\mathcal{K})$.
- $\dim_k \left(\text{Tor}_{n-\dim(\mathcal{K})-1}^{k[x_1,\dots,x_n]}(k[\mathcal{K}], k)\right) \neq 0$.

Theorem 3.2.3 (Reisner)
Let k be a field or $\mathbb{Z}$. Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then $k[\mathcal{K}]$ is Cohen-Macaulay if and only if

$$\widetilde{H}_i\left(\text{lk}_{\mathcal{K}}(I); k\right) = 0 \qquad \text{for all } I \in \mathcal{K} \text{ and } 0 \leq i \leq \dim(\text{lk}_{\mathcal{K}}(I)) - 1$$

Theorem 3.2.4 (Munkres, Stanley)
Let k be a field or $\mathbb{Z}$. Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then $k[\mathcal{K}]$ is Cohen-Macaulay if and only if

$$\widetilde{H}_i\left(|\mathcal{K}|, |\mathcal{K}| \setminus \{x\}; k\right) = \widetilde{H}_i(\mathcal{K}; k) = 0 \qquad \text{for all } x \in |\mathcal{K}| \text{ and } 0 \leq i \leq \dim(\mathcal{K}) - 1$$

Corollary 3.2.5
Let $\mathcal{K}$ be a triangulation of S^n. Let k be a field. Then $k[\mathcal{K}]$ is Cohen-Macaulay.

Proposition 3.2.6
Let k be a field. Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then $\mathcal{K}$ is Cohen Macaulay over k if and only if $I_{\mathcal{K}^\vee}$ admits a linear resolution.

Definition 3.2.7 (Sequential Cohen Macaulay Complexes)
Let $\mathcal{K}$ be a simplicial complex. Let R be a commutative ring. We say that $\mathcal{K}$ is sequentially Cohen Macaulay over R if

$$\mathcal{K}^{\langle i \rangle} = \langle F \in \mathcal{K} \mid \dim(F) \geq i \rangle$$

is Cohen Macaulay for all i.

Proposition 3.2.8
Let $\mathcal{K}$ be a simplicial complex. Let k be a field. Then $\mathcal{K}$ is sequentially Cohen Macaulay over k if and only if there is a filtration

$$0 = M_0 \subset \cdots \subset M_r = k[\mathcal{K}]$$

of graded $k[\mathcal{K}]$-modules such that $\frac{M_j}{M_{j-1}}$ is Cohen Macaulay and $\dim(M_1/M_0) < \cdots < \dim(M_r/M_{r-1})$.

Proposition 3.2.9
Let $\mathcal{K}$ be a simplicial complex. Let k be a field. Then $\mathcal{K}$ is Cohen Macaulay over k if and only if $\mathcal{K}$ is sequentially Cohen Macaulay and pure.

Proposition 3.2.10
Let k be a field. Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then $\mathcal{K}$ is sequentially Cohen Macaulay over k if and only if $I_{\mathcal{K}^\vee}$ admits a componentwise linear resolution.

3.3 Shellable Simplicial Complexes

3.4 Gorenstein Complexes

Definition 3.4.1 (Gorenstein Complexes)
Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex. We say that $\mathcal{K}$ is Gorenstein over R if $R[\mathcal{K}]$ is a Gorestein ring.

This is equivalent to saying that $\text{Tor}^{R[x_1,\dots,x_n]}_{m-\dim(\mathcal{K})-1}(R[\mathcal{K}],R)\cong R$ since Gorenstein rings are Cohen-Macaulay rings of type 1.

Proposition 3.4.2 (Stanley's Combinatorial Criterion)
Let k be a field. Let $\mathcal{K}$ be a Cohen-Macaulay simplicial complex over $\{1,\dots,n\}$. Then $\mathcal{K}$ is Gorenstein over k if and only if

$$\dim_k(\widetilde{H}_{\dim(\text{lk}(\sigma))}(\text{lk}(\sigma);k))\leq 1$$

Proposition 3.4.3
Let $\mathcal{K}$ be a simplicial complex. Then $\mathcal{K}$ is Gorenstein over $\mathbb{Z}$ if and only if it is Gorenstein over k for all fields k.

Definition 3.4.4 (The Core of a Simplicial Complex)
Let $\mathcal{K}$ be a simplicial complex. Define the core of $\mathcal{K}$ to be

$$\text{core}(mK)=\mathcal{K}_{\{v\in V(\mathcal{K})\;|\;\text{st}(v)\neq\mathcal{K}\}}$$

Definition 3.4.5 (Gorenstein* Complexes)
Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex. We say that $\mathcal{K}$ is a Gorenstein* complex if the following are true.
- $\mathcal{K}$ is Gorenstein over R.
- $\text{core}(\mathcal{K})=\mathcal{K}$.

$A=\bigoplus_{i=0}^d$ is a Poincare algebra if the map $A^i\to\text{Hom}(A^{d-i},A^d)$ given by $a\mapsto(b\mapsto ab)$ is an isomorphism.

Proposition 3.4.6
Let R be a commutative ring. Let $\mathcal{K}$ be a simplicial complex. Then the following are equivalent.
- $\mathcal{K}$ is a Gorenstein* complex.
- For any $\sigma\in\mathcal{K}$, we have
$$H^i(\text{lk}(\sigma))\cong\begin{cases}R & \text{if } i=0 \text{ or } \dim(\text{lk}(\sigma))\\ 0 & \text{otherwise}\end{cases}$$
- For any $x\in|\mathcal{K}|$, we have
$$H^i(|\mathcal{K}|,|\mathcal{K}|\setminus\{x\};R)\cong\widetilde{H}^i(\mathcal{K};R)\cong\begin{cases}R & \text{if } i=\dim(\mathcal{K})\\ 0 & \text{otherwise}\end{cases}$$
- $\text{Tor}_*^{R[x_1,\dots,x_n]}(R[\mathcal{K}],R)$ is a Poincare duality algebra.

4 Lattice Ideals and Affine Semigroups

Definition 4.0.1 (Lattice Ideal)
Let $L \subseteq \mathbb{Z}^n$ be a lattice. Let R be a commutative ring. Define the lattice ideal of L in $R[x_1, \dots, x_n]$ to be
$$I_L = \langle x^u - x^v \mid u, v \in \mathbb{N}^n \text{ and } u - v \in L \rangle$$

Proposition 4.0.2
Let k be a field. Let S be a finitely generated commutative monoid. Let $s_1, \dots, s_n$ be a set of generators of S. Let $\phi : \mathbb{Z}^n \to G(S)$ be the group homomorphism defined by $e_i \mapsto s_i$. Then there is a k-algebra isomorphism
$$k[S] \cong \frac{k[x_1, \dots, x_n]}{I_{\ker(\phi)}}$$

Proposition 4.0.3
Let k be a field. Let S be a finitely generated commutative monoid. Let $s_1, \dots, s_n$ be a set of generators of S. Then we have
$$\dim(k[S]) = n - \text{rank}(\ker(\phi))$$

Definition 4.0.4 (Affine Semi-group)
Let S be a monoid. We say that S is an affine semi-group if there exists an injective monoid homomorphism $S \to \mathbb{N}^d$ for some $d \in \mathbb{N}$.

Proposition 4.0.5
Let k be a field. Let S be a finitely generated commutative monoid. Let $s_1, \dots, s_n$ be a set of generators of S. Let $\phi : \mathbb{Z}^n \to G(S)$ be the group homomorphism defined by $e_i \mapsto s_i$. Then the following are equivalent.
- $I_{\ker(\phi)}$ is prime.
- $k[S]$ is an integral domain.
- $G(S)$ is a free abelian group.
- S is an affine semi-group.

Part II

Toric Topology

5 Polyhedral Products

5.1 The Homotopy Theory of Polyhedral Products

Definition 5.1.1 (Polyhedral Products)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A}) = \{(X_1,A_1),\dots,(X_n,A_n)\}$ be pairs of spaces.

- Let $\sigma \in \mathcal{K}$. Define
$$(\mathbf{X},\mathbf{A})^\sigma = \left\{(x_1,\dots,x_n) \in \prod_{i=1}^n X_i \;\middle|\; x_i \in A_i \text{ for } i \notin I\right\}$$
- Define the polyhedral product of $(\mathbf{X},\mathbf{A})$ corresponding to $\mathcal{K}$ to be
$$(\mathbf{X},\mathbf{A})^\mathcal{K} = \bigcup_{I\in\mathcal{K}} (\mathbf{X},\mathbf{A})^\sigma$$

Proposition 5.1.2
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A}) = \{(X_1,A_1),\dots,(X_n,A_n)\}$ be pairs of spaces. Then we have
$$(\mathbf{X},\mathbf{A})^\mathcal{K} = \operatorname*{colim}_{\sigma\in\mathsf{Cat}(\mathcal{K})} (\mathbf{X},\mathbf{A})^\sigma$$

In particular, the assignment $(-)^\mathcal{K} : (\mathbf{Top}^2)^n \to \mathbf{Top}$ is functorial. It is also a homotopy functor.

Proposition 5.1.3
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A})$ and $(\mathbf{Y},\mathbf{B})$ be n pairs of spaces. If (X_i,A_i) and (Y_i,B_i) are homotopy equivalent for $1 \le i \le n$, then there is a homotopy equivalence
$$(\mathbf{X},\mathbf{A})^\mathcal{K} \simeq (\mathbf{Y},\mathbf{B})^\mathcal{K}$$

Example 5.1.4
Let $(\mathbf{X},\mathbf{A}) = \{(X_1,A_1),\dots,(X_n,A_n)\}$ be pairs of spaces. Then the following are true.

- We have $(\mathbf{X},\mathbf{A})^{\Delta^{n-1}} = \prod_{i=1}^n X_i$.
- If $A_i = X_i$ for all i, then we have $(\mathbf{X},\mathbf{A})^{\Delta^{n-1}} = \prod_{i=1}^n X_i$.
- We have $(\mathbf{X},\mathbf{A})^\emptyset = \prod_{i=1}^n A_i$.

Example 5.1.5
Let $\mathbf{X} = \{X_1,\dots,X_n\}$ be spaces. Then the following are true.

- If $\mathcal{K}$ is a discrete set of n points, then we have $(\mathbf{X},*)^\mathcal{K} = \bigvee_{i=1}^n X_i$.
- If $A_i = X_i$ for all i, then we have $(\mathbf{X},\mathbf{A})^{\Delta^{n-1}} = \prod_{i=1}^n X_i$.
- We have $(\mathbf{X},\mathbf{A})^\emptyset = \prod_{i=1}^n A_i$.

Proposition 5.1.6
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $\mathcal{L}$ be a subcomplex of $\mathcal{K}$. Let $(\mathbf{X},\mathbf{A})$ be n pairs of spaces. Then the inclusion map induces a map
$$(\mathbf{X},\mathbf{A})^\mathcal{L} \to (\mathbf{Y},\mathbf{B})^\mathcal{K}$$

Proposition 5.1.7
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $I \subseteq \{1,\dots,n\}$ be a subset. Let $(X,A,*)$ be a pointed pair of spaces. Then the following are true.
- The map $X^{|I|} \hookrightarrow X^n$ given by $(x_i)_{i\in I} \to (y_1,\dots,y_n)$ where
$$y_j = \begin{cases} x_j & \text{if } j \in I \\ * & \text{otherwise} \end{cases}$$
restricts to an injective map $(X,A)^{\mathcal{K}} \hookrightarrow (X,A)^{\mathcal{L}}$.
- The map $X^n \to X^{|I|}$ given by $(x_1,\dots,x_n) \mapsto (x_i)_{i\in I}$ restricts to a surjective map $(X,A)^{\mathcal{L}} \to (X,A)^{\mathcal{K}}$.
- The composition $(X,A)^{\mathcal{K}} \to (X,A)^{\mathcal{L}} \to (X,A)^{\mathcal{K}}$ is the identity.

Let $f : \{1,\dots,n\} \to \{1,\dots,m\}$ be a function. Let X be a monoid in **Top**. Then f induces a map
$$X^n \to X^m$$
given by
$$(x_1,\dots,x_n) \mapsto (y_1,\dots,y_m)$$
where
$$y_j = \begin{cases} \prod_{i\in f^{-1}(j)} x_i & \text{if } f^{-1}(j) \neq \emptyset \\ 1_X & \text{otherwise} \end{cases}$$

Proposition 5.1.8
Let $\mathcal{K},\mathcal{L}$ be a simplicial complexes. Let (X,A) be a monoid in $\textbf{Top}^2$. Let $f : \mathcal{K} \to \mathcal{L}$ be a simplicial map. Then f induces a map
$$(X,A)^{\mathcal{K}} \to (X,A)^{\mathcal{L}}$$

Proposition 5.1.9
Let $\mathcal{K},\mathcal{L}$ be a simplicial complex on $\{1,\dots,n\}$. Let $(\mathbf{X},\mathbf{A}) = \{(X_1,A_1),\dots,(X_n,A_n)\}$ be pairs of spaces. Let $f_i : (X_i,A_i) \to (Y_i,B_i)$ be a map of pair for $1 \leq i \leq n$). Then we have
$$(\mathbf{X},\mathbf{A})^{\mathcal{K}*\mathcal{L}} = (\mathbf{X},\mathbf{A})^{\mathcal{K}} \times (\mathbf{X},\mathbf{A})^{\mathcal{L}}$$

Example 5.1.10
Let $\mathcal{K}$ be a simplicial complex. Then we have
$$\text{cc}(\mathcal{K}) = (I,1)^{\mathcal{K}}$$

Proposition 5.1.11
Let $\mathcal{K}$ be a simplicial complex. Let $p : (E,E') \to (B,B')$ be a map such that p and $p|_{E'}$ are fibrations with fibers F and F' respectively. Suppose that one of the following are true.
- $F = F'$.
- $B = B'$.

Then we have that
$$(F,F')^{\mathcal{K}} \to (E,E')^{\mathcal{K}} \to (B,B')^{\mathcal{K}}$$
is a fibration sequence.

Proposition 5.1.12
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,m\}$. Let G be a topological group. Let $p : (E,E') \to (B,B')$ be a map such that p and $p|_{E'}$ are fiber bundles with fibers F and F' respectively with structure group G. Suppose that one of the following are true.
- $F = F'$.
- $B = B'$.

Then we have that
$$(F, F')^{\mathcal{K}} \to (E, E')^{\mathcal{K}} \to (B, B')^{\mathcal{K}}$$
is a fiber bundle with structure group G^m.

Proposition 5.1.13
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, m\}$. Let G be a topological group. Then the following are true.

- $EG^m \times_{G^m} (EG, G)^{\mathcal{K}} \simeq (BG, *)^{\mathcal{K}}$.
- We have that
$$(EG, G)^{\mathcal{K}} \to (BG, *)^{\mathcal{K}} \to BG^m$$
is a fibration sequence.
- We have that
$$G^m \to (EG, G)^{\mathcal{K}} \to (BG, *)^{\mathcal{K}}$$
is a fiber bundle.

Definition 5.1.14 (Polyhedral Smash Products)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $(\mathbf{X}, \mathbf{A})$ be pairs of spaces.

- Let $\sigma \in \mathcal{K}$. Define
$$(\mathbf{X}, \mathbf{A})^{\wedge\sigma} = \left\{ (x_1, \dots, x_n) \in \bigwedge_{i=1}^{n} X_i \;\middle|\; x_i \in A_i \text{ for } i \notin I \right\}$$
- Define the polyhedral smash product of $(\mathbf{X}, \mathbf{A})$ corresponding to $\mathcal{K}$ to be
$$(\mathbf{X}, \mathbf{A})^{\mathcal{K}} = \bigcup_{I \in \mathcal{K}} (\mathbf{X}, \mathbf{A})^{\wedge\sigma}$$

Theorem 5.1.15 (Bahri, Bendersky, Cohen, Gitler)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $(\mathbf{X}, \mathbf{A})$ be n-pairs of CW complexes. Suppose that the inclusion maps $A_j \to X_j$ are null homotopic for $1 \leq j \leq n$. Then there is a homotopy equivalence
$$(\mathbf{X}, \mathbf{A})^{\wedge\mathcal{K}} \simeq \bigvee_{\sigma \in \mathcal{K}} |\mathrm{lk}_{\mathcal{K}}(\sigma)| * (\mathbf{X}_J, \mathbf{A}_J)^{\wedge\sigma}$$

Theorem 5.1.16 (Bahri, Bendersky, Cohen, Gitler)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $(\mathbf{X}, \mathbf{A})$ be n-pairs of CW complexes. Then there is a homotopy equivalence
$$\Sigma(\mathbf{X}, \mathbf{A})^{\mathcal{K}} \simeq \Sigma \left(\bigvee_{J \subseteq \{1, \dots, n\}} (\mathbf{X}_J, \mathbf{A}_J)^{\wedge\mathcal{K}_J} \right)$$
that is natural in $(\mathbf{X}, \mathbf{A})$, where $(\mathbf{X}_J, \mathbf{A}_J) = \{(X_j, A_j) \mid j \in J\}$.

Corollary 5.1.17
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $(\mathbf{X}, \mathbf{A})$ be n-pairs of CW complexes. Suppose that A_i are null homotopic for all i. Then there is a homotopy equivalence
$$\Sigma(\mathbf{X}, \mathbf{A})^{\mathcal{K}} \simeq \Sigma \left(\bigvee_{\sigma \in \mathcal{K}} \mathbf{X}^{\wedge\sigma} \right)$$

Corollary 5.1.18
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $(\mathbf{X}, \mathbf{A})$ be n-pairs of CW complexes. Suppose

that X_i are null homotopic for all i. Then there is a homotopy equivalence

$$\Sigma(\mathbf{X},\mathbf{A})^{\mathcal{K}} \simeq \Sigma\left(\bigvee_{J\notin\mathcal{K}} |\mathcal{K}_J| * \mathbf{A}^{\wedge J}\right)$$

5.2 Polyhedral Products Arising from a Cone

Theorem 5.2.1 (Grbic–Theriault, Iriye–Kishimoto)
Let $\mathcal{K}$ be a shifted simplicial complex on $\{1,\dots,n\}$. Let $\mathbf{A}$ be n pointed CW complexes. Then there is a homotopy equivalence

$$(CA,A)^{\mathcal{K}} \simeq \bigvee_{J\notin\mathcal{K}} |\mathcal{K}_J| * \mathbf{A}^{\wedge J}$$

Example 5.2.2
Let $\mathbf{A}$ be n pointed CW complexes. Then there is a homotopy equivalence

$$(CA,A)^{\text{sk}_i\Delta^{n-1}} \simeq \bigvee_{k=i+2}^{n}\left(\bigvee_{1\leq j_1<\cdots<j_k\leq n} \left(\Sigma^{i+1}A_{j_1}\wedge\cdots\wedge A_{j_k}\right)^{\vee\binom{k-1}{i+1}}\right)$$

5.3 Other Results

Proposition 5.3.1
Let $\mathcal{K}$ be a simplicial complex. Let X be a space. Then there is a ring isomorphism

$$H^*(X^{\mathcal{K}},\mathbb{Z}) \cong \lim_{\sigma\in\mathcal{K}} H^*(X^\sigma,\mathbb{Z})$$

Proposition 5.3.2
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then we have

$$H^*((\mathbb{CP}^\infty,*)^{\mathcal{K}}) \cong \mathbb{Z}[\mathcal{K}]$$

Proposition 5.3.3
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the inclusion map $(\mathbb{CP}^\infty,*)^{\mathcal{K}} \hookrightarrow (\mathbb{CP}^\infty)^n$ induces the quotient map

$$\mathbb{Z}[x_1,\dots,x_n]\to\mathbb{Z}[\mathcal{K}]$$

where $\deg(x_i)=2$.

6 Moment Angle Complexes

6.1 Basic Definitions

Definition 6.1.1 (Moment Angle Complexes)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Define the moment angle complex of $\mathcal{K}$ to be

$$\mathcal{Z}_{\mathcal{K}} = (D^2, S^1)^{\mathcal{K}}$$

Proposition 6.1.2
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following diagram is a pullback

$$\begin{tikzcd} \mathcal{Z}_{\mathcal{K}} \arrow[r] \arrow[d] & (D^2)^n \subseteq \mathbb{C}^n \arrow[d, "\mu"] \\ \mathrm{cc}(\mathcal{K}) \arrow[r, hook] & I^n \end{tikzcd}$$

where $D^2 \subseteq \mathbb{C}$ is the closed unit disc and $\mu : (D^2)^n \to I^n$ is the map given by $(z_1, \dots, z_n) \mapsto (|z_1|^2, \dots, |z_n|^2)$.

Corollary 6.1.3
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then $\mathcal{Z}_{\mathcal{K}}$ is a T^n-equivariant subspace of $(D^2)^n$.

Proposition 6.1.4
Let $\mathcal{K}, \mathcal{L}$ be simplicial complexes. Let $f : \mathcal{K} \to \mathcal{L}$ be a simplicial map. Let (X, A) be a pair of spaces. Then f induces an map

$$\mathcal{Z}_{\mathcal{K}} \to \mathcal{Z}_{\mathcal{L}}$$

that is equivariant with respect to the map of tori.

Example 6.1.5
The following are true.
- $\mathcal{Z}_{\Delta^{n-1}} = (D^2)^n$.
- $\mathcal{Z}_{\partial\Delta^{n-1}} \cong S^{2n-1}$.

Proposition 6.1.6
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, m\}$. Then $\mathcal{Z}_{\mathcal{K}}$ is a topological $(m+n+1)$-manifold if and only if the following are true.
- $H_*(\mathrm{lk}_{\mathcal{K}}(\sigma)) \cong H_*(S^{n-1-\dim(\sigma)})$ for all $\sigma \in \mathcal{K}$.
- $H_*(\mathcal{K}) \cong H_*(S^n)$.

Corollary 6.1.7
Let $\mathcal{K}$ be a Gorenstein* simplicial complex over $\mathbb{Z}$. Then $\mathcal{Z}_{\mathcal{K}}$ is a topological manifold.

Corollary 6.1.8
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, m\}$ that is a triangulation of S^n. Then $\mathcal{Z}_{\mathcal{K}}$ is a compact topological manifold of dimension $m + n + 1$.

6.2 Homotopy Theory of Moment Angle Complexes

Corollary 6.2.1
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then there is a homotopy equivalence

$$\Sigma\mathcal{Z}_{\mathcal{K}} \simeq \bigvee_{J \notin \mathcal{K}} \Sigma^{2+|J|} |\mathcal{K}_J|$$

Proposition 6.2.2
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then there is a factorization

$$\begin{tikzcd} (\mathbb{CP}^\infty, *)^{\mathcal{K}} \arrow[rr, hook] \arrow[dr, "h"'] & & (\mathbb{CP}^\infty)^n \\ & ET^n \times_{T^n} \mathcal{Z}_{\mathcal{K}} \arrow[ur, "p"'] & \end{tikzcd}$$

where h is a homotopy equivalence and p is a fibration with fiber $\mathcal{Z}_{\mathcal{K}}$.

Corollary 6.2.3
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then we have

$$H^*_{T^n}(\mathcal{Z}_{\mathcal{K}}) \cong \mathbb{Z}[\mathcal{K}]$$

Proposition 6.2.4
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, m\}$. Then

$$\mathcal{Z}_{\mathcal{K}} \to (BS^1, *)^{\mathcal{K}} = (\mathbb{CP}^\infty)^{\mathcal{K}} \to BT^m = (\mathbb{CP}^\infty)^m$$

is a fibration sequence.

Proposition 6.2.5
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are true.

- $\mathcal{Z}_{\mathcal{K}}$ is 2-connected.
- $\pi_k(\mathcal{Z}_{\mathcal{K}}) \cong \pi_k((\mathbb{CP}^\infty, *)^{\mathcal{K}})$ for $k \geq 3$.

Looping the fibration sequence gives

$$\Omega\mathcal{Z}_{\mathcal{K}} \to \Omega(\mathbb{CP}^\infty)^{\mathcal{K}} \to T^m$$

which is principle, and so we have a splitting

$$\Omega(\mathbb{CP}^\infty)^{\mathcal{K}} \simeq \Omega\mathcal{Z}_{\mathcal{K}} \times T^m$$

This homotopy equivalence does not preserve monoid structures. It is given by the m generators of $\pi_2((\mathbb{CP}^\infty)^{\mathcal{K}}) \cong \mathbb{Z}^m$ given by $\widehat{\mu}_i : S^2 \to \mathbb{CP}^\infty \hookrightarrow \bigvee_{i=1}^m \mathbb{CP}^\infty \to (\mathbb{CP}^\infty)^{\mathcal{K}}$. Take the adjoint to get $\mu_i : S^1 \to \Omega(\mathbb{CP}^\infty)^{\mathcal{K}}$. Then we have the samelson product of these elements are 0 if and only if $\{i, j\} \in \mathcal{K}$.

6.3 Cohomology of Moment Angle Complexes

Consider the decomposition of D^2 into the following CW complex structure:

- One 0-cell $1 \in D^2$.
- One 1-cell S^1.
- One 2-cell D^2.

Let $n \in \mathbb{N}$. This gives the product complex D^{2n} a CW complex structure, whose cells are a product of 1-cells and 2-cells. We parameterize the cells in the following way. For $J, I \subseteq \{1, \dots, n\}$ with $J \cap I = \emptyset$, we denote $\chi(J, I)$ as the $(|J| + |I|)$-cell corresponding to the cell with the factor of a 1-cell in the jth position for $j \in J$, the factor of a 0-cell in the ith position for $i \in I$, and the factor of a 0-cell for $k \notin I \cup J$.

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then $\mathcal{Z}_{\mathcal{K}}$ is a union of cells in D^{2n}, and so is a cellular subcomplex of D^{2n}. Namely, for each $I \in \mathcal{K}$, we have that $\chi(J, I) \subseteq \mathcal{Z}_{\mathcal{K}}$ for all $J \subseteq \{1, \dots, n\} \setminus I$.

Q: How does the cellular cochain split into a bicomplex

Definition 6.3.1 (Bigrading in the Cochain Complex)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Define a bigrading in $C^*(\mathcal{Z}_{\mathcal{K}}; \mathbb{Z})$ as follows. For each basis element $\chi(J, I)$, define $\deg(\chi(J, I)) = (-|J|, 2(|I| + |J|))$.

Lemma 6.3.2
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following are true.

- The $(2j)$th components for each $j\in\mathbb{N}$ defines a cochain complex $C^{-i,2j}(\mathcal{Z}_\mathcal{K};\mathbb{Z})$ in i.
- $(C^{-*,*}(\mathcal{Z}_\mathcal{K};\mathbb{Z}),d_{CW},0)$ is a double cochain complex.
- The cellular cochain complex of $\mathcal{Z}_\mathcal{K}$ splits into a direct sum
$$C^*(\mathcal{Z}_\mathcal{K};\mathbb{Z})=\text{Tot}^\oplus(C^{-*,*}(\mathcal{Z}_\mathcal{K}))=\bigoplus_{j\in\mathbb{N}}C^{-*,2j}(\mathcal{Z}_\mathcal{K};\mathbb{Z})$$
where
$$C^k(\mathcal{Z}_\mathcal{K};\mathbb{Z})=\bigoplus_{2j-i=k}C^{-i,2j}(\mathcal{Z}_\mathcal{K};\mathbb{Z})$$

The consequence of the splitting is that cohomology also accquires a splitting
$$H^n(\mathcal{Z}_\mathcal{K};\mathbb{Z})=\bigoplus_{2j-i=n}H^{-i}(C^{-*,2j}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))$$

Proposition 6.3.3
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following are true.

- There are natural isomorphisms
$$H^{-p}(C^{-*,2q}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))\cong H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n\mathbb{Z}[\mathcal{K}]u_i\right)\right)_{2q}=H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n\mathbb{Z}[\mathcal{K}]u_i\right)_{2q}\right)$$
where $\Lambda\left(\bigoplus_{i=1}^n R[\mathcal{K}]u_i\right)$ is the differential graded algebra whose homological grading is given by $\deg(u_i)=-1$, internal grading is given by $\deg(v_i)=2$ and differential is given by $d(u_i)=v_i$ (and $d(v_i)=0$ since we think of the DGA as a DGA over the ring $\mathbb{Z}[\mathcal{K}]$).
- There above natural isomorphisms induces natural isomorphisms
$$H^k(\mathcal{Z}_\mathcal{K};\mathbb{Z})\cong\bigoplus_{2q-p=k}H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n\mathbb{Z}[\mathcal{K}]u_i\right)\right)_{2q}$$
- The above natural isomorphisms induce a natural isomorphism of rings
$$H^*(\mathcal{Z}_\mathcal{K};\mathbb{Z})\cong H_*\left(\Lambda\left(\bigoplus_{i=1}^n\mathbb{Z}[\mathcal{K}]u_i\right)\right)$$

Note that $H^n(\mathcal{Z}_\mathcal{K};\mathbb{Z})$ is not isomorphic to $H_{-n}\left(\Lambda\left(\bigoplus_{i=1}^n\mathbb{Z}[\mathcal{K}]u_i\right)\right)$ despite the suggestive notation. This is a mismatch of algebraic objects since the left is an abelian group while the right hand side is a $\mathbb{Z}[\mathcal{K}]$-module. In order to have an isomorphism of abelian groups we must further decompose each homology $\mathbb{Z}[\mathcal{K}]$-modules further into $\mathbb{Z}$-modules by considering $\mathbb{Z}[\mathcal{K}]$ as a $\mathbb{Z}$-module. But this results in a shift of degrees.

Let $K_\bullet(v_1,\dots,v_n;\mathbb{Z}[\mathcal{K}])$ be the Koszul complex over $\mathbb{Z}[v_1,\dots,v_n]$. Notice that the cochain complex defined by $K^n=K_n(v_1,\dots,v_n;\mathbb{Z}[\mathcal{K}])$ is precisely our differential graded algebra $\Lambda\left(\bigoplus_{i=1}^n\mathbb{Z}[\mathcal{K}]u_i\right)$. Then the cohomology of $\mathcal{Z}_\mathcal{K}$ acquires an additional description:
$$H^n(\mathcal{Z}_\mathcal{K};\mathbb{Z})\cong\bigoplus_{2q-p=n}\left(H_p\left(K_\bullet(v_1,\dots,v_n;\mathbb{Z}[\mathcal{K}])\right)\right)_{2q}$$
Finally, since $K_\bullet(v_1,\dots,v_n)$ is a free resolution for $\mathbb{Z}$ as a $\mathbb{Z}[v_1,\dots,v_n]$-module, we know that
$$H_p(K_\bullet(v_1,\dots,v_n;R[\mathcal{K}]))=\text{Tor}_p^{\mathbb{Z}[v_1,\dots,v_n]}(\mathbb{Z}[\mathcal{K}],\mathbb{Z})$$
The Koszul algebra gives the Tor groups the structure of an algebra, and so the cohomology ring acquires the description
$$H^*(\mathcal{Z}_\mathcal{K};\mathbb{Z})\cong\text{Tor}_*^{\mathbb{Z}[v_1,\dots,v_n]}(\mathbb{Z}[v_1,\dots,v_n],\mathbb{Z})$$

But again note that the nth cohomology group of $\mathcal{Z}_\mathcal{K}$ is not isomorphic to the nth Tor group of $\mathbb{Z}[\mathcal{K}]$ and $\mathbb{Z}$ by virtually the same reason as above.

Corollary 6.3.4
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the above homotopy equivalence descends to a natural isomorphism

$$\bigoplus_{J\subseteq\{1,\dots,n\}} C^{k-|J|-1}(\mathcal{K}_J;\mathbb{Z}) \cong C^*(\mathcal{Z}_\mathcal{K};\mathbb{Z})$$

given by

$$\alpha_L \mapsto \varepsilon(L,J)\chi(J\setminus L, L)$$

where α_L is the cochain basis of Hochster's formula given by the simplex $L\subseteq J$ and

$$\varepsilon(L,J) = \prod_{j\in L}\varepsilon(j,J)$$

and $\varepsilon(j,J) = (-1)^{r-1}$ if j is in the rth position of J.

Proposition 6.3.5
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following are true.

- There above natural isomorphisms induces natural isomorphisms

$$H^{-i}(C^{-*,2j}(\mathcal{Z}_\mathcal{K};\mathbb{Z})) \cong \bigoplus_{\substack{J\subseteq\{1,\dots,n\}\\ |J|=j}} \widetilde{H}^{j-i-1}(\mathcal{K}_J;\mathbb{Z})$$

  where by convention we set $\widetilde{H}^{-1}(\mathcal{K}_\emptyset) = \mathbb{Z}$.
- The above natural isomorphisms induces natural isomorphisms

$$H^k(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \bigoplus_{J\subseteq\{1,\dots,n\}} \widetilde{H}^{k-|J|-1}(\mathcal{K}_J;\mathbb{Z})$$

Proposition 6.3.6
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then the following are true.

- The ring structure in Hochster's formula is equal to the relative Kunneth theorem for

$$H^*(|\mathcal{K}_I * \mathcal{K}_J|) = H^*(\Sigma(|\mathcal{K}_I|\wedge|\mathcal{K}_J|)) = \widetilde{H}^{*-1}(|K_I|\times|K_J|, |K_I\vee K_J|)$$

  via the inclusion $\mathcal{K}_{I\amalg J}\hookrightarrow \mathcal{K}_I * \mathcal{K}_J$.
- The natural isomorphism in the above prp induces a natural isomorphism of rings

$$H^*(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \bigoplus_{J\subseteq\{1,\dots,n\}} \widetilde{H}^*(\mathcal{K}_J;\mathbb{Z})$$

While the cohomology groups can be easily computed using Hochster's formula, the ring structure can be seen more easily using the DGA.

Example 6.3.7
Let $\mathcal{K}$ be the standard polygon with vertices given by $\{1,\dots,5\}$ and edges given by $\{\{1,2\},\dots,\{4,5\},\{1,5\}\}$. We compute that

- $H^0(\mathcal{Z}_\mathcal{K};\mathbb{Z}) = \bigoplus_{J\subseteq\{1,\dots,5\}} \widetilde{H}^{0-|J|-1}(\mathcal{K}_\mathcal{J};\mathbb{Z}) = \widetilde{H}^{-1-|\emptyset|}(\mathcal{K}_\emptyset;\mathbb{Z}) = \mathbb{Z}$.
- $H^1(\mathcal{Z}_\mathcal{K};\mathbb{Z}) = H^2(\mathcal{Z}_\mathcal{K};\mathbb{Z}) = H^5(\mathcal{Z}_\mathcal{K};\mathbb{Z}) = H^6(\mathcal{Z}_\mathcal{K};\mathbb{Z}) = 0$.

Also we have

$$\begin{aligned}
H^3(\mathcal{Z}_\mathcal{K};\mathbb{Z}) &= \bigoplus_{J\subseteq\{1,\dots,5\}} \widetilde{H}^{2-|J|}(\mathcal{K}_\mathcal{J};\mathbb{Z}) \\
&= \widetilde{H}^0(\mathcal{K}_{\{1,3\}};\mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{1,4\}};\mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{2,4\}};\mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{2,5\}};\mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{3,5\}};\mathbb{Z}) \\
&= \mathbb{Z}[\alpha_{\{3\}}] \oplus \mathbb{Z}[\alpha_{\{4\}}] \oplus \mathbb{Z}[\alpha_{\{4\}}] \oplus \mathbb{Z}[\alpha_{\{5\}}] \oplus \mathbb{Z}[\alpha_{\{5\}}] \\
&= \mathbb{Z}[u_1v_3] \oplus \mathbb{Z}[u_1v_4] \oplus \mathbb{Z}[u_2v_4] \oplus \mathbb{Z}[u_2v_5] \oplus \mathbb{Z}[u_3v_5]
\end{aligned}$$

$$\begin{aligned}
H^4(\mathcal{Z}_\mathcal{K};\mathbb{Z}) &= \bigoplus_{J\subseteq\{1,\dots,5\}} \widetilde{H}^{3-|J|}(\mathcal{K}_\mathcal{J};\mathbb{Z}) \\
&= \widetilde{H}^0(\mathcal{K}_{\{1,2,4\}};\mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{2,3,5\}};\mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{1,3,4\}};\mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{2,4,5\}};\mathbb{Z}) \oplus \widetilde{H}^0(\mathcal{K}_{\{1,3,5\}};\mathbb{Z}) \\
&= \mathbb{Z}[\alpha_{\{4\}}] \oplus \mathbb{Z}[\alpha_{\{5\}}] \oplus \mathbb{Z}[\alpha_{\{1\}}] \oplus \mathbb{Z}[\alpha_{\{2\}}] \oplus \mathbb{Z}[\alpha_{\{3\}}] \\
&= \mathbb{Z}[-u_1u_2v_4] \oplus \mathbb{Z}[-u_2u_3v_5] \oplus \mathbb{Z}[-u_3u_4v_1] \oplus \mathbb{Z}[-u_4u_5v_2] \oplus \mathbb{Z}[u_1u_5v_3]
\end{aligned}$$

and

$$\begin{aligned}
H^7(\mathcal{Z}_\mathcal{K};\mathbb{Z}) &= \bigoplus_{J\subseteq\{1,\dots,5\}} \widetilde{H}^{6-|J|}(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \\
&= \widetilde{H}^1(\mathcal{K};\mathbb{Z}) \\
&= \mathbb{Z}[\alpha_{\{1,2\}}] \\
&= \mathbb{Z}[-u_3u_4u_5v_1v_2]
\end{aligned}$$

where wee identified the generators in Hochster's formula to generators in the DGA via the given formulas. Denote the generators of H^3 by $x_1,\dots,x_5$ and denote the generators of H^4 by $y_1,\dots,y_5$ and denote the generator of H^7 by z. Then we have the relations $x_1y_4 = x_2y_2 = -x_3y_5 = x_4y_3 = x_5y_1 = z$, and 0 otherwise. Thus relabelling the generators give

$$H^*(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \frac{\Lambda_\mathbb{Z}[x_1,\dots,x_5,y_1,\dots,y_5]}{(x_iy_i - x_jy_j, x_iy_j, x_ix_j, y_i, y_j \mid 1 \leq i \neq j \leq 5)}$$

where $\deg(x_i) = 3$ and $\deg(y_i) = 4$. Notice that this is precisely the cohomology ring of $(S^3 \times S^4)^{\#5}$.

There are three descriptions of the cohomology groups:

$$H^k(\mathcal{Z}_\mathcal{K};\mathbb{Z}) \cong \bigoplus_{2q-p=k} H_{-p}\left(\Lambda\left(\bigoplus_{i=1}^n \mathbb{Z}[\mathcal{K}]u_i\right)\right)_{2q} = \bigoplus_{J\subseteq\{1,\dots,n\}} \widetilde{H}^{k-|J|-1}(\mathcal{K}_J;\mathbb{Z})$$

For $L \subseteq J$ such that α_L is a cochain basis in the Hochster formula, we have

$$[\varepsilon(L,J)\chi(J\setminus L, L)] \leftrightarrow [\varepsilon(L,J)u_{J\setminus L}v_L] \leftrightarrow \alpha_L$$

Proposition 6.3.8

Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots.n\}$. Then the following are true.

- rank$(H^0(C^{-*,0}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))) = 1$.
- rank$(H^0(C^{-*,2q}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))) = 0$ for $q \neq 0$.
- rank$(H^{-p}(C^{-*,2q}(\mathcal{Z}_\mathcal{K};\mathbb{Z})))$ for $p > q$ or $q > n$.
- rank$(H^{-p}(C^{-*,2q}(\mathcal{Z}_\mathcal{K};\mathbb{Z})))$ for $p \geq q > 0$ or $q - p > \dim(\mathcal{K}) + 1$.
- rank$(H^{-1}(C^{-*,4}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))) = \binom{n}{2} - f_1$ (recall that f_1 is the number of edges of $\mathcal{K}$).
- rank$(H^{-(n-\dim(\mathcal{K})+1)}(C^{-*,2n}(\mathcal{Z}_\mathcal{K};\mathbb{Z}))) = $ rank$(\widetilde{H}^{\dim(\mathcal{K})}(\mathcal{K}))$.

Proposition 6.3.9

Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots.n\}$. Then the following are true.

- rank$(H^0(\mathcal{Z}_\mathcal{K};\mathbb{Z})) = 1$.

- $\operatorname{rank}(H^1(\mathcal{Z}_\mathcal{K};\mathbb{Z})) = \operatorname{rank}(H^2(\mathcal{Z}_\mathcal{K};\mathbb{Z})) = 0$.
- $\operatorname{rank}(H^3(\mathcal{Z}_\mathcal{K};\mathbb{Z})) = \binom{n}{2} - f_1$ (recall that f_1 is the number of edges of $\mathcal{K}$).
- $\operatorname{rank}(H^{n+\dim(\mathcal{K})+1}(\mathcal{Z}_\mathcal{K};\mathbb{Z})) = \operatorname{rank}(\widetilde{H}^{\dim(\mathcal{K})}(\mathcal{K}))$.

Lemma 6.3.10
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. If $\mathcal{K}$ is a triangulation of S^{m-1}, then the top cohomology $H^{n+m}(\mathcal{Z}_\mathcal{K};\mathbb{Z})$ is generated by $[\chi(\{1,\dots,n\} \setminus I, I)]$ for any $I \in \mathcal{K}$ and such that $|I| = n$.

If we use the Koszul complex as the cohomology model for $\mathcal{Z}_\mathcal{K}$, then the top cohomology is generated by $u_{\{1,\dots,n\}\setminus I}v_I$ for any $I \in \mathcal{K}$ such that $|I| = n$.

6.4 Geometry of Moment Angle Complexes

Definition 6.4.1 (Coordinate Subspaces)
Let $I \subseteq \{1,\dots,n\}$. Define the coordinate subspace of I to be

$$L_I = \{(z_1,\dots,z_n) \in \mathbb{C}^n \mid z_j = 0 \text{ for all } j \in I\}$$

Definition 6.4.2 (Coordinate Subspace Arrangement)
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Define the coordinate subspace arrangement of $\mathcal{K}$ to be

$$U(\mathcal{K}) = \mathbb{C}^n \setminus \bigcup_{\sigma\in\mathcal{K}} L_\sigma$$

Proposition 6.4.3
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then we have $U(\mathcal{K}) = (\mathbb{C},\mathbb{C}^\times)^\mathcal{K}$.

Theorem 6.4.4
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. The map $(\mathbb{C},\mathbb{C}^\times) \to (D^2,S^1)$ induces a T^n-equivariant deformation retract

$$U(\mathcal{K}) \simeq \mathcal{Z}_\mathcal{K}$$

7 Geometric Structures of Moment Angle Complexes

7.1 Moment Angle Polytopes and Intersection of Quadrics

Let $P \subseteq \mathbb{R}^n$ be a polytope defined by

$$P = \{x \in \mathbb{R}^n \mid \langle a_i, x\rangle + b_i \geq 0 \text{ for } 1 \leq i \leq m\}$$

such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Let

$$A = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix} \in M_{m\times n}(\mathbb{R}) \quad \text{and} \quad b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} \in \mathbb{R}^m$$

Recall that

$$P = \{x \in \mathbb{R}^n \mid Ax + b \geq 0\}$$

Definition 7.1.1 (Moment Angle Polytopes)
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Let $i_P : \mathbb{R}^n \to \mathbb{R}^m$ be the map $i_P(x) = Ax - b$. Let $\mu : \mathbb{C}^m \to \mathbb{R}^m$ be the map $\mu(z_1, \dots, z_m) = (|z_1|^2, \dots, |z_m|^2)$. Define the moment angle polytope $\mathcal{Z}_{A,b}$ to be the pullback

$$\begin{tikzcd} \mathcal{Z}_{A,b} \arrow[r] \arrow[d] & \mathbb{C}^n m \arrow[d, "\mu"] \\ P \arrow[r, hook, "i_P"] & \mathbb{R}^m \end{tikzcd}$$

Proposition 7.1.2
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Then there is a homeomorphism

$$\mathcal{Z}_{A,b} \cong \left\{ z = (z_1, \dots, z_m) \in \mathbb{C}^m \;\middle|\; \Gamma \begin{pmatrix} |z_1|^2 \\ \vdots \\ |z_m|^2 \end{pmatrix} = \Gamma b \right\}$$

where $\Gamma = \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_{m-n} \end{pmatrix}$ and $\gamma_1, \dots, \gamma_{m-n}$ is a basis for $\ker(A^T)$.

This makes $Z_{A,b}$ an intersection of quadrics in $\mathbb{R}$, as well as an intersection of Hermitian quadrics (Hermitian since $|z_i|^2 = z_i\overline{z_i}$).

Definition 7.1.3 (Intersection of Quadrics)
Let $\Gamma \in M_{(m-n)\times n}(\mathbb{R})$ be a matrix. Let $\delta \in \mathbb{R}^{m-n}$. Define the intersection of quadrics associated to (Γ, δ) by

$$\mathcal{Z}_{\Gamma,\delta} = \left\{ z = (z_1, \dots, z_m) \in \mathbb{C}^m \;\middle|\; \Gamma \begin{pmatrix} |z_1|^2 \\ \vdots \\ |z_m|^2 \end{pmatrix} = \delta \right\}$$

Lemma 7.1.4
Let $\Gamma \in M_{(m-n)\times n}(\mathbb{R})$ be a matrix. Let $\delta \in \mathbb{R}^{m-n}$. Then we have

$$\frac{\mathcal{Z}_{\Gamma,\delta}}{T^m} \cong \left\{ y = (y_1, \dots, y_m) \in \mathbb{R}^m \;\middle|\; \Gamma y = \delta \right\}$$

Definition 7.1.5 (Associated Polytope)
Let $\Gamma \in M_{(m-n)\times n}(\mathbb{R})$ be a matrix. Let $\delta \in \mathbb{R}^{m-n}$. Define the associated polytope of $\mathcal{Z}_{\Gamma,\delta}$ to be $\mathcal{Z}_{A,b}$ where
- The columns $a_1, \dots, a_m$ of A form a basis for $\ker(\Gamma^T)$.
- $b \in \mathbb{R}^m$ is such that $\Gamma b = \delta$.

Theorem 7.1.6
There is a one to one correspondence

$$\frac{\left\{\mathcal{Z}_{A,b} \;\middle|\; \substack{A\in M_{m\times n}(\mathbb{R}), b\in\mathbb{R}^m \text{ such that} \\ \text{the columns } a_1,\dots,a_m \text{ of } A \text{ span } \mathbb{R}^n}\right\}}{\text{Isomorphism of } \mathbb{R}^n} \overset{1:1}{\longleftrightarrow} \frac{\left\{\mathcal{Z}_{\Gamma,\delta} \;\middle|\; \substack{\Gamma\in M_{(m-n)\times n}(\mathbb{R}), \delta\in\mathbb{R}^{m-n} \text{ such that} \\ \text{the columns } \gamma_1,\dots,\gamma_m \text{ of } \Gamma \text{ span } \mathbb{R}^{m-n}}\right\}}{\text{Linear isomorphism of } \mathbb{R}^{m-n}}$$

Proposition 7.1.7
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Then P is simple if and only if $\mathcal{Z}_{A,b}$ is a smooth manifold of real dimension $m + n$.

7.2 Relation to Moment Angle Complexes

Proposition 7.2.1
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Then there is a homotopy equivalence

$$\mathcal{Z}_{\mathcal{K}_P} \simeq \mathcal{Z}_{A,b}$$

Proposition 7.2.2
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. If P is simple, then there is a T^m-equivariant homeomorphism

$$\mathcal{Z}_{\mathcal{K}_P} \cong \mathcal{Z}_{A,b}$$

There is a canonical choice of intersection of quadrics in this case.

Proposition 7.2.3
Let $P \subseteq \mathbb{R}^n$ be a polytope defined by $A \in M_{m\times n}(\mathbb{R})$ and $b \in \mathbb{R}^m$ such that $a_1, \dots, a_m$ span $\mathbb{R}^n$. Suppose that P is simple. Then there is a T^m-equivariant diffeomorphism

$$\mathcal{Z}_{\mathcal{K}_P} \cong \left\{ z = (z_1, \dots, z_m) \in \mathbb{C}^m \;\middle|\; \sum_{k=1}^m |z_k|^2 = 1, \sum_{k=1}^m g_k |z_k|^2 = 0 \right\}$$

where $g_1, \dots, g_m \in \mathbb{R}^{m-n-1}$ is a combinatorial gale diagram of P^*.

Theorem 7.2.4 (de Medrano)
Let $P \subseteq \mathbb{R}^n$ be a simple polytope. Suppose that $\mathcal{Z}_{\mathcal{K}_P}$ is a non-empty and non-degenerate intersection of 3 quadrics. Then exactly one of the following hold.
- P is combinatorially equivalent to a product of three simplicies, and there is a diffeomorphism
$$\mathcal{Z}_{\mathcal{K}_P} \cong S^{n_1} \times S^{n_2} \times S^{n_3}$$
where n_1, n_2, n_3 are odd.
- There is a diffeomorphism
$$\mathcal{Z}_{\mathcal{K}_P} \cong \#_{k=3}^{n+1} (S^k \times S^{2n+3-k})^{\# q_k}$$
for some q_k determined by the Gale diagram of P.

8 Real Moment Angle Complexes

8.1 Basic Definitions

Definition 8.1.1 (Real Moment Angle Complexes)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Define the real moment angle complex of $\mathcal{K}$ to be

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}} = (D^1, S^0)^{\mathcal{K}}$$

Proposition 8.1.2
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following diagram is a pullback

$$\begin{tikzcd} \mathbb{R}\mathcal{Z}_{\mathcal{K}} \arrow[r] \arrow[d] & (D^1)^n \subseteq \mathbb{C}^n \arrow[d, "\mu"] \\ \text{cc}(\mathcal{K}) \arrow[r, hook] & I^n \end{tikzcd}$$

where $D^1 \subseteq \mathbb{R}$ is the closed unit disc and $\mu : (D^2)^n \to I^n$ is the map given by $(x_1, \dots, x_n) \mapsto (x_1^2, \dots, x_n^2)$.

Example 8.1.3
We have

$$\mathcal{Z}_{\partial\Delta^{n-1}} \cong S^{n-1}$$

Proposition 8.1.4
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$ that is a triangulation of S^m. Then $\mathcal{Z}_{\mathcal{K}}$ is a compact topological manifold of dimension $m + 1$.

Example 8.1.5
Let $\mathcal{K}$ be the boundary of an n-gon. Then there is a homeomorphism

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}} \cong \Sigma_{1+(n-4)2^{n-3}}$$

Corollary 8.1.6
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then there is a homotopy equivalence

$$\Sigma\mathbb{R}\mathcal{Z}_{\mathcal{K}} \simeq \bigvee_{J \notin \mathcal{K}} \Sigma^2 |\mathcal{K}_J|$$

9 Fat Wedge Filtrations of Polyhedral Products

9.1 Basic Definitions

Definition 9.1.1 (The Classical Fat Wedge Filtration)
Let $X_1, \dots, X_n$ be pointed spaces. Define the fat wedge filtration of $\prod_{i=1}^n X_i$ at the kth step by

$$T^k(X_1, \dots, X_n) = \left\{ (x_1, \dots, x_n) \in \prod_{i=1}^n X_i \;\middle|\; |\{i \mid x_i = *\}| \geq n - k \right\}$$

Definition 9.1.2 (Fat Wedge Filtration of Polyhedral Products)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $(\mathbf{X}, \mathbf{A}, *)$ be n-pairs of pointed spaces. Define the fat wedge filtration of the polyhedral product at the kth step by

$$(\mathbf{X}, \mathbf{A})^{\mathcal{K}, k} = (\mathbf{X}, \mathbf{A})^{\mathcal{K}} \cap T^k(X_1, \dots, X_n)$$

9.2 Decomposition of the Fat Wedge Filtration

We consider the fat wedge filtration of $\mathbb{R}\mathcal{Z}_{\mathcal{K}}$.

Proposition 9.2.1
Let $\mathcal{K}$ be a simpicial complex on $\{1, \dots, n\}$. Then we have

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}}^i = \bigcup_{I \subseteq \{1, \dots, n\}, |I| = i} \mathbb{R}\mathcal{Z}_{\mathcal{K}_I}$$

Recall that cubical complex of a simplicial complex. We know that

$$(D^1, S^0)^\sigma = \bigcup_{\substack{\mu \subseteq \tau \subseteq \text{Vert}(\mathcal{K}) \\ \tau \setminus \mu = \sigma}} C_{\mu \subseteq \tau}$$

and

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}} = \bigcup_{\sigma \in \mathcal{K}} \bigcup_{\substack{\mu \subseteq \tau \subseteq \text{Vert}(\mathcal{K}) \\ \tau \setminus \mu = \sigma}} C_{\mu \subseteq \tau}$$

Proposition 9.2.2
Let $n \in \mathbb{N}^+$. Let $\mathcal{K}$ be a simplicial complex on n. Let $J \subseteq \mathcal{K}$. Then there is a relative homeomorphism

$$(|\text{Cone}(\text{Bd}(\mathcal{K}_J))|, |\text{Bd}(\mathcal{K}_J)|) \to (\mathbb{R}\mathcal{Z}_{\mathcal{K}_J}, \mathbb{R}\mathcal{Z}_{\mathcal{K}_J}^{|J|-1})$$

Proof
Let $i_c : |\Delta^n| \to I^{n+1}$ denote the cubical embedding. For each $J \subseteq \{1, \dots, n\}$, the embedding restricts to a map $i_c^J : |\mathcal{K}_J| \to I^J$ where $I^J = (I, 0)^J$ is the moment angle complex of I. For each $\mu \subseteq \tau \subseteq J$, denote $C_{\mu \subseteq \tau}^J$ the $C_{\mu \subseteq \tau}$ face of I^J (so $C_{\mu \subseteq \tau}^J$ still lives inside I^J but some coordinates are 0). We compute that

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}_J}^{|J|-1} = \bigcup_{A \subseteq J, |A| = |J|-1} \bigcup_{\sigma \in \mathcal{K}_A} \bigcup_{\substack{\mu \subseteq \tau \subseteq A \\ \tau \setminus \mu = \sigma}} C_{\sigma \subseteq \tau}^A = |\text{Bd}(\mathcal{K}_J)|$$

and similarly

$$\mathbb{R}\mathcal{Z}_{\mathcal{K}_J} = \bigcup_{\sigma \in \mathcal{K}_J} \bigcup_{\substack{\mu \subseteq \tau \subseteq J \\ \tau \setminus \mu = \sigma}} C_{\mu \subseteq \tau}^J = |\text{Cone}(\text{Bd}(\mathcal{K}_J))|$$

■

Note: $f:(X,A)\to(Y,B)$ is a relative homeomorphism if $f:X\to Y$ is a homeomorphism and $\widetilde{f}:X/A\to Y/B$ is a homeomorphism.

Let us verify this for the case $\mathcal{K}=\partial\Delta^2$. Pick $I=\{1\}$. Let $i_c:|\mathcal{K}|\to I^n$ be the map that is the cubical embedding. Now the relative version of the identity

$$(D^1,S^0)^\sigma=\bigcup_{\substack{\mu\subseteq\tau\subseteq\text{Vert}(\mathcal{K})\\ \tau\setminus\mu=\sigma}}C_{\mu\subseteq\tau}$$

says that if $\sigma\in\mathcal{K}_I$, then

$$(D^1,S^0)^\sigma=\bigcup_{\substack{\mu\subseteq\tau\subseteq\text{Vert}(\mathcal{K})\\ \tau\setminus\mu=\sigma}}C^I_{\mu\subseteq\tau}$$

where we consider $(D^1,S^0)^\sigma$ as a subspace of $\mathcal{Z}_\mathcal{K}$ by considering $\mathcal{Z}_{\mathcal{K}_I}$ as a subspace of $\mathcal{Z}_\mathcal{K}$, and on the right $C^I_{\mu\subseteq\tau}$ is the usual meaning but restricted to the cube $(D^1)^I$ of I^3.

Restricting this map to $\mathcal{K}_I$, this the map sending $\{1\}$ to $C^I_{\{1\},\{1\}}$ which is the point $-1\in C^I_{\{1\},\{1\}}=D^1$, but is the point $(-1,1,1)$ in I^3.

This induces a relative homeomorphism

$$\coprod_{J\subseteq\{1,\dots,n\},|J|=i}(|\text{Cone}(\text{Bd}(\mathcal{K}_J))|,|\text{Bd}(\mathcal{K}_J)|)\to(\mathbb{R}\mathcal{Z}^i_\mathcal{K},\mathbb{R}\mathcal{Z}^{i-1}_\mathcal{K})$$

Let $\varphi_{\mathcal{K}_J}$ denote the map $|\text{Bd}(\mathcal{K}_J)|\to\mathbb{R}\mathcal{Z}^{i-1}_{\mathcal{K}_J}$. Let $j_{\mathcal{K}_I}$ denote the inclusion $\mathbb{R}\mathcal{Z}^{i-1}_{\mathcal{K}_I}\hookrightarrow\mathbb{R}\mathcal{Z}^{i-1}_\mathcal{K}$.

Proposition 9.2.3
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Then there is a pushout

$$\begin{tikzcd}
\coprod_{I\subseteq\{1,\dots,n\},|I|=i}|\text{Bd}(\mathcal{K}_I)| \arrow[r, "\coprod_{I\subseteq\{1,\dots,n\},|I|=i}(j_{\mathcal{K}_I}\circ\varphi_{\mathcal{K}_I})"] \arrow[d, hook] & \mathbb{R}\mathcal{Z}^{i-1}_\mathcal{K} \arrow[d] \\
\coprod_{I\subseteq\{1,\dots,n\},|I|=i}|\text{Cone}(\text{Bd}(\mathcal{K}_I))| \arrow[r] & \mathbb{R}\mathcal{Z}^i_\mathcal{K}
\end{tikzcd}$$

for $1\leq i\leq n$.

Lemma 9.2.4
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $\mathbf{X}=(X_1,\dots,X_n)$ be spaces. Then $(C\mathbf{X}_I,\mathbf{X}_I)^{\mathcal{K}_I}$ is a retract of $(C\mathbf{X},\mathbf{X})^\mathcal{K}$.

Proposition 9.2.5
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $\mathbf{X}=(X_1,\dots,X_n)$ be spaces. Then we have

$$(C\mathbf{X},\mathbf{X})^{\mathcal{K},i}=\bigcup_{I\subseteq\{1,\dots,n\},|I|=i}\mathcal{Z}_{\mathcal{K}_I}(C\mathbf{X}_I,\mathbf{X}_I)$$

Proposition 9.2.6
Let $\mathcal{K}$ be a simplicial complex on $\{1,\dots,n\}$. Let $\mathbf{X}=(X_1,\dots,X_n)$ be spaces. Then there is a relative homeomorphism

$$(|\text{Cone}(\text{Bd}(\mathcal{K}_I))|,|\text{Bd}(\mathcal{K}_I)|)\times(X^{\times I},T^{i-1}(\mathbf{X}_I))\to((C\mathbf{X}_I,\mathbf{X}_I)^{\mathcal{K}_I},(C\mathbf{X}_I,\mathbf{X}_I)^{\mathcal{K}_I,i-1})$$

9.3 Trivial Fat Wedge Filtrations

Definition 9.3.1 (Trivial Real Fat Wedge Filtration)
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $\mathbf{X} = (X_1, \dots, X_n)$ be spaces. We say that the fat wedge filtration of $\mathbb{R}\mathcal{Z}_\mathcal{K}$ is trivial if $\varphi_{\mathcal{K}_I}$ is trivial for all $I \subseteq \{1, \dots, n\}$.

Proposition 9.3.2
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. If $\dim(\mathcal{K}) \geq n - 2$, then the fat wedge filtration of $\mathbb{R}\mathcal{Z}_\mathcal{K}$ is trivial.

Theorem 9.3.3
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. If the fat wedge filtration of $\mathbb{R}\mathcal{Z}_\mathcal{K}$ is trivial, then the BBCG decomposition desuspends and we have a homotopy equivalence

$$(C\mathbf{X}, \mathbf{X})^\mathcal{K} \simeq \bigvee_{\emptyset \neq I \subseteq \{1,\dots,n\}} |\Sigma\mathcal{K}_I| \wedge \left(\bigwedge_{i \in I} X_i \right)$$

Provable: the fat wedge filtration of $\mathcal{Z}_\mathcal{K}$ is a cone decomposition. Hence trivialness can also be defined.

Corollary 9.3.4
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. If the fat wedge filtration of $\mathbb{R}\mathcal{Z}_\mathcal{K}$ is trivial, then the fat wedge filtration of $\mathcal{Z}_\mathcal{K}$ is trivial.

Theorem 9.3.5
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Then the following are equivalent.

- The fat wedge filtration of $\mathcal{Z}_\mathcal{K}$ is trivial.
- $\mathcal{Z}_\mathcal{K}$ is a co-H-space.
- The BBCG decomopsition desuspends to give

$$\mathcal{Z}_\mathcal{K} \simeq \bigvee_{\emptyset \neq I \subseteq \{1,\dots,n\}} \Sigma^{|I|+1} |\mathcal{K}_I|$$

Theorem 9.3.6
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. If $\mathcal{K}^\vee$ is sequentially Cohen Macaulay, then the fat wedge filtration of $\mathbb{R}\mathcal{Z}_\mathcal{K}$ is trivial.

Theorem 9.3.7
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. If $\mathcal{K}$ is $\lceil \frac{\dim(\mathcal{K})}{2} \rceil$-neighbourly, then the fat wedge filtration of $\mathbb{R}\mathcal{Z}_\mathcal{K}$ is trivial.

10 Quasitoric Manifolds

10.1 The Davis-Januszkiewicz Classification Theorem

Definition 10.1.1 (Quasitoric Manifolds)
Let M be a smooth manifold of dimension $2n$. Let T^n be a torus acting smoothly on M. We say that M is a quasitoric manifold if the following are true.

- For each $p \in M$, there is a T^n-equivariant chart $(U, \varphi : U \to \mathbb{C}^n)$ where $\mathbb{C}^n$ is equipped with the standard torus action.
- There exists a convex polytope P such that there is a homeomorphism

$$M/T^n \cong P$$

Toric symplectic manifolds are quasitoric manifolds. Moment angle manifolds are NOT quasitoric manifolds.

Definition 10.1.2 (Orbit Type Strata)
Let M be a quasitoric manifold. Let P be the underlying convex polytope. Let $H \leq T^n$ be a subgroup. Define the orbit type strata of H to be

$$M_{(H)} = \{x \in M \mid \text{Stab}_{T^n}(x) = H\}$$

Proposition 10.1.3
Let M be a quasitoric manifold. Let P be the underlying convex polytope. Then there is a bijection

$$\{F \in \mathcal{F}(P) \mid F \text{ has codimension } k\} \overset{1:1}{\leftrightarrow} \left\{ N \subseteq M \;\middle|\; \begin{array}{c} N \text{ is an orbit type strata of subtori of} \\ T^n \text{ of dimension } k \end{array} \right\}$$

given as follows. Let $\pi : M \to P$ be the quotient map. Then we have

$$F_i \mapsto \pi^{-1}(\text{relint}(F))$$

Definition 10.1.4 (Primitive Vectors)
Let $v = (v_1, \dots, v_n) \in \mathbb{Z}^n$. We say that v is primitive if $v \neq \lambda u$ for any $\lambda \in \mathbb{Z} \setminus \{0\}$ and $u \in \mathbb{Z}^n$.

Definition 10.1.5 (The Characteristic Matrix)
Let M be a quasitoric manifold. Let P be the underlying convex polytope. Define the characteristic matrix Λ as follows. For each facet F_i of P for $1 \leq i \leq m$, the facet corresponds to a a circle $S_i^1 \subseteq T^n$ by the above. Every circle $S_i^1 \subseteq T^n$ corresponds to a unique primitive vector $v_i = (v_{1i}, \dots, v_{1n}) \in \mathbb{Z}^n$ up to sign. Then

$$\Lambda = \begin{pmatrix} v_{11} & \cdots & v_{1m} \\ \vdots & \ddots & \vdots \\ v_{n1} & \cdots & v_{nm} \end{pmatrix} \in M_{n \times m}(\mathbb{Z})$$

Definition 10.1.6 (Characteristic Pairs)
Let P be a simple polytope of dimension n. Let $F_1, \dots, F_m$ be the facets of P. Let $\Lambda \in M_{n \times m}(\mathbb{Z})$ be a matrix. Let $c_1, \dots, c_m$ be the columns of Λ. We say that (P, Λ) is a characteristic pair if the following are true.

- c_i is a primitive vector for all $1 \leq i \leq m$.
- If $F_{i_1} \cap \cdots \cap F_{i_n}$ is a vertex of P, then

$$\det \begin{pmatrix} | & & | \\ c_{i_1} & \cdots & c_{i_n} \\ | & & | \end{pmatrix} = \pm 1$$

Definition 10.1.7 (The Associated Quasitoric Manifold)
Let (P, Λ) be a characteristic pair. Suppose that $\dim(P) = n$. Let $F_1, \dots, F_m$ be the facets of P. Let $c_1, \dots, c_m$ be the columns of Λ. Define the associated quasi-toric manifold of (P, Λ) by

$$M(P, \Lambda) = \frac{P \times T^n}{\sim}$$

where $(p, t_1) \sim (p, t_2)$ if $t_2^{-1} t_1 \in \langle c_i \mid p \in F(p) \rangle \cong T^k \leq T^n$ and $F(p)$ is the unique facet such that $p \in \text{relint}(F(p))$ for any $p \in P$.

Theorem 10.1.8 (Davis-Januszkiewicz Classification Theorem)
There is a one to one bijection

$$\frac{\{M \mid M \text{ is a quasitoric manifold}\}}{\cong} \overset{1:1}{\leftrightarrow} \frac{\{(P, \Lambda) \text{ is a characteristic pair}\}}{\sim}$$

where $(P, \Lambda) \sim (Q, \Gamma)$ if P and Q are combinatorially equivalent and there exists $A \in \text{GL}(n, \mathbb{Z})$ with $\det(A) = \pm 1$ such that $A\Lambda$ and Q are equal up to permutation of columns.

10.2 Basic Structures

Proposition 10.2.1
Let M be a quasitoric manifold with characteristic pair (P, Λ). Let $F_1, \dots, F_m$ be the facets of P. Then there is an isomorphism of graded rings

$$H^*(M; \mathbb{Z}) \cong \frac{\mathbb{Z}[x_1, \dots, x_m]}{I_{\text{SR}} + I_\Lambda}$$

where
- $\deg(x_i) = 2$ for $1 \leq i \leq m$.
- I_{SR} is the Stanley-Reisner Ideal given by

$$I_{\text{SR}} = (xj_1 \cdots x_{j_t} \mid F_{j_1} \cap \cdots \cap F_{j_t} = \emptyset)$$

- I_Λ is the linear relations given by

$$I_\Lambda = (\Lambda(x_1, \dots, x_n) = 0)$$

Recall that we denote $(h_0(P), \dots, h_n(p))$ by the h-vector of P.

Corollary 10.2.2
Let M be a quasitoric manifold with characteristic pair (P, Λ). Then we have

$$\text{rank}(H^{2k}(M; \mathbb{Z})) = h_k(P)$$

10.3 Bundles over Quasitoric Manifolds

Proposition 10.3.1
Let M be a quasitoric manifold. Then $TM \oplus \mathbb{R}^k$ admits a complex structure for some $k \in \mathbb{N}$.

The chern class of $TM \oplus \mathbb{R}^k$ is given by

$$c(TM \oplus \mathbb{R}^k) = \prod_{i=1}^{m} (1 + x_i) \in H^*(M) \cong \frac{\mathbb{Z}[x_1, \dots, x_m]}{I_{\text{SR}} + I_\Lambda}$$

There is a diffeo $Z_P/(G(P) \cong T^{m-n}) \cong M(P, \Lambda)$ for any P. (from this perspective one can see that projective toric manifolds are quasi-toric manifolds). Also this map $Z_P \to M(P, \Lambda)$ is a principle $G(P)$-bundle.

If $M(P,\Lambda) = V_P$ is a projective toric variety, the classical construction $\mathbb{C}^m \setminus Z(\Sigma_P)$ exhibits Z_P algebraically: there is an equivariant homotopy equivalence $\mathbb{C}^m \setminus Z(\Sigma_P) \simeq Z_P$.

11 Papers

11.1 Stanton, Theriault: Polyhedral products associated to pseudomanifolds

Define
$$\mathcal{P} = \left\{ X \in \mathbf{Top} \;\middle|\; X \simeq \prod_{i=1}^{n} \Omega^{j_i} S^{k_i} \right\}$$
Define
$$\mathcal{W} = \left\{ X \in \mathbf{Top} \;\middle|\; X \simeq \bigvee_{i=1}^{n} S^{k_i} \right\}$$

- If A is a retract of X and $X \in \mathcal{P}$, then $A \in \mathcal{P}$.
- If $\Sigma A_1, \dots, \Sigma A_n \in \mathcal{W}$ and $\mathcal{K}$ is one-dimensional, then $\Omega(C\mathbf{A}, \mathbf{A})^{\mathcal{K}} \in \mathcal{P}$.
- If $\Sigma A_1, \dots, \Sigma A_n \in \mathcal{W}$ and $\mathcal{K}$ is two-dimensional and $H^*(|L|)$ is torsion free for all full subcomplexes L with complete 1-skeleton, then $\Omega(C\mathbf{A}, \mathbf{A})^{\mathcal{K}} \in \mathcal{P}$.
- If $X \in \mathcal{W}$, then $\Omega X \in \mathcal{P}$.
- If X is simply connected with cells in two consecutive dimensions and $H_*(X)$ is torsion free, then $X \in \mathcal{W}$.
- $\mathcal{P}$ is closed under pushouts over full subcomplexes of the simplicial complex

Theorem 11.1.1
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $\sigma \in \mathcal{K}$ be a maximal face such that $\dim(\sigma) \geq 2$. Let $\tau \in \partial\sigma$ such that $|\tau| = |\sigma| - 1$ and τ is a maximal face of $\mathcal{K} \setminus \{\sigma\}$. Let $\mathbf{A}$ be n spaces. Then the induced map of polyhedral products
$$(\mathbf{CA}, \mathbf{A})^{\partial\sigma \setminus \{\tau\}} \to (\mathbf{CA}, \mathbf{A})^{\partial\sigma}$$
is null homotopic.

Theorem 11.1.2
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, n\}$. Let $\sigma \in \mathcal{K}$ be a maximal face such that $\dim(\sigma) \geq 2$. Let $\tau \in \partial\sigma$ such that $|\tau| = |\sigma| - 1$ and τ is a maximal face of $\mathcal{K} \setminus \{\sigma\}$. Let $\mathbf{A}$ be n spaces. Then map of polyhedral products
$$(\mathbf{CA}, \mathbf{A})^{\mathcal{K} \setminus \{\sigma\}} \to (\mathbf{CA}, \mathbf{A})^{\mathcal{K}}$$
admits a right homotopy inverse and we have a homotopy equivalence
$$(\mathbf{CA}, \mathbf{A})^{\mathcal{K} \setminus \{\sigma\}} \simeq \left((\mathbf{CA}, \mathbf{A})^{\partial\sigma} \rtimes \prod_{i \notin \sigma} A_i \right) \vee (\mathbf{CA}, \mathbf{A})^{\mathcal{K}}$$

Theorem 11.1.3
Let $\mathcal{K}$ be a simplicial complex that is a pseudomanifold of dimension n. Suppose that the maximal faces of $\mathcal{K}$ is given by $\sigma_1, \dots, \sigma_l$. Suppose that each connected component of $D(\mathcal{K})$ contains a vertex of degree strictly less than $n+1$. Let $\mathbf{A}$ be n spaces. Then there is a homotopy equivalence
$$(\mathbf{CA}, \mathbf{A})^{\mathrm{sk}_{n-1}(\mathcal{K})} \simeq \bigvee_{i=1}^{l} \left((\mathbf{CA}, \mathbf{A})^{\partial\sigma_i} \rtimes \prod_{j \notin \sigma_i} A_i \right) \vee (\mathbf{CA}, \mathbf{A})^{\mathcal{K}}$$
($D(\mathcal{K})$ is the dual graph of $\mathcal{K}$)

Theorem 11.1.4
Let $\mathcal{K}$ be a simplicial complex that is a pseudomanifold of dimension n. Suppose that each connected component of $D(\mathcal{K})$ contains a vertex of degree strictly less than $n+1$. Let $\mathbf{A}$ be

spaces such that $H_*(A_i)$ is torsion free for all i. Then $H_*((\mathbf{CA}, \mathbf{A})^{\mathcal{K}})$ is torsion free if and only if $H_*((\mathbf{CA}, \mathbf{A})^{\mathrm{sk}_{n-1}\mathcal{K}})$ is torsion free.

Theorem 11.1.5
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, m\}$ that is a pseudomanifold of dimension n. Suppose that each connected component of $D(\mathcal{K})$ contains a vertex of degree strictly less than $n+1$. Let $\mathbf{A}$ be spaces such that $\Sigma A_i \in \mathcal{W}$ for all i. If $n = 1$, or $n = 2$, or $n = 3$ and $H_*(|\mathcal{L}|)$ is torsion free for all subcomplexes $\mathcal{L}$ of $\mathcal{K}$ with complete 1-skeleton, then $\Omega(\mathbf{CA}, \mathbf{A})^{\mathcal{K}} \in \mathcal{P}$.

Theorem 11.1.6
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, m\}$ that does not have a complete 1-skeleton. Let $\mathbf{A}$ be spaces such that $\Sigma A_i \in \mathcal{W}$ for all i. If $\Omega(\mathbf{CA}, \mathbf{A})^{\mathcal{K}_{\{1,\dots,m\}\setminus\{i\}}} \in \mathcal{P}$ for all i, then $\Omega(\mathbf{CA}, \mathbf{A})^{\mathcal{K}} \in \mathcal{P}$.

Theorem 11.1.7
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, m\}$ that is a pseudomanifold of dimension n. Then the following are true.

- $\mathcal{K}_{\{1,\dots,m\}\setminus\{i\}}$ is a pesudomanifold of dimension n.
- Each connected component of $D(\mathcal{K}_{\{1,\dots,m\}\setminus\{i\}})$ contains a vertex of degree strictly less than $n+1$.

Theorem 11.1.8
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, m\}$ that is a pseudomanifold of dimension 2 or 3. Suppose that the following are true.

- $H_*(|\mathcal{L}|)$ is torsion free for all full subcomplexes $\mathcal{L}$ of $\mathcal{K}$ with complete 1-skeleton.
- A_i are spaces such that $\Sigma A_i \in \mathcal{W}$ for all i.
- $\mathcal{K}$ does not have a complete 1-skeleton.

Then we have $\Omega(\mathbf{CA}, \mathbf{A})^{\mathcal{K}} \in \mathcal{P}$.

Corollary 11.1.9
Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, m\}$ that is the triangulation of a compact connected closed orientable surface. If $A_1, \dots, A_m$ are spaces such that $\Sigma A_i \in \mathcal{W}$ for all i, then we have $\Omega(\mathbf{CA}, \mathbf{A})^{\mathcal{K}} \in \mathcal{P}$.

Let $I \subseteq k[x_0, \dots, x_n]$ be a homogeneous ideal. We say that $k[x_0, \dots, x_n]/I$ is Golod if all products and higher Massey products in $\mathrm{Tor}_*^{k[x_0,\dots,x_n]}(R, k)$ vanish.

Let $\mathcal{K}$ be a simplicial complex. We say that $\mathcal{K}$ is Golod over k. If $k[\mathcal{K}]$ is Golod.

We have $\mathcal{K}$ is Golod if and only if all products and higher Massey products in $H^*(\mathcal{Z}_{\mathcal{K}})$ vanish (Indeed the Tor homology algebra of $k[\mathcal{K}]$ is $H^*(\mathcal{Z}_{\mathcal{K}})$).

Let $\mathcal{K}$ be a simplicial complex on $\{1, \dots, m\}$. We say that $\mathcal{K}$ is minimally non-Golod if $\mathcal{K}_{\{1,\dots,m\}\setminus\{i\}}$ is Golod for all i. Example: If $\mathcal{Z}_m K$ is a suspension or a co-H-space.

Proposition 11.1.10
Let $\mathcal{K}$ be a pesudomanifold of dimension $2n+1$. If $\mathcal{K}$ is $(n+1)$-neighbourly, then the BBCG decomposition for $\Sigma\mathcal{Z}_{\mathcal{K}_{\{1,\dots,m\}\setminus\{i\}}}$ desuspends for all i.

Corollary 11.1.11
Let $\mathcal{K}$ be a pesudomanifold of dimension $2n+1$. If $\mathcal{K}$ is $(n+1)$-neighbourly, then $\mathcal{K}$ is either Golod or minimally non-Golod.

Corollary 11.1.12
Let $\mathcal{K}$ be a triangulation of S^n. Then $\mathcal{K}$ is Golod if and only if $\mathcal{K} = \partial\Delta^{n+1}$.

Theorem 11.1.13
Let $\mathcal{K}$ be a triangulation of S^n on $\{1, \dots, m\}$. Then we have

$$H_*(\mathcal{Z}_{\mathcal{K}}^{m+n}) \cong \bigoplus_{I \notin \mathcal{K}} H_*(\Sigma^{1+|I|}|\mathcal{K}_I|) \quad \text{and} \quad H_*(\mathcal{Z}_{\mathcal{K}}^{m+n}) \cong \bigoplus_{I \notin \mathcal{K}, I \neq \{1,\dots,m\}} H_*(\Sigma^{1+|I|}|\mathcal{K}_I|)$$

Theorem 11.1.14
Let $\mathcal{K}$ be a triangulation of S^n on $\{1, \dots, m\}$. If the BBCG decomposition $\Sigma\mathcal{Z}_{\mathcal{K}_{\{1,\dots,m\}\setminus\{i\}}}$ desuspends for all i, then the following are true.

- $\mathcal{K}$ is Golod if and only if $\mathcal{K} = \partial\Delta^{n+1}$.
- $\mathcal{K}$ is minimally non-Golod if and only if $\mathcal{K} \neq \partial\Delta^{n+1}$.
- There is a homotopy equivalence

$$\mathcal{Z}_{\mathcal{K}}^{m+n} \simeq \bigvee_{I \notin \mathcal{K}, I \neq \{1,\dots,m\}} \Sigma^{1+|I|}|\mathcal{K}_I|$$

Corollary 11.1.15
Let $\mathcal{K}$ be a triangulation of S^{2n+1} on $\{1, \dots, m\}$. Suppose that $\mathcal{K}$ is $(n+1)$-neighbourly. Then the following are true.

- $\mathcal{K}$ is Golod if and only if $\mathcal{K} = \partial\Delta^{2n+2}$.
- $\mathcal{K}$ is minimally non-Golod if and only if $\mathcal{K} \neq \partial\Delta^{2n+2}$.
- There is a homotopy equivalence

$$\mathcal{Z}_{\mathcal{K}}^{m+2n+1} \simeq \bigvee_{I \notin \mathcal{K}, I \neq \{1,\dots,m\}} \Sigma^{1+|I|}|\mathcal{K}_I|$$

- For all $I \notin \mathcal{K}$ such that $I \neq \{1, \dots, m\}$, we have $|\mathcal{K}_I|$ is either contractible or homotopy equivalent to S^n. Thus $\mathcal{Z}_{\mathcal{K}}^{m+n} \in \mathcal{W}$.
- $\Omega\mathcal{Z}_{\mathcal{K}} \in \mathcal{P}$.

Lemma 11.1.16
Let $\mathcal{K}$ be a triangulation of S^3. Then $H_*(\mathcal{Z}_{\mathcal{K}})$ is torsion free.

Theorem 11.1.17
Let $\mathcal{K}$ be a triangulation of S^3. Then $\Omega\mathcal{Z}_{\mathcal{K}} \in \mathcal{P}$.

11.2 Cai: On products in a real moment-angle manifold

Let $\mathcal{K}$ be a simplicial complex such that $|\mathcal{K}|$ is a polytope. We say that $\mathcal{K}$ is a homology n-manifold if for each $\sigma \in \mathcal{K}$, we have $H_*(\text{lk}_{\mathcal{K}}(\sigma)) \cong H_*(S^{n-1-\dim(\sigma)})$.

Let $\mathcal{K}$ be a homology n-manifold. We say that $\mathcal{K}$ is a generalized homology n-sphere if $H_*(\mathcal{K}) \cong H_*(S^n)$.

Theorem 11.2.1
Let $\mathcal{K}$ be a simplicial complex. Then $\mathcal{K}$ is a generalized homology n-sphere if and only if $\mathcal{Z}_{\mathcal{K}}$ is a manifold.

Theorem 11.2.2
Let $\mathcal{K}$ be a simplicial complex. Then the following are true.

- $\mathbb{R}\mathcal{Z}_{\mathcal{K}}$ is a homology n-manifold if and only if $\mathcal{K}$ is a generalized homology $(n-1)$-sphere.

- Let $n \geq 3$. Let $\mathbb{R}\mathcal{Z}_{\mathcal{K}}$ be a homology n-manifold. Then $\mathbb{R}\mathcal{Z}_{\mathcal{K}}$ is a topological manifold if and only if $|\mathcal{K}|$ is simply connected.

Theorem 11.2.3
Let $\mathcal{K}$ be a simplicial complex. Then $\mathcal{Z}_{\mathcal{K}}$ is a topological $(m+n+1)$-manifold if and only if $\mathcal{K}$ is a generalized homology n-sphere.

11.3 Iriye, Kishimoto: Fat wedge filtrations and decomposition of polyhedral products

Definition 11.3.1 (Shellings)
Let $\mathcal{K}$ be a simplicial complex. A shelling of $\mathcal{K}$ is an ordering of the facets $F_1, \dots, F_t$ of $\mathcal{K}$ such that

$$\langle F_k \rangle \cap \langle F_1, \dots, F_{k-1} \rangle$$

is a pure simplicial complex and $(|F_k| - 2)$-dimensional. We say that $\mathcal{K}$ is shellable if it admits a shelling.

Proposition 11.3.2 (Bruggesser, Mani 1971)
Let $P \subseteq \mathbb{R}^n$ be a convex polytope. Then ∂P is shellable.

Proposition 11.3.3
Let $\mathcal{K}$ be a simplicial complex. If $\mathcal{K}$ is shellable, then $k[\mathcal{K}]$ is SCM for all fields k.

11.4 Amelotte, Briggs: Homotopy types of moment-angle complexes associated to almost linear resolutions

Definition 11.4.1 (Quasi-Koszul Modules)
Let k be a field. Let R be a standard graded algebra over k. Let M be a finitely generated graded R-module. We say that M is a quasi-Koszul module if $\text{Ext}_R^*(M, k)$ is generated by $\text{Ext}_R^0(M, k)$ as a module over the $\text{Ext}_R^*(k, k)$-subalgebra $\text{Ext}_R^1(k, k)$.

Proposition 11.4.2
Let k be a field. Let R be a standard graded algebra over k. Let M be Koszul module. Then M is quasi-Koszul.

Proposition 11.4.3
Let k be a field. Let R be a standard graded algebra over k. Let M be a finitely generated graded R-module. If M is generated in a single degree, then M is Koszul if and only if it is quasi-Koszul.

Proposition 11.4.4
Let k be a field. Let R be a standard graded algebra over k. Let M be a finitely generated graded R-module. Then M is quasi-Koszul if and only if

$$J^2 P_n \cap \ker(d_n) = J \ker(d_n)$$

where $P_\bullet$ is a minimal graded projective resolution of M.

Definition 11.4.5 (The Quasi-Linear Strand)
Let k be a field. Let R be a standard graded algebra over k. Let M be a finitely generated graded R-module. Define the quasi-ilnear strand of M to be the $\text{Ext}_R^1(k, k)$-module generated by $\text{Ext}_R^0(M, k)$.

Clearly M is a quasi-Koszul module if and only if the quasi-linear strand equals all of $\text{Ext}_R^*(M, k)$.

Theorem 11.4.6
Let k be a field. Let $\mathcal{K}$ be a simplicial complex. Then the quasi-linear strand of $k[\mathcal{K}]$ is contained in
$$\operatorname{im}(h : \pi_*(\mathcal{Z}_\mathcal{K}) \to H_*(\mathcal{Z}_\mathcal{K}; k))$$

Corollary 11.4.7
Let $\mathcal{K}$ be a simplicial complex. If $I_\mathcal{K}$ satisfies $N_{d,p}$ over $\mathbb{F}_p$ for all p prime, then there is a homotopy equivalence
$$\mathrm{sk}_{2d+r}\mathcal{Z}_\mathcal{K} \simeq \bigvee_{i=1}^{n} S^{k_i}$$
for some n and some $k_i \in \mathbb{N}$.

Definition 11.4.8 (Homology Missing Face Complex)
Let k be a field. Let $\mathcal{K}$ be a simplicial complex over $\{1, \dots, n\}$. We say that $\mathcal{K}$ is a homology missing face complex if $\widetilde{H}_*(\mathcal{K}_U; k)$ is generated as a k-module by the missing faces of $\mathcal{K}$ for all $U \subseteq \{1, \dots, n\}$ (meaning the images in homology of the inclusion $\mathcal{K}_I \hookrightarrow \mathcal{K}_U$) for minimal non faces I.

Lemma 11.4.9
Let k be a field. Let $\mathcal{K}$ be a simplicial complex over $\{1, \dots, n\}$. Then $\mathcal{K}$ is an HMF complex over k if and only if $I_\mathcal{K}$ is a quasi-Koszul $k[x_1, \dots, x_n]$-module.

Corollary 11.4.10
Let k be a field. Let $\mathcal{K}$ be an HMF simplicial complex over k. Then there is a homotopy equivalence
$$\mathcal{Z}_\mathcal{K} \simeq \bigvee_{i=1}^{n} S^{k_i}$$
for some n and some $k_i \in \mathbb{N}$.

Corollary 11.4.11
Let $\mathcal{K}$ be a simplicial complex. If $I_\mathcal{K}$ is componentwise linear over $\mathbb{F}_p$ for all p prime, then there is a homotopy equivalence
$$\mathcal{Z}_\mathcal{K} \simeq \bigvee_{i=1}^{n} S^{k_i}$$
for some n and some $k_i \in \mathbb{N}$.

Corollary 11.4.12
Let k be a field. Let $\mathcal{K}$ be a simplicial complex. If $I_\mathcal{K}$ is componentwise linear over k, then $\mathcal{K}$ is Golod over k.

Proposition 11.4.13
Let $\mathcal{K}$ be a flag complex. Let $r > 2$. Then there is a homotopy equivalence
$$\mathrm{sk}_{r+4}\mathcal{Z}_\mathcal{K} \simeq \bigvee_{i=1}^{n} S^{k_i}$$
for some n and some $k_i \in \mathbb{N}$ if and only if the underlying graph has no chordless cycles of length $\leq r + 2$.

Definition 11.4.14 (Almost Quasi-Koszul Modules)
Let k be a field. Let R be a standard graded algebra over k. Let M be a finitely generated graded

R-module. We say that M is almost quasi-Koszul if the quasi-linear strand of M contains

$$\bigoplus_{i=1}^{\mathrm{pd}_R(M)-1} \mathrm{Ext}_R^i(M, k)$$

Almost linear implies almost quasi-Koszul Componentwise almost linear implies almost quasi-Koszul

Proposition 11.4.15
Let k be a field. Let $\mathcal{K}$ be a simplicial complex over $\{1, \dots, n\}$. Suppose that $\mathcal{K}$ is Gorenstein* over k. Then the following are equivalent.
- $I_{\mathcal{K}}$ is an almost quasi-Koszul module.
- $I_{\mathcal{K} \setminus \{i\}}$ is a quasi-Koszul module for all $1 \leq i \leq n$.
- $\widetilde{H}_*(\mathcal{K}_U; k)$ is generated as a k-module by the missing faces of $\mathcal{K}$ for all $U \subseteq \{1, \dots, n\}$ (meaning the images in homology of the inclusion $\mathcal{K}_I \hookrightarrow \mathcal{K}_U$) for minimal non faces I.

Proposition 11.4.16
Let k be a field. Let $\mathcal{K}$ be a simplicial complex over $\{1, \dots, n\}$. Suppose that $\mathcal{K} \neq \partial \Delta^{n-1}$ is Gorenstein* over k. Then $I_{\mathcal{K}}$ has an almost linear resolution if and only if $\dim(\mathcal{K})$ is odd and $\mathcal{K}$ is $\lfloor n/2 \rfloor$-neighbourly.

Notice that for sphere triangulations, this implies that $I_{\mathcal{K}}$ having an almost linear resolution is independent of the base field.

Theorem 11.4.17
Let $\mathcal{K}$ be a simplicial complex over $\{1, \dots, n\}$. Suppose that $\mathcal{K} \neq \partial \Delta^{n-1}$ is Gorenstein* over $\mathbb{Z}$. If $I_{\mathcal{K}}$ is an almost quasi-Koszul module, then the following are true.
- $\mathcal{Z}_{\mathcal{K}}$ is an $n + \dim(\mathcal{K}) + 1$ dimensional manifold and there is a homotopy equivalence
$$\mathrm{sk}_{n+\dim(\mathcal{K})} \mathcal{Z}_{\mathcal{K}} \simeq \bigvee_{i=1}^{m} S^{k_i}$$
for some m and some $k_i \in \mathbb{N}$.
- $\mathcal{Z}_{\mathcal{K}}$ is rationally homotopy equivalent to a connected sum of products of two spheres
- There is a homotopy equivalence
$$\Omega \mathcal{Z}_{\mathcal{K}} \simeq \prod_{i=1}^{m} S^{k_i} \times \prod_{j=1}^{l} \Omega S^{k_j}$$
for some n, l and some $k_i, k_j \in \mathbb{N}$.

Corollary 11.4.18
Let $\mathcal{K}$ be a simplicial complex over $\{1, \dots, n\}$. Suppose that $\mathcal{K} \neq \partial \Delta^{n-1}$ is Gorenstein* over $\mathbb{Z}$. If $I_{\mathcal{K}}$ is an almost quasi-Koszul module, then $\mathcal{K}$ is minimally non-Golod.

Part III

Random Sections

12 Profinite Groups

12.1 Basic Properties

Definition 12.1.1 (Profinite Groups)
Let G be a topological group. We say that G is a profinite group if there exists an inverse system $X : \mathcal{I} \to {}_G\textbf{Top}$ consisting of finite groups with the discrete topology such that

$$G = \lim_{\mathcal{I}} X$$

In particular, this means that G is equipped with the weak topology.

Proposition 12.1.2
Let G be a topological group. Then G is profinite if and only if G is compact and totally disconnected.

Corollary 12.1.3
Let G be a profinite group. Then the following are true.
- G is Haudorff.
- G is locally compact.

Proposition 12.1.4
Let G be a compact topological group. Then G/G_0 is a profinite group.

Proposition 12.1.5
Let G be a profinite group. Let $H \leq G$ be a subgroup. Then the following are true.
- If H is closed, then H is profinite.
- If H is closed and normal, then G/H is profinite.
- H is open if and only if H is closed and $[G : H] < \infty$.

12.2 Profinite Completion

13 Equivariant Algebraic Topology

13.1 G-Homotopy

Definition 13.1.1 (G-Homotopy) Let G be a topological group and let X, Y be G-spaces. Let $f, g : X \to Y$ be G-equivariant maps. A G-homotopy from f to g is a homotopy $H : X \times I \to Y$ from f to g such that for each $t \in I$, the map

$$H(-, t) : X \to Y$$

is G-equivariant map.

13.2 Equivariant CW Complexes

13.3

Definition 13.3.1 (The Homotopy Quotient (Borel Construction))
Let G be a topological group. Let X be a G-space. Define the homotopy quotient of X to be the orbit space

$$EG \times_G X = \frac{EG \times X}{G}$$

where the action of G on $EG \times X$ is given by $g \cdot (e, x) = (g \cdot e, g^{-1} \cdot x)$.

Proposition 13.3.2
Let G be a topological group. Let X be a G-space. Then

$$X \hookrightarrow EG \times_G X \to BG$$

is a fibration.

Lemma 13.3.3
Let G be a topological group. Let X be a G-space. If G acts freely on X, then there is a homotopy equivalence

$$EG \times_G X \simeq \frac{X}{G}$$

induced by the projection map $EG \times_G X \to X$.

14

There is a one to one bijection between representations of $\pi_1(X)$ and locally constant sheaves (called local systems).

Definition 14.0.1
Let X be a space. Let $x_0 \in X$. Define the functor $\text{Fib}_{x_0} : \text{Cov}(X) \to {}_{\pi_1(X,x_0)}\textbf{Sets}$ as follows.

- For $p : \widetilde{X} \to X$ a covering space, define $\text{Fib}_{x_0}(p) = p^{-1}(x_0)$.
- For $f : \widetilde{X}_1 \to \widetilde{X}_2$ a map of covering spaces, define $\text{Fib}_{x_0}(f)$ to be $f|_{p^{-1}(x_0)}$.

Theorem 14.0.2
Let X be a connected and locally simply connected space. Let $x_0 \in X$. Then

$$\text{Fib}_{x_0} : \text{Cov}(X) \to {}_{\pi_1(X,x_0)}\textbf{Sets}$$

is fully faithful and essentially surjective.

Theorem 14.0.3 (Classification of Local Systems)
Let X be a space (locally compact, Hausdorff, second countable, locally path connected, semi locally path connected, path connected). Then there is an equivalent of categories

$$\text{Func}(\textbf{Open}^{\text{op}}, \text{Vect}_{\mathbb{C}}) \simeq \text{Rep}(\pi_1(X))$$

Ref: Galois Groups and Fundamental Groups

Theorem 14.0.4 (Classification Theorem for Flat Connections)
Let M be a connected Riemannian manifold. Let G be a Lie group. Then there is a one to one correspondence

$$\frac{\{(P, A) \mid P \to M \text{ is a principal } G\text{-bundle and } A \text{ is a flat connection}\}}{\sim} \overset{1:1}{\longleftrightarrow} \text{Rep}(\pi_1(X), G)$$

where $(P_1, A_1) \sim (P_2, A_2)$ if there exists bundle isomorphism $\varphi : P_1 \to P_2$ such that $\varphi^* A_2 = A_1$.

Ref: Differential Geometry; Taubes Bijection described in: Geometry of Characteristic classes; Morita

There is a natural bijection between principal bundles and vector bundles by taking the associated bundle. If $G = GL(V)$ for some finite dimensional vector space V over the field $k = \mathbb{R}$ or $\mathbb{C}$, the sheaf of sections of the associated bundle is a locally constant sheaf.

Also: Upgrade to diffeomorphism.

15 Infinite Galois Theory

Proposition 15.0.1
Let F be a field. Let K/F be a Galois extension. Then the following are true.

- The inverse limit of the system $\text{Gal}(L/F)$ for L/F a finite Galois extension is $\text{Gal}(K/F)$.
- $\text{Gal}(K/F)$ is a profinite group.
- The projection maps $\text{Gal}(K/F) \to \text{Gal}(L/F)$ for L/F a finite Galois extension is a surjective homomorphism.
- Let $F < L < K$ be a field extension. Then K/L is a Galois extension and $\text{Gal}(K/L)$ is a closed subgroup of $\text{Gal}(K/F)$.

Theorem 15.0.2 (Krull)

Let F be a field. Let K/F be a field extension. Then there is an inclusion reversing bijection

$$\begin{Bmatrix} \text{Closed subgroups of} \\ \text{Aut}_F(K) \end{Bmatrix} \overset{1:1}{\longleftrightarrow} \begin{Bmatrix} \text{Intermediary subfields} \\ F<L<K \end{Bmatrix}$$

given as follows.

- For a subgroup $H \leq \text{Aut}_F(K)$, $H \mapsto \text{Fix}_F(H)$.
- For an intermediary subfield M, $M \mapsto \text{Aut}_L(K)$.

Proposition 15.0.3

Let F be a field. Let K/F be a Galois extension. Then the above bijection restricts to a bijection

$$\begin{Bmatrix} \text{Normal Subgroups} \\ \text{of Aut}_K(L) \end{Bmatrix} \overset{1:1}{\longleftrightarrow} \begin{Bmatrix} \text{Normal Extensions} \\ K<M \end{Bmatrix}$$

16 Local Cohomology

16.1

Definition 16.1.1 (The I-Torsion of a Module)
Let R be a Noetherian commutative ring. Let M be an R-module. Let I be an ideal of R. Define the I-torsion of M to be

$$\Gamma_I(M) = \{m \in M \mid I^t m = 0 \text{ for some } t \in \mathbb{N}\}$$

Definition 16.1.2 (I-Torsion Modules)
Let R be a Noetherian commutative ring. Let M be an R-module. Let I be an ideal of R. We say that M is I-torsion if $M = \Gamma_I(M)$.

Definition 16.1.3 (The Torsion Functor)
Let R be a Noetherian commutative ring. Let I be an ideal of R. Define the I-torsion functor

$$\Gamma_I : {}_R\mathbf{Mod} \to {}_R\mathbf{Mod}$$

by the following.

- For each R-module M, the functor sends M to the I-torsion $\Gamma_I(M)$.
- For each R-module homomorphism $f : M \to N$, the functor sends f to $\Gamma_I(f) = f|_{\Gamma_I(M)}$.

Lemma 16.1.4
Let R be a Noetherian commutative ring. Let I be an ideal of R. Then Γ_I is a left exact functor.

Definition 16.1.5 (Local Cohomology)
Let R be a Noetherian commutative ring. Let M be an R-module. Let I be an ideal of R. Define the nth local cohomology of M with support in I to be

$$H_I^n(M) = (R^n\Gamma_I)(M)$$

Proposition 16.1.6
Let R be a Noetherian commutative ring. Let M be an R-module. Let I, J be ideals of R. If $\sqrt{I} = \sqrt{J}$, then the equality $\Gamma_I(M) = \Gamma_J(M)$ induces an equality

$$H_I^n(M) = H_J^n(M)$$

for all $n \in \mathbb{N}$.

Thus we only need to consider the local cohomology with support in radical ideals.

Lemma 16.1.7
Let R be a Noetherian commutative ring. Let M be an R-module. Let I be an ideal of R. Then $H_I^n(M)$ is I-torsion for all $n \in \mathbb{N}$.

Lemma 16.1.8
Let R be a Noetherian commutative ring. Let M_i be R-modules for each $i \in J$. Let I be an ideal of R. Then there is a natural isomorphism

$$H_I^n\left(\bigoplus_{i \in J} M_i\right) \cong \bigoplus_{i \in J} H_J^n(M_i)$$

Proposition 16.1.9
Let R be a Noetherian commutative ring. Let M be an R-module. Let I be an ideal of R.Then there are natural isomorphisms

$$\varinjlim_{i\in\mathbb{N}} \operatorname{Ext}_R^n(R/I^i, M) \cong H_I^n(M)$$

for all $n \in \mathbb{N}$.

17 Morita Theory

17.1

Definition 17.1.1 (Morita Equivalence)
Let R, S be rings. We say that R and S are Morita equivalent if there exists an equivalence of categories
$$_R\mathbf{Mod} \simeq {_S\mathbf{Mod}}$$

Lemma 17.1.2
Let R, S be rings. Then R and S are Morita equivalent if and only if there exists an equivalence of categories
$$\mathbf{Mod}_R \simeq \mathbf{Mod}_S$$

Lemma 17.1.3
Let R, S be rings. Suppose that R and S are Morita equivalent. Let $F : {_R\mathbf{Mod}} \to {_S\mathbf{Mod}}$ be an equivalence. Then F is additive.

17.2 Progenerators

Definition 17.2.1 (Progenerators)
Let R be a ring. Let P be an R-module. We say that P is a progenerator of R if the following are true.
- P is finitely generated.
- P is projective.

Proposition 17.2.2
Let R, S be rings. Let $F : {_R\mathbf{Mod}} \to {_S\mathbf{Mod}}$ and $G : {_S\mathbf{Mod}} \to {_R\mathbf{Mod}}$ be additive functors. Then F and G are inverses if and only if there exists an (S, R)-bimodule P such that the following are true.
- $_SP$ is a progenerator.
- P_R is a progenerator.
- There are natural isomorphisms $F \cong P \otimes_R -$ and $G \cong \text{Hom}(_SP, -)$.

Theorem 17.2.3 (Morita)
Let R, S be rings. Then the following are equivalent.
- R and S are Morita equivalent.
- There exists a progenerator R-module P_R such that $S \cong \text{End}_R(P_R)$.
- There exist an (R, S)-bimodule M and a (S, R)-bimodule N such that $M \otimes_S N \cong R$ as (R, R)-bimodules and $N \otimes_R M \cong S$ as (S, S)-bimodules.

17.3 Properties of Morita Equivalence

Proposition 17.3.1
Let R, S be rings. If R and S are Morita equivalent, then the following are true.
- There is a ring isomorphism $Z(R) \cong Z(S)$.
- $R/J(R)$ is Morita equivalent to $S/J(S)$.

Proposition 17.3.2
Let R, S be rings. Let M be an R-module. Suppose that R and S are Morita equivalent via the functor $F : {_R\mathbf{Mod}} \to {_S\mathbf{Mod}}$. Then the following are true.
- M is injective if and only if $F(M)$ is injective.
- M is projective if and only if $F(M)$ is projective.
- M is simple if and only $F(M)$ is simple.

- M is semisimple if and only if $F(M)$ is semisimple.
- M is finitely generated if and only if $F(M)$ is finitely generated.
- M is Noetherian if and only if $F(M)$ is Noetherian.
- M is Artinian if and only if $F(M)$ is Artinian.

Proposition 17.3.3
Let R, S be rings. Suppose that R and S are Morita equivalent. Then the following are equivalent.
- R is simple if and only if S is simple.
- R is semisimple if and only if S is semisimple.
- R is Noetherian if and only if S is Noetherian.
- R is Artinian if and only if S is Artinian.

Proposition 17.3.4
Let R, S be rings. Then R and S are Morita equivalent if and only if ${}_R\mathbf{FGMod} \simeq {}_S\mathbf{FGMod}$.

18 Deformation of Manifold Structures

18.1

Recall:

$$\begin{tikzcd}
\text{Smooth Manifolds} \arrow[r, phantom, "\subset"] & \text{PL Manifolds} \arrow[r, phantom, "\subset"] \arrow[d, leftrightarrow] & \text{Topological Manifolds} \\
 & \text{Combinatorial Manifolds} \arrow[r, phantom, "\subset"] & \text{Triangulated Manifolds} \arrow[u]
\end{tikzcd}$$

is every triangulated manifold a combinatorial manifold? manifold Hauptvermutung:

Definition 18.1.1 (Smooth Embedding of a Simplex)
Let $\Delta^k \subseteq \mathbb{R}^{k+1}$ be the standard k-simplex. Let M be a smooth manifold. Let $\sigma : \Delta^k \to M$ be a map. We say that σ is a smooth embedding if the following are true.
- f is injective.
- f induces a homeomorphism from Δ^k to $\operatorname{im}(f)$.
- For all facets $F \subseteq \Delta^k$, the induced map $\sigma|_{F^\circ}$ is an immersion of smooth manifolds.

Definition 18.1.2 (Smooth Triangulations)
Let M be a smooth manifold with boundary. Let K be a simplicial complex that is a triangulation of M. We say that K is a smooth triangulation if the following are true.
- For each simplex $\sigma \in K$, the map σ is a smooth embedding.
- There exists a subcomplex $K' \subseteq K$ such that K' is a triangulation of ∂M.

Lemma 18.1.3
Let M be a smooth manifold with boundary. Let K be a smooth triangulation of M. Then M admits a PL structure.

Theorem 18.1.4 (Whitehead, 1940)
Let M be a smooth manifold with boundary. Then M admits a smooth triangulation. Moreover, if K and K' are both smooth triangulations, then $|K|$ and $|K'|$ are PL-homeomorphic.

18.2 Microbundles

Define the tangent microbundle τ_M for a topological n-manifold M. Stable topological microbundles over M are classified by $BTop = \varinjlim_k B\text{Top}(k)$, so the stable tangent microbundle determines a classifying map $M \to BTop$. Since $\text{Top}(k)/\text{PL}(k) \to Top/PL$ is a homotopy equivalence for $k \geq 5$, a lift of $M \to BTop$ through $BPL \to BTop$ is equivalent to a PL structure on the unstable tangent microbundle τ_M.

Lemma 18.2.1
The following are true.
- There is a homotopy equivalence
$$\text{hofib}(BPL \to BTop) \simeq K(\mathbb{Z}/2\mathbb{Z}, 3)$$
- $BPL \to BTop$ is a principal fibration.
- The principal fibration induces a fibration $BTop \to K(\mathbb{Z}/2\mathbb{Z}, 4)$.

18.3 The Tangent Microbundle

Theorem 18.3.1 (Hirsch-Mazur Theorem)
Let M be a topological manifold of dimension $n \geq 5$. Then τ_M admits a PL structure if and only if M admits a PL structure.

18.4 The Kirby-Siebenmann Class

Definition 18.4.1 (The Kirby-Siebenmann Class)
Let M be a topological manifold. Define the Kirby-Siebenmann class of M to be the element

$$\kappa(M) \in H^4(M; \mathbb{Z}/2\mathbb{Z}) \cong [M, K(\mathbb{Z}/2\mathbb{Z}, 4)]$$

given by the element $M \stackrel{\tau_M}{\to} BTop \to K(\mathbb{Z}/2\mathbb{Z}, 4)$.

Proposition 18.4.2 (Kirby-Siebenmann Structure Theorem)
Let M be a compact topological manifold of dimension $n \geq 5$. Then the following are true.

- M admits a PL structure if and only if $\kappa(M) = 0$.
- If $\kappa(M) = 0$, then the set of concordance classes of PL structure on M is a torsor over $H^3(M; \mathbb{Z}/2\mathbb{Z})$.

18.5 The Low Dimensional Case

Theorem 18.5.1 (Rado's Theorem)
Let M be topological manifold of dimension 2. Then M admits a unique smooth structure, unique PL structure and unique triangulation.

Theorem 18.5.2 (Moise's Theorem)
Let M be topological manifold of dimension 3. Then M admits a unique smooth structure, unique PL structure and unique triangulation.

Theorem 18.5.3 (Cairns-Hirsch)
Let M be piecewise linear manifold of dimension 4. Then M admits a unique smooth structure and a unique PL structure.

Theorem 18.5.4
Let M be a topological manifold of dimension 4. Then the following are true.

- If $\kappa(M) \neq 0$, then M admits no PL structure and no smooth structure.
- If $\kappa(M) = 0$ and Q_M is not diagonalizable over $\mathbb{Z}$, then M admits no PL structure and no smooth structure.

Theorem 18.5.5
The following are true.

- There exists uncountably many pairwise non-diffeomorphic smooth structures on $\mathbb{R}^4$.
- For $n \neq 4$, there exists a unique smooth structure on $\mathbb{R}^n$.

Theorem 18.5.6 (Manolescu 2016)
For all $n \geq 5$, there exists a compact topological manifold of dimension n that admits no triangulation.

Theorem 18.5.7
Let M be PL manifold of dimension $n \leq 6$. Then M admits a smooth structure.

$$\underset{\text{Manifolds}}{\text{Smooth}} \subset \underset{\text{Manifolds}}{\text{PL}} \subset \underset{\text{Manifolds}}{\text{Topological}}$$

Summarizing: All inclusions are equalities for $\dim \leq 3$. The first inclusion is strict for $\dim = 4$.

19 Rational Homotopy Theory

19.1 Homology with Coefficients in a Subring of $\mathbb{Q}$

Theorem 19.1.1 (Whitehead-Serre Theorem)
Let X, Y be simply connected spaces. Let R be a subring of $\mathbb{Q}$. Let $f : X \to Y$ be a map. Then the following are equivalent.

- $\pi_*(f) \otimes_\mathbb{Z} R$ is an isomorphism of graded abelian groups.
- $H_*(f; R)$ is an isomorphism of graded rings.
- $H_*(\Omega f; R)$ is an isomorphism of graded rings.

Lemma 19.1.2
Let X be an n-connected space. Let R be a subring of $\mathbb{Q}$. Suppose that one of the following are true.

- $n \geq 1$.
- $n = 0$ and $\pi_1(X)$ is abelian.

Then the Hurewicz homomorphism induces an isomorphism

$$\pi_{n+1}(X) \otimes_\mathbb{Z} R \cong H_{n+1}(X; R)$$

Proposition 19.1.3
Let A be an abelian group. Let $n \in \mathbb{N}^+$. Let R be a subring of $\mathbb{Q}$. Then we have $A \otimes_\mathbb{Z} R = 0$ if and only if $H_*(K(A, n); R) \cong H_*(*; R)$.

19.2 P-Local Modules

Definition 19.2.1 (P-Local Groups)
Let A be an abelian group. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. We say that A is P-local if for all $q \in \mathbb{N} \setminus P$, the map $\cdot q : A \to A$ is an isomorphism.

Definition 19.2.2 (P-Localization Map)
Let A be an abelian group. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. Define the P-localization map of A to be the map

$$A \to A \otimes_\mathbb{Z} S_P^{-1}\mathbb{Z}$$

where $S_P = \{n \in \mathbb{Z} \mid \gcd(n, p) = 1 \text{ for all } p \in P\}$.

Proposition 19.2.3
Let A be an abelian group. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. Let $S_P = \{n \in \mathbb{Z} \mid \gcd(n, p) = 1 \text{ for all } p \in P\}$. Then the following are equivalent.

- A is P-local.
- A is a $S_P^{-1}\mathbb{Z}$-module by extension of scalars.
- The P-localization map $A \to A \otimes_\mathbb{Z} S_P^{-1}\mathbb{Z}$ is an isomorphism.

When $P = \emptyset$, then $S_P^{-1}\mathbb{Z} = \mathbb{Q}$ is just rationalization.

19.3 P-Local Spaces

Definition 19.3.1 (P-Local Space)
Let X be a simply connected space. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. We say that X is P-local if $\pi_n(X)$ is a P-local group for all $n \geq 2$.

Proposition 19.3.2
Let X be a simply connected space. Let $P \subseteq \mathbb{N}$ be a set of prime numbers. Then the following are equivalent.

- X is P-local.
- $H_n(X,*)$ is P-local for all $n\geq 0$.
- $H_n(\Omega X,*)$ is P-local for all $n\geq 0$.

Lemma 19.3.3
Let A be an abelian group. Let $n\geq 1$. Let $P\subseteq\mathbb{N}$ be a set of prime numbers. If A is P-local, then we have
$$H_*(X;\mathbb{Z}/p\mathbb{Z})\cong H_*(*,\mathbb{Z}/p\mathbb{Z})$$
for all $p\notin P$.

Definition 19.3.4 (P-Localization)
Let X be a simply connected space. Let $P\subseteq\mathbb{N}$ be a set of prime numbers. A P-localization of X is a space X_P together with a map $i_p:X\to X_P$ such that f induces an isomorphism
$$\pi_n(X)\otimes_{\mathbb{Z}}S_P^{-1}\mathbb{Z}\to\pi_n(X_P)$$
for all $n\geq 2$.

Proposition 19.3.5
Let X be a simply connected space. Let $P\subseteq\mathbb{N}$ be a set of prime numbers. Suppose that X_P exists. Then the following are true.

- Suppose that Z is a P-local space and $f:X\to Z$ is a map. Then there exists a map $g:X_P\to Z$ such that $g\circ i_p=f$.
- Suppose that Z is a P-local space and $f_1,f_2:X\to Z$ are map. Suppose that $g_1,g_2:X_P\to Z$ are maps such that $g_1\circ i_p=f_1$ and $g_2\circ i_p=f_2$. Suppose that $H:X\times I\to Z$ is a homotopy from f_1 to f_2. Then there exists a homotopy $K:X_P\times I\to Z$ such that $K\circ i_p\times\mathrm{id}_Z=H$.

Proposition 19.3.6
Let X be a simply connected space. Let $P\subseteq\mathbb{N}$ be a set of prime numbers. Then X_P exists and is unique up to homotopy equivalence relative to X.

19.4 Rational Homotopy Equivalence

Definition 19.4.1 (Rational Model)
Let X be a simply connected space. A rational model of X is a P-localization of X where $P=\emptyset$. It is denoted by $X_{\mathbb{Q}}$.

Proposition 19.4.2
Let X be a simply connected space. Then the following are true.

- $X_{\mathbb{Q}}$ is homotopy equivalent to a CW complex.
- $C_*(X_{\mathbb{Q}})$ has all differentials as 0.
- $H_*(X_{\mathbb{Q}})$ is a free graded $\mathbb{Z}$-module.

20 Topics in Simplicial Homology

20.1

Proposition 20.1.1
Let S be a simplicial complex. Let $\sigma \in S$. Let $p \in |\sigma|^\circ$. Then we have

$$H^k_{\text{sing}}(|S|, |S| \setminus p) \cong \widetilde{H}^{k-|\sigma|}_{\Delta}(\text{lk}(\sigma))$$

21 The Nature of Mathematical Physics

In mathematical physics, physical observations are given a mathematical formalism. A recurring theme is that mathematics often predict new phenomena beyond the original observation.

Mechanics

Mechanics is the part of physics that study the motion of bodies. Statics is the subject that studies equilibriums. Kinematics is concerned with general properties of motions regardless of the source. Dynamics study motion caused by forces or interactions. In regards to the system that is studied, mechanics can also be divided into material points, rigid bodies, elastic bodies and fluids.

Mechanics have received a number of revolutionary ideas that transformed the subject. Completely. In the modern day there are three main branches of mechanics: classical mechanics, relativistic mechanics and quantum mechanics.

Classical Mechanics

Theoretically, there are three different formalisms of classical mechanics that are related: Newtonian, Lagrangian and Hamiltonian. Newtonian mechanics is the easiest to formulate. Lagrangian is based on the least action principle, which is more sophisticated but is efficient for mechanical systems that are under constraints. Hamiltonian mechanics connects the bridge between classical mechanics and quantum mechanics.

For a mathematical formalism of classical mechanics, we look for a mathematical model that describes the following physically observed phenomena.

- Our space is three dimensional and time is one dimensional.
- There exists coordinate systems called inertial such that all laws of nature at all moments of time are the same in all inertial coordinates, and all coordinate systems in uniform rectilinear motion with respect to an inertial one are themselves inertial.
- The initial state of a mechanical system uniquely determines all of its motion.

The limitations of classical mechanics come from Maxwell's equations, which says that electromagnetic waves travel at a fixed speed regardless of which inertial frame one uses. In particular, classical mechanics is a good enough estimate for motions of bodies that are significantly less than the speed of light, and whose bodies are macroscopic objects.

Relativistic Mechanics

For a mathematical formalism of relativistic mechanics, we look for a mathematical model that describes the following physically observed phenomena.

- The Principle of Relativity: The laws of physics are the same in all inertial frames.
- The speed of light is the same for all inertial frames, regardless of the motion of the source or observer.

A number of consequences appear because the speed of light being constant.

- Time dilation: Moving clocks run slower.
- Length contraction: Moving objects are shortened along the direction of motion.
- Relativity of simultaneity: Two events simultaneous in one frame are not simultaneous in another.
- Speed do not add linearly, nothing with mass can have speed that is the speed of light.
- Mass energy equivalence: $E = mc^2$.
- Instantaneous interactions are impossible, and so must propagate at finite speed.

Special relativity introduces the framework that is Lorentz geometry for the universe. This framework is one that is compatible with the observation that is the speed of light is the same for all inertial frames. In particular, it deduces the relativity of simultaneity, which means that instantaneous interactions are impossible. This introduces the concept of fields and field theory.

The limitations of relativistic mechanics come from general relativity, which says that gravity should be a result of the geometry, not a force. Relativistic mechanics does not take this to account. Also, classical field theory cannot handle the destruction and creation of particles.

General Relativity

Quantum Mechanics

Quantum Field Theory

Statistical Mechanics

Thermal Dynamics

Quantum Information

Condensed Matter Physics

Cosmology

22 K Theory

22.1

Definition 22.1.1 (The 0th K-Group)
Let R be a ring. Define the 0th K-group of R to be the set

$$K_0(R) = \frac{\{P \mid P \text{ is a finitely generated projective } R\text{-module }\}}{\cong}$$

together with addition given by $[P_1] + [P_2] = [P_3]$ if there is a short exact sequence

$$0 \longrightarrow P_1 \longrightarrow P_3 \longrightarrow P_2 \longrightarrow 0$$

Example 22.1.2
Suppose that R is a PID. Then there is an isomorphism

$$K_0(R) \cong \mathbb{Z}$$

given by $[n] \mapsto [R^n]$.

Definition 22.1.3 (The Change of Rings Homomorphism)
Let R, S be rings. Let $f : R \to S$ be a ring homomrphism. Define the change of rings homomorphism to be the map $K_0(f) : K_0(R) \to K_0(S)$ given by

$$[P] \mapsto [S \otimes_R P]$$

Recall the Grothendieck completion of an abelian semi-group $G(-)$.

Proposition 22.1.4
Let R be a ring. Let $P(R) = \{P \mid P \text{ is a finitely generated projective } R\text{-modules }\}$. Then there is a natural isomorphism

$$K_0(R) \cong G(P(R), \oplus)$$

Proposition 22.1.5 (Morita Equivalence)
Let R be a ring. Then there is a natural isomorphism

$$K_0(R) \cong K_0(M_n(R))$$

given by $[P] \mapsto R^n \otimes_R P$ where R^n is considered as an $(M_n(R), R)$-bimodule.

Proposition 22.1.6
Let R, S be rings. Then there is a natural isomrophism

$$K_0(R \times S) \cong K_0(R) \times K_0(S)$$

induced by the product of the projection maps.

Example 22.1.7
Let F be a field. Let V be a vector space over F of countable dimension. Then there is an isomorphism

$$K_0(\text{End}_F(V)) \cong \{1\}$$

Definition 22.1.8 (The Reduced 0th K-Group)

Let R be a ring. Define the reduced 0th K-group to be the quotient

$$\widetilde{K}_0(R) = \frac{K_0(R)}{\langle [R^m] - [R^n] \mid m, n \in \mathbb{N} \rangle}$$

Lemma 22.1.9
Let R be a ring. Then there is a natural isomorphism

$$\widetilde{K}_0(R) \cong \langle [R] \rangle \leq \widetilde{K}_0(R)$$

Proposition 22.1.10
Let G be a group. Let R be an integral domain. Then there is an isomorphism

$$K_0(R[G]) \cong \widetilde{K}_0(R[G]) \oplus \mathbb{Z}$$

give by $[P] \mapsto ([P], \dim_{\text{Frac}(R)}(\text{Frac}(R) \otimes_{R[G]} P))$

Conjecture 22.1.11 (Farrell-Jones Conjecture)
Let R be a regular ring. Let G be a torsion free group. Then the inclusion map induces a natural isomorphism

$$K_0(R) \cong K_0(R[G])$$

23 Ergodic Theory

23.1

Definition 23.1.1 (Measure Preserving Systems)
Let $(X, \mathcal{F}, \mu)$ be a measure space. Let $T : X \to X$ be a measurable function. We say that T is a measure preserving function if
$$\mu(T^{-1}(E)) = \mu(E)$$
for all $E \in \mathcal{F}$. In this case, we call $(X, \mathcal{F}, \mu, T)$ a measure preserving system.

Definition 23.1.2 (Probability Preserving System)
Let $(X, \mathcal{F}, \mu, T)$ be a measure preserving system. We say that it is a probability preserving system if $(X, \mathcal{F}, \mu)$ is a probability space.

Theorem 23.1.3 (Poincare Recurrence Theorem)
Let $(X, \mathcal{F}, \mu, T)$ be a probability preserving system. Let $E \in \mathcal{F}$. Then we have
$$\mu(\{x \in E \mid \exists N \text{ such that } T^n(x) \notin E \text{ for all } n > N\}) = 0$$

In other words, most points in E go back to E infinitely often after applying a certain number of T's.

Definition 23.1.4 (Measure Theoretically Isomorphic)
Let $(X, \mathcal{F}, \mu, T)$ and $(Y, \mathcal{E}, \sigma, S)$ be measure preserving systems. Let $\pi : X \to Y$ be a measurable function. We say that π is a measure theoretically isomorphism if the following are true.

- There exists $A \subseteq X$ and $B \subseteq Y$ such that $\mu(X \setminus A) = \mu(Y \setminus B) = 0$ and $\pi|_A : A \to B$ is invertible with a measurable inverse.
- For all $E \in \mathcal{E}$, we have $\mu(\pi^{-1}(E)) = \sigma(E)$.
- $S \circ \pi = \pi \circ T$.

23.2

Definition 23.2.1 (Invariant Sets)
Let $(X, \mathcal{F}, \mu, T)$ be a measure preserving system. Let $E \in \mathcal{F}$. We say that E is invariant if
$$T^{-1}(E) = E$$

Definition 23.2.2 (Ergodic Systems)
Let $(X, \mathcal{F}, \mu, T)$ be a measure preserving system. We say that the system is ergodic if for all invariant sets E, we have $\mu(E) = 0$ or $\mu(X \setminus E) = 0$. In this case, we say that μ is an ergodic measure.

24 The Logical Chain of Classical Field Theory and Gauge Theory

Step 1: The Field

We observe a physical quantity distributed over spacetime — a temperature, a velocity, an electromagnetic strength. Mathematically this is a fiber bundle $\pi : E \to M$, where M is spacetime and the fiber E_x is the space of possible values of the quantity at x. A **section** $\phi \in \Gamma(E)$ is one complete specification of the quantity at every point — a possible physical configuration.

Step 2: The Lagrangian as a Choice of Physics

A Lagrangian density $\Lambda \in \Omega^n(J^k E)$ is a choice of action functional

$$S[\phi] = \int_\Omega (j^k\phi)^*\Lambda$$

Different Lagrangians encode different physical theories. The Euler-Lagrange equations — the equations of motion — are derived from Λ by the variational principle. The physical sections are the critical points of S.

Step 3: Observing a Symmetry

You compute the Euler-Lagrange equations and notice that the Lagrangian Λ is invariant under some subgroup of automorphisms of E. An **automorphism** of $\pi : E \to M$ is a fiber-preserving diffeomorphism $\Phi : E \to E$. Such an automorphism acts on sections by

$$\phi \mapsto \Phi \circ \phi$$

and the Lagrangian being invariant means

$$S[\Phi \circ \phi] = S[\phi] \quad \text{for all } \phi \in \Gamma(E)$$

At this stage, the symmetry group acts the same way at every point — it is **global**. You observe this as an empirical fact about your Lagrangian.

Symmetries of the action induces certain laws of physics, such as conservation of energy, momentum, charge etc. This is Noether's theorem.

Another note: Rather than starting with a Lagrangian and observe the symmetries of its action, we may also start with a set of symmetries that we would like to enforce, apply it to all differential forms and retain only the invariants, and see what Lagrangian you would get.

Step 4: The Absolute vs Relative Observation

The global symmetry exists because the field takes values in a space that has **no preferred origin or frame** — the fiber E_x has a group of self-maps (rotations, phase shifts, color rotations) that are physically unobservable. Different observers, or different conventions, would describe the same physical situation using sections related by this group action. The global symmetry is an artifact of having made consistent conventional choices everywhere at once. We need not restrict ourselves to consider symmetries of the entire space, and this is supported by physics:

Argument 1: Special Relativity

Special relativity says there is no notion of simultaneity — two spacelike separated points have no canonical notion of "the same moment." A global transformation is therefore not a physically coherent operation in a relativistic theory. If you believe in special relativity, you are forced to allow your symmetry to act differently at different points. The global symmetry is then a special case of the local one, not the fundamental object.

Argument 2: The Gauge Principle

The difference induced by the symmetry of the action is not observable. The phase is therefore a **convention** — a choice made by an observer at x to describe the field. If the phase is a convention, then different observers at different points of spacetime should be free to make different conventional choices independently. There is no physical mechanism that would force their conventions to agree globally. The true symmetry group of the theory is therefore not the global group G but the local group $mathcalG = \mathrm{Map}(M, G)$ of all smooth G-valued functions on spacetime.

This is the **gauge principle**: any redundancy in the description of a physical system should be a local redundancy. It is a postulate, not a theorem. Its justification is that it works — every time it is applied, it produces a physically correct and predictive theory.

Step 5: Promoting to Local Symmetry

This forces you to demand invariance not just under global G-transformations

$$\phi(x) \mapsto g \cdot \phi(x), \quad g \in G \text{ constant}$$

but under local G-transformations

$$\phi(x) \mapsto g(x) \cdot \phi(x), \quad g : M \to G \text{ smooth}$$

The Lagrangian, built from ordinary derivatives $\partial_\mu \phi$, immediately fails to be locally invariant. The reason is fundamental: $\partial_\mu \phi$ compares field values at nearby points x and $x + dx$, which under local symmetry live in fibers with **independent conventions**. Subtracting them is meaningless — it is comparing quantities measured in different units.

Step 6: The Mathematics Is Now Forced

To make sense of differentiation under local symmetry, we need a way to **compare vectors in nearby fibers** compatibly with the G-structure. But this is precisely the definition of a **connection**. Where does this connection live? The answer is that the principal bundle is not introduced from outside — it is **already contained inside the vector bundle** E.

The fibers E_x are vector spaces, but the isomorphism $E_x \cong \mathbb{R}^r$ is not canonical — it requires a choice of basis. A **frame** at x is an ordered basis for E_x. The set of all frames at x carries a natural free transitive right action of $GL(r, \mathbb{R})$, making it a $GL(r, \mathbb{R})$-torsor. Taking the disjoint union over all $x \in M$ produces the **frame bundle**

$$P := \bigsqcup_{x \in M} \{\text{ordered bases for } E_x\}$$

a principal $GL(r, \mathbb{R})$-bundle canonically constructed from E.

Now, the observation of a global G-symmetry in Step 3 is precisely the observation that the physically relevant frames are not arbitrary frames, but only those related by G-transformations. The structure group **reduces** from $GL(r, \mathbb{R})$ to G, producing a principal G-bundle

$$P_G \subset P$$

canonically determined by the symmetry. The original vector bundle is recovered as the associated bundle $E = P_G \times_G V$.

The connection demanded by local symmetry lives on P_G, because P_G is the bundle whose fibers are G-torsors — they encode exactly the gauge freedom at each point. A connection on P_G transports G-frames between nearby fibers, and therefore transports everything associated to those frames, including sections of $E = P_G \times_G V$.

Step 7: What Follows From The Maths

Since the principal bundle is just a collection of frames at each fiber. A local section of the principal bundle is a choice of local coordinates. The mathematical fact that non-trivial principal bundles have no global sections can be interpreted as the fact that there is generally no globally consistent choice of coordinates because the symmetries now act locally.

A connection in the principal bundle gives a covariant derivative in the associated bundle. This gives a way to compare phases across different time space.

25 Bordism and Cobordism

25.1 Classical Bordism

Definition 25.1.1 (Bordant Manifolds)
Let M, N be smooth compact manifolds without boundary. A bordism between M and N is a smooth compact manifold X such that $\partial X = M \amalg N$. In this case, we say that M and N are bordant.

Proposition 25.1.2
Let M, N be smooth compact manifolds without boundary. If M and N are diffeomorphic, then M and N are bordant.

Definition 25.1.3 (The Bordism Groups)
Define the nth bordism group to be the set

$$\Omega_n = \{[M] \mid [M] \text{ is a bordism class }\}$$

together with addition given by $[M] + [N] = [M \amalg N]$.

Proposition 25.1.4
The following are true regarding the bordism groups.
- $\Omega_0 = \langle [*] \rangle \cong \mathbb{Z}/2\mathbb{Z}$.
- $\Omega_1 = 0$.
- $\Omega_2 = \langle [\mathbb{RP}^2] \rangle \cong \mathbb{Z}/2\mathbb{Z}$.

Definition 25.1.5 (The Bordism Ring)
Define the bordism ring to be the abelian group

$$\Omega = \bigoplus_{i=0}^{\infty} \Omega_n$$

together with multiplication given by $[M] \cdot [N] = [M \times N]$.

Proposition 25.1.6
There is an isomorphism of commutative graded rings

$$\Omega \cong \mathbb{Z}/2\mathbb{Z}[x_i \mid i \neq 2^k - 1 \text{ for some } k]$$

Proposition 25.1.7
Let M, N be a smooth compact manifolds without boundary of dimension n. Then M and N are bordant if and only if

$$\langle w_{i_1}(M) \cdots w_{i_k}(M), [M] \rangle = \langle w_{i_1}(N) \cdots w_{i_k}(N), [N] \rangle$$

for all partitions $(i_1, \dots, i_k)$ of n.

25.2 Oriented Bordism

Definition 25.2.1 (Oriented Bordant Manifolds)
Let M, N be smooth oriented compact manifolds without boundary. An oriented bordism between M and N is a smooth oriented compact manifold X such that $\partial X = M \amalg \overline{N}$ where $\overline{N}$ has the inverse orientation of M.

Definition 25.2.2 (The Oriented Bordism Groups)
Define the nth bordism group to be the set

$$\Omega_n^{SO} = \{[M] \mid [M] \text{ is an oriented bordism class } \}$$

together with addition given by $[M] + [N] = [M \amalg N]$.

Lemma 25.2.3
Let M be a smooth oriented compact manifold without boundary. Then we have $-[M] = [\overline{M}]$.

Definition 25.2.4 (The Oriented Bordism Ring)
Define the bordism ring to be the abelian group

$$\Omega^{SO} = \bigoplus_{i=0}^{\infty} \Omega_n^{SO}$$

together with multiplication given by $[M] \cdot [N] = [M \times N]$.

Proposition 25.2.5
There is an isomorphism of commutative graded rings

$$\Omega^{SO} \otimes \mathbb{Q} \cong \mathbb{Q}[x_{4k} \mid k \in \mathbb{N}]$$

given by $[\mathbb{CP}^{2k}] \mapsto x_{4k}$.

Proposition 25.2.6
Let $t \in (\Omega^{SO})^{\text{tor}}$. Then we have $2t = 0$.

Proposition 25.2.7
There is an isomorphism of commutative graded rings

$$\frac{\Omega^{SO}}{(\Omega^{SO})^{\text{tor}}} \cong \mathbb{Z}[x_{4k} \mid k \in \mathbb{N}]$$

Proposition 25.2.8
Let M, N be a smooth oriented compact manifolds without boundary of dimension n. Then M and N are oriented bordant if and only if

$$\langle w_{i_1}(M) \cdots w_{i_k}(M), [M] \rangle = \langle w_{i_1}(N) \cdots w_{i_k}(N), [N] \rangle$$

for all partitions $(i_1, \dots, i_k)$ of n and

$$\langle p_{j_1}(M) \cdots p_{j_k}(M), [M] \rangle = \langle p_{j_1}(N) \cdots p_{j_k}(N), [N] \rangle$$

for all partitions $(j_1, \dots, j_k)$ of $n/4$

Proposition 25.2.9
The following are true regarding the oriented bordism groups.

- $\Omega_0^{SO} = \langle * \rangle \cong \mathbb{Z}$.
- $\Omega_1^{SO} = \Omega_2^{SO} = \Omega_3^{SO} = 0$.
- $\Omega_4^{SO} = \langle [\mathbb{CP}^2] \rangle \cong \mathbb{Z}$.
- $\Omega_5^{SO} = \langle [Y] \rangle \cong \mathbb{Z}/2\mathbb{Z}$ where Y is the Dold manifold.
- $\Omega_6^{SO} = \Omega_7^{SO} = 0$.
- $\Omega_8^{SO} = \langle [\mathbb{CP}^2 \times \mathbb{CP}^2], [\mathbb{CP}^4] \rangle \cong \mathbb{Z}^2$.
- $\Omega_n^{SO} \neq 0$ for all $n \geq 9$.

25.3 Framed Bordism

Recall that the stable normal bundle of a manifold is the equivalence class η_M of stably equivalent vector bundles.

Definition 25.3.1 (Framings)
Let M be a smooth manifold. A framing of M consists of the following data.

- A choice of embedding $\iota : M \hookrightarrow \mathbb{R}^k$.
- A choice of trivialization $\phi : N_{\mathbb{R}^k/M} \xrightarrow{\cong} e^k$.

In this case, we say that (M, ϕ) is a framed manifold.

25.4

Definition 25.4.1 (Unoriented Bordisms)
Let X, M, N be smooth compact manifolds. Let $M_1 \to X$ and $M_2 \to X$ be maps. We say that the maps are bordant if there exists a bordism W of M_1 and M_2 and a map $W \to X$ such that the following diagram is commutative:

$$\begin{tikzcd} M_1 \amalg M_2 \arrow[r, hook] \arrow[dr] & W \arrow[d, dashed, "\exists"] \\ & X \end{tikzcd}$$

Definition 25.4.2 (The Unoriented Bordism Groups)
Let X be a space. Define the nth unoriented bordism group of X to be the set

$$O_n(X) = \{[M \to X] \mid [M \to X] \text{ is a bordism class } \}$$

Notice that the bordism group of manifolds is a special case:

$$O_n(*) \cong = \Omega_n^O$$

Definition 25.4.3 (The Unoriented Bordism Module)
Let X be a space. Define the unoriented bordism ring of X to be the abelian group

$$O_*(X) = \bigoplus_{n=0}^{\infty} O_n(X)$$

together with the Ω-module structure given by ???

Definition 25.4.4 (The Unoriented Bordism Group of a Pair)
Let $(X, A), (M, \partial M), (N, \partial N)$ be pairs of manifolds. Let $(M_1, \partial M_1) \to (X, A)$ and $(M_2, \partial M_2) \to (X, A)$ be maps. A bordism between $(M_1, \partial M_1)$ and $(M_2, \partial M_2)$ consists of the following data.

- A bordism W between ∂M_1 and ∂M_2.
- A manifold Y such that $\partial Y = M_1 \amalg M_2 \amalg W$.
- A map $f : (Y, W) \to (X, A)$ such that the following diagram commutes:

$$\begin{tikzcd} M_1 \arrow[r, hook] \arrow[dr] & Y \arrow[d, dashed, "\exists f"] & M_2 \arrow[l, hook'] \arrow[dl] \\ & X & \end{tikzcd}$$

Proposition 25.4.5
$X \to O_*(X)$ defines a generalized homology theory on manifolds.

Part IV

Sites and Grothendieck Topologies

26 Sites, Sieves and Grothendieck Topologies

26.1 Sites on a Category

We now want to generalize the notion of sheaves to general categories, not just spaces. However, there is no notion of open sets in an arbitrary category. Grothendieck realized that the essence of sheaves comes from the data of the open covers $U = \bigcup_{i\in I} U_i$ for each open set U in a space X.

Definition 26.1.1 (Covers) Let $\mathcal{C}$ be a category. Let $U \in \mathcal{C}$ be an object. A cover of U is a collection of morphisms

$$\{U_i \to U \mid i \in I\}$$

Definition 26.1.2 (Coverages (nLab)) Let $\mathcal{C}$ be a category. A coverage $\text{Cov}(\mathcal{C})$ of $\mathcal{C}$ consists a collection of covers for each $U \in \mathcal{C}$, such that the following are true.

- If $\{f_i : U_i \to U \mid i \in I\}$ is a cover of U and $g : V \to U$ is a morphism in $\mathcal{C}$, then there exists a cover $\{h_j : V_j \to V\}$ of V such that every $g \circ h_j : V_j \to U$ factors through the cover of U. Or in other words, for each $h_j : V_j \to V$, there exists U_i and $f_i : U_i \to U$ such that the following diagram commutes:

$$\begin{tikzcd} V_j \arrow[r, dashed, "k"] \arrow[d, "h_j"'] & U_i \arrow[d, "f_i"] \\ V \arrow[r, "g"'] & U \end{tikzcd}$$

Definition 26.1.3 (Coverages (Stacks)) Let $\mathcal{C}$ be a category. A coverage $\text{Cov}(\mathcal{C})$ of $\mathcal{C}$ consists a collection of covers for each $U \in \mathcal{C}$, such that the following are true.

- If $V \to U$ is an isomorphism then $\{V \to U\} \in \text{Cov}(\mathcal{C})$
- If $\{U_i \to U \mid i \in I\} \in \text{Cov}(\mathcal{C})$ and $\{V_{i,j} \to U_i \mid j \in J_i\} \in \text{Cov}(\mathcal{C})$ for each $i \in I$, then $\{V_{i,j} \to U_i \to U \mid i,j \in I\} \in \text{Cov}(\mathcal{C})$
- If $\{f_i : U_i \to U \mid i \in I\} \in \text{Cov}(\mathcal{C})$ and $g : V \to U$ is a morphism, then the pullback

$$\begin{tikzcd} U_i \times_U V \arrow[r, dashed, "\text{proj.}"] \arrow[d, "\text{proj.}"'] & U_i \arrow[d, "g"] \\ V \arrow[r, "f_i"'] & U \end{tikzcd}$$

exists for all $i \in I$ and $\{U_i \times_U V \to U\} \in \text{Cov}(\mathcal{C})$.

Definition 26.1.4 (Sites (nLab)) A site is a category $\mathcal{C}$ together with a coverage $\text{Cov}(\mathcal{C})$.

Definition 26.1.5 (Continuous Functors) Let $\mathcal{C}$ and $\mathcal{D}$ be two sites. Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. We say that F is continuous if the following are true.

- F preserves fiber products
- F sends covers to covers

27 Sheaf Cohomology of Sites

27.1 Sheaves on Sites

Definition 27.1.1 (Sheaves on Sites) Let $(\mathcal{C}, \text{Cov}(\mathcal{C}))$ be a site. Let $\mathcal{F}$ be a presheaf of sets on $\mathcal{C}$. We say that $\mathcal{F}$ if a sheaf on $\mathcal{C}$ if for all $\{f_i : U_i \to U \mid i \in I\} \in \text{Cov}(\mathcal{C})$, the following diagram

$$\mathcal{F}(U) \xrightarrow{p} \prod_{i \in I} \mathcal{F}(U_i) \overset{q}{\underset{r}{\rightrightarrows}} \prod_{i,j \in I} \mathcal{F}(U_i \times_U U_j)$$

is an equalizer diagram, where p is the product of the maps $\mathcal{F}(f_i)$, q is the product of the maps $\mathcal{F}(\text{proj}_i : U_i \times_U U_j \to U_i)$ and r is the product of the maps $\mathcal{F}(\text{proj}_j : U_i \times_U U_j \to U_j)$.

Definition 27.1.2 (Abelian Sheaves) Let $\mathcal{C}$ be a site. Define the category of abelian sheaves

$$\mathbf{Ab}(\mathcal{C})$$

over $\mathcal{C}$ to consist of the following data.
- The objects are the abelian sheaves on $\mathcal{C}$
- The morphisms are the natural transformations of functors
- Composition is given by the composition of natural transformations

Theorem 27.1.3 Let $\mathcal{C}$ be a site. Then $\mathbf{Ab}(\mathcal{C})$ has enough injectives.

TBA: category of abelian sheaves on $\mathcal{C}$ has enough injectives

TBA: A bunch of adjunctions analogous to that of sheaves.

27.2 Sheaf Cohomology of Sites

Definition 27.2.1 (Global Sections Functor) Let $\mathcal{C}$ be a site. Define the global section functor

$$\Gamma(-, \mathcal{C}) : \mathbf{Ab}(\mathcal{C}) \to \mathbf{Ab}$$

by the following data.
- A sheaf $\mathcal{F}$ on $\mathcal{C}$ is sent to $\Gamma(\mathcal{F}, \mathcal{C}) = \mathcal{F}(\mathcal{C})$.
- For a morphism of sheaves $\phi : \mathcal{F} \to \mathcal{G}$, the global section functor sends it to the group homomorphism $\mathcal{F}(\mathcal{C}) \to \mathcal{G}(\mathcal{C})$

Proposition 27.2.2 Let $\mathcal{C}$ be a site. Then the global section functor $\Gamma(-, \mathcal{C}) : \mathbf{Ab}(\mathcal{C}) \to \mathbf{Ab}$ is left exact.

Definition 27.2.3 (Sheaf Cohomology of Sites) Let $\mathcal{C}$ be a site. Define the sheaf cohomology functor of $\mathcal{C}$ to be

$$H^k(\mathcal{C}, -) = R^k\Gamma : \mathbf{Ab}(\mathcal{C}) \to \mathbf{Ab}$$

the right derived functors of $\Gamma(-, \mathcal{C})$.

28 Etale Cohomology

To be removed:

Definition 28.0.1 ((Small) Etale Sites) Let X be a scheme. The etale site X_{et} over X consists of the following data.

- The underlying category is the full subcategory $\mathbf{Sch}/X$ of schemes over X consisting of etale morphisms $U \to X$
- The coverage consists of the surjective families of etale morphisms $\{U_i \to U \mid i \in I\}$.

Finite etale site, the flat site, the big etale site

Definition 28.0.2 (Etale Cohomology) Let X be a scheme. Define the etale cohomology of X to be the sheaf cohomology of the category X_{et}. Explicitly, this means that

$$H^k_{\text{et}}(X, -) = H^k(X_{\text{et}}, -) : \mathbf{Ab}(X_{\text{et}}) \to \mathbf{Ab}$$

Part V

Old Sheaf Theory Sections

29 The General Theory of Sheaves

29.1 Basic Definition of Presheaves

As with how we equipped to each variety V its coordinate ring $k[V]$ which are functions on V, we want to equip to each spectrum some ring which are functions on them. It will not make sense that we can define functions on spectrums immediately.

In its full generality, sheaves are designed to encode all local information into one compact notation. It often has uses in complex geometry for its differential forms and smooth functions. In algebraic geometry it is fundamental for redefining the notion of a variety.

Definition 29.1.1 (The Category of Open Sets) Let $(X, \mathcal{T})$ be a topological space. Define the category $\textbf{Open}(X)$ to consist of the following data.

- The objects $\text{Ob}(\textbf{Open}(X)) = \mathcal{T}$ are the open sets of X
- For $U, V \subseteq X$ two open sets of X, the morphisms

$$\text{Hom}_{\textbf{Open}(X)}(U, V) = \begin{cases} \{\iota : U \hookrightarrow V \text{ the inclusion map}\} & \text{if } U \subseteq V \\ \emptyset & \text{otherwise} \end{cases}$$

- Composition is given as the composition of functions.

Recall from Category Theory 1 that a presheaf on $\mathcal{C}$ is just a contravariant functor $\mathcal{C} \to \textbf{Set}$. Using the category of open sets we can now define presheaves on spaces.

Definition 29.1.2 (Presheaves on a Space)
Let X be a space. Let $\mathcal{C}$ be a category. A $\mathcal{C}$-presheaf on X is a contravariant functor

$$\mathcal{F} : \textbf{Open}(X) \to \mathcal{C}$$

As per the definition of a functor, for each inclusion map $\iota : V \to U$ of open sets in X, passing through a presheaf $\mathcal{F}$ gives a unique morphism $\mathcal{F}(\iota) : \mathcal{F}(U) \to \mathcal{F}(V)$. We often denote $\mathcal{F}(\iota)$ as res^U_V or $(-)|_V$.

Example 29.1.3 (The Constant Presheaf)
Let X be a space. Let S be a set. Define the constant presheaf with value S to be the presheaf $\underline{S}_{\text{pre}}$ defined as follows.

- For each open set $U \subseteq X$, we have
$$\underline{S}_{\text{pre}}(U) = S$$
- For each inclusion $V \subseteq U$ of open sets, define
$$\text{res}^U_V = \text{id}_S$$

Definition 29.1.4 (The Stalk of a Presheaf) Let $\mathcal{F}$ be a presheaf on a topological space $(X, \mathcal{T})$. Let $p \in X$. Consider the full subcategory $\mathcal{J}_p$ of $\textbf{Open}(X)$ consisting of open sets in X that contain p. Define the stalk of $\mathcal{F}$ at p to be the colimit

$$\mathcal{F}_p = \operatorname*{colim}_{\mathcal{J}_p} \mathcal{F}(V_p)$$

Definition 29.1.5 (Morphism of Presheaves)
Let X be a topological space. Let $\mathcal{C}$ be a category. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{C}$-presheaves on X. A morphism $\varphi : \mathcal{F} \to \mathcal{G}$ is a natural transformation of functors.

Notice that the natural transformation ϕ here takes every open set U and maps it to a group homomorphism $\phi(U) : \mathcal{F}(U) \to \mathcal{G}(U)$ if $\mathcal{F}$ and $\mathcal{G}$ are presheaves with values in **Grp**.

Definition 29.1.6 (Induced Map of Stalks)
Let X be a topological space. Let $\mathcal{C}$ be a category. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{C}$-presheaves on X. Let $p \in X$. Define the induced map of stalks

$$\phi_p : \mathcal{F}_p \to \mathcal{G}_p$$

to be the map induced by the universal product of colimits.

Definition 29.1.7 (Isomorphism of Presheaves)
Let X be a topological space. Let X be a topological space. Let $\mathcal{C}$ be a category. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{C}$-presheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves. We say that φ is an isomorphism if φ is a natural isomorphism. In this case, we say that $\mathcal{F}$ and $\mathcal{G}$ are isomorphic.

29.2 Sheaves with Values in a Set

Definition 29.2.1 (Sheaves) Let X be a space. A **Set**-sheaf on X is a **Set**-presheaf $\mathcal{F} : \textbf{Open} \to \textbf{Sets}$ satisfying two additional properties
- Identity: If $\{U_i | i \in I\}$ is an open cover of U and $\phi_1, \phi_2 \in \mathcal{F}(U)$ and $\phi_1|_{U_i} = \phi_2|_{U_i}$ for all i, then $\phi_1 = \phi_2$
- Gluing: If $\{U_i | i \in I\}$ is an open cover of U and $\phi_i \in \mathcal{F}(U_i)$ for all $i \in I$ such that $\phi_i|_{U_i \cap U_j} = \phi_j|_{U_i \cap U_j}$ for all $i, j \in I$, then there is exists some $\phi \in \mathcal{F}(U)$ such that $\phi|_{U_i} = \phi_i$ for all $i \in I$.

Example 29.2.2
Let X, Y be spaces. Define a presheaf $\mathcal{F}$ on X by

$$\mathcal{F}(U) = \{f : U \to Y \mid f \text{ is continuous }\}$$

for $U \subseteq X$ an open set, together with restriction maps. Then $\mathcal{F}$ is a sheaf.

Proof
Identity: Suppose that $f, g \in \mathcal{F}(U)$ for some open set $U \subseteq X$ and $f|_{U_i} = g|_{U_i}$ for $\{U_i \mid i \in I\}$ an open cover of U. Then for any $x \in U$, there exists $i \in I$ such that $x \in U_i$, and so

$$f(x) = f|_{U_i}(x) = g|_{U_i}(x) = g(x)$$

Hence $f = g$.

Gluing: Suppose that $f_i \in \mathcal{F}(U_i)$ for $\{U_i \mid i \in I\}$ an open cover of an open set $U \subseteq X$ such that $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$. Define $f \in \mathcal{F}(U)$ by $f(x) = f_i(x)$ for $x \in U_i$. This is well defined since $\{f_i\}$ agree on intersections. It remains to show that f is continuous. Let $W \subseteq Y$ be an open set. I claim that $f^{-1}(W) = \bigcup_{i \in I} f_i^{-1}(W)$. Let $x \in f^{-1}(W)$. Then $f(x) \in W$. Since $x \in W \subseteq U = \bigcup_{i \in I} U_i$, there exists $i \in I$ such that $x \in U_i$. Hence $f_i(x) = f(x) \in W$. Thus $x \in f_i^{-1}(W)$. Conversely, suppose that $x \in f_i^{-1}(W)$. Then clearly $f(x) = f_i(x) \in W$ so that $x \in f^{-1}(W)$. Thus $f^{-1}(W) = \bigcup_{i \in I} f_i^{-1}(W)$. Since each f_I is continuous and W is an open set, $f_i^{-1}(W)$ is open and so the arbitrary union of $f_i^{-1}(W)$ is open. Hence $f^{-1}(W)$ is open. ∎

Example 29.2.3 (Locally Constant Sheaf)
Let X be a space. Let S be a set. Define the locally constant presheaf $\underline{S}$ of X with values in S as follows.

- For each open set $U \subseteq X$, define

$$\underline{S}(U) = \{f : U \to S \mid f \text{ is continuous where } S \text{ has the discrete topology } \}$$

- For each inclusion $V \subseteq U$ of open subsets, define the induced map $\underline{S}(U) \to \underline{S}(V)$ by $f \mapsto f|_V$.

The following are true regarding the locally constant sheaf.

- $\underline{S}$ is a sheaf.
- Let $U \subseteq X$ be an open set. Let $\mathcal{B}$ be the set of connected components of U. Then

$$\underline{S}(U) \cong \prod_{C \in \mathcal{B}} S$$

- For all $p \in X$, $\underline{S}_p = S$.

Proof

Let Y be a connected component of U. Let $f \in \underline{S}(Y)$. I claim that f is constant. Let $y_0 \in Y$. Define $A = \{y \in Y \mid f(y) = f(y_0)\} = f^{-1}(f(y_0))$ and $B = \{y \in Y \mid f(y) \neq f(y_0)\}$. Clearly $Y = A \cup B$ and $A \cap B = \emptyset$. Both A and B are open sets since S has a discrete topology. Since Y is connected and A is non-empty, this implies that $B = \emptyset$. Hence $Y = A$ and f is constant on Y. Thus there is a bijection $\underline{S}(Y) \cong S$.

Define a function $\underline{S}(U) \to \prod_{C \in \mathcal{B}} S$ as follows. For each connected component $C \in \mathcal{B}$ of U, choose $x_C \in C$. Then the map is defined by $f \mapsto (f(x_C))$. This map is injective since $f(x_C) = g(x_C)$ implies that $f(C) = g(C)$ for all connected components $C \in \mathcal{B}$. Hence $f = g$. It is surjective since for any choice of sequence $(x_C)_{C \in \mathcal{B}}$, the map $f : U \to S$ defined by $f(x) = x_C$ for $x \in C$ is continuous because S is discrete and $\{x_C\}$ is a closed set, and $f^{-1}(x_C) = C$ is a connected component and so is a closed set. ■

Proposition 29.2.4 (Gluing Sheaves)
Let X be a space. Let $\{U_i \mid i \in I\}$ be an open cover of X. Let $\mathcal{F}_i$ be a sheaf of sets on U_i. Let $\phi_{i,j} : \mathcal{F}_i|_{U_i \cap U_j} \to \mathcal{F}_j|_{U_i \cap U_j}$ be a collection of morphism of sheaves for ranging $i, j \in I$ such that the following are true.

- For each $i, j \in I$, $\phi_{i,j}$ is an isomorphism.
- For each $i, j, k \in I$, we have $\phi_{j,k} \circ \phi_{i,j} = \phi_{i,k}$.

Then there exists a unique (up to unique isomoprhism) sheaf of sets $\mathcal{F}$ on X such that there are isomorphisms $\mathcal{F}|_{U_i} \xrightarrow{\cong} \mathcal{F}_i$ for all $i \in I$.

Proposition 29.2.5
Let X be a space. Let $\mathcal{S}$ be a subcategory of $\mathbf{Open}(X)$. If $\mathcal{S} \subseteq \mathbf{Open}(X)$ is filtered, then there is an equivalence of categories

$$\mathbf{Sh}(X) = \text{Func}(\mathcal{S}^{\text{op}}, \mathbf{Set})$$

An example of $\mathcal{S}$ would be: any base $\mathcal{B}$ of the topology of X.

29.3 Sheaves with Values in a Category

Presheaves are easily generalized to take values in an arbitrary category because they are simply defined to be contravariant functors.

Definition 29.3.1 (Presheaves with Values in a Category) Let X be a space. Let $\mathcal{C}$ be a category. A presheaf on X with values in $\mathcal{C}$ is a contravariant functor

$$\mathcal{F} : \mathbf{Open}(X) \to \mathcal{C}$$

However, it is hard to define sheaves with values on an arbitrary category $\mathcal{C}$. This is because in our definition of sheaves on sets, we made use of the fact that morphisms in **Set** have a well defined notion of restriction. We will define sheaves on a general category by first considering sheaves with an underlying set.

Theorem 29.3.2 Let X be a space. Let $\mathcal{F}$ be a presheaf with values in **Set**. Then the following are equivalent characterizations.

- $\mathcal{F}$ is a sheaf
- For any open covering $U = \bigcup_{i\in I} U_i$, the sequence
$$0 \longrightarrow \mathcal{F}(U) \xrightarrow{f} \prod_{i\in I} \mathcal{F}(U_i) \xrightarrow{g} \prod_{i,j\in I} \mathcal{F}(U_i \cap U_j)$$
where f is given by the product of the unique maps $\mathcal{F}(U) \to \mathcal{F}(U_i)$ and g is given by sending $(s_i)_{i\in I}$ to $(s_i|_{U_i\cap U_j} - s_j|_{U_i\cap U_j})_{i,j\in I}$ is exact.
- For any open covering $U = \bigcup_{i\in I} U_i$, the following diagram
$$\mathcal{F}(U) \xrightarrow{f} \prod_{i\in I} \mathcal{F}(U_i) \overset{g}{\underset{h}{\rightrightarrows}} \prod_{i,j\in I} \mathcal{F}(U_i \cap U_j)$$
is an equalizer diagram, where f is given by the product of the unique maps $\mathcal{F}(U) \to \mathcal{F}(U_i)$, g is given by sending $(s_i)_{i\in I}$ to $(s_i|_{U_i\cap U_j})_{i,j\in I}$, h is given by sending $(s_i)_{i\in I}$ to $(s_j|_{U_i\cap U_j})_{i,j\in I}$.

Proof
We first note that the second and third condition are equivalent. Therefore we just have to show that the first condition is equivalent to the third one. Let $\mathcal{F}$ be an abelian sheaf. We want to show that $\text{im}(f) = \{x \in \prod_{i\in I} F(U_i) \mid g(x) = h(x)\}$. Therefore we begin with $x \in \text{im}(f)$. Then $f(x) = (x_i)_{i\in I}$. But we have that $g((x_i)_{i\in I}) = (x_i|_{U_i\cap U_j})_{i,j\in I}$ and $h((x_i)_{i\in I}) = (x_j|_{U_i\cap U_j})_{i,j\in I}$. We want to show that for all $i, j \in I$, we have that $x_i|_{U_i\cap U_j} = x_j|_{U_i\cap U_j}$. But x_i is just the restriction of x to U_i and x_j is the restriction of x to U_j. But by definition of a sheaf, restricting to V_i and further to V_{ij} is the same as directly restricting to V_{ij}. Similarly, restricting to V_j and further to V_{ij} is the same as directly restricting to V_{ij}. We conclude that applying g and h to $(x_i)_{i\in I}$ give the same element. Thus $x \in \{x \in \prod_{i\in I} F(U_i) \mid g(x) = h(x)\}$.

Now let $(y_i)_{i\in I} \in \{x \in \prod_{i\in I} F(U_i) \mid g(x) = h(x)\}$. This means that $y_i|_{U_i\cap U_j} = y_j|_{U_i\cap U_j}$ for all $i, j \in I$. By the gluing axiom, there exists $y \in C(U)$ such that $y|_{U_i} = y_i$. Thus y is in the image of f and we conclude that $y \in \text{im}(f)$.

Finally, we also need to show that f is injective. Suppose that $x, y \in \mathcal{F}(U)$ is such that $x_i = y_i$ for all $i \in I$. Then by the identity axiom we conclude that $x = y$. Thus f is injective. Thus the equalizer condition is satisfied.

Now suppose that the equalizer condition is satisfied. Then f is injective. This means that for $x, y \in \mathcal{F}(U)$ such that $x_i = y_i$ for all $i \in I$, then $x = y$. But this is precisely the identity axiom. Suppose that $\text{im}(f) = \text{Eq}(g, h)$. Let $s_i \in \mathcal{F}(U_i)$ such that

$$s_i|_{U_i\cap U_j} = g(s_i) = h(s_i) = s_j|_{U_i\cap U_j}$$

for all $i, j \in I$. Then $(s_i)_{i\in I} \in \text{Eq}(g, h) = \text{im}(f)$. Which means that there exists some $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$. But this is the content of the gluing axiom. Since the identity and the gluing axiom are satisfied, we conclude that $\mathcal{F}$ is a sheaf. ■

Motivated by the above characterization, we define sheaves with values on a category with finite products as follows.

Definition 29.3.3 (Sheaves with Values in a Category) Let X be a space. Let $\mathcal{C}$ be a category that admits all products. Let $\mathcal{F}$ be a presheaf on X with values in $\mathcal{C}$. We say that $\mathcal{F}$ is a sheaf with values in $\mathcal{C}$ if for every open covering $U = \bigcup_{i\in I} U_i$, the following diagram
$$\mathcal{F}(U) \xrightarrow{f} \prod_{i\in I} \mathcal{F}(U_i) \overset{g}{\underset{h}{\rightrightarrows}} \prod_{(i,j)\in I\times I} \mathcal{F}(U_i \cap U_j)$$
is an equalizer diagram, where f, g, h are defined as follows.

- The morphism f is the product

$$\prod_{i\in I}\text{res}^U_{U_i}:\mathcal{F}(U)\to\mathcal{F}(U_i)$$

- The morphism g is the product

$$\prod_{(i,j)\in I\times I}\text{res}^{U_i}_{U_i\cap U_j}:\mathcal{F}(U_i)\to\mathcal{F}(U_i\cap U_j)$$

- The morphism h is the product

$$\prod_{(i,j)\in I\times I}\text{res}^{U_j}_{U_i\cap U_j}:\mathcal{F}(U_j)\to\mathcal{F}(U_i\cap U_j)$$

Note: the double product is ordered so that the equalizing condition is the same as the gluing condition.

Proposition 29.3.4
Let X be a space. Let $\mathcal{C}$ be a category that admits all products. Let $F:\mathcal{C}\to\mathbf{Set}$ be a functor such that the following are true.

- F is faithful.
- F preserves limits.
- A morphism $f\in\mathcal{C}$ is an isomorphism if and only if $F(f)\in\mathbf{Set}$ is an isomorphism.

Let $\mathcal{F}$ be a presheaf on X with values in $\mathcal{C}$. Then $\mathcal{F}$ is a sheaf with values in $\mathcal{C}$ if and only if $F\circ\mathcal{F}$ is a sheaf of sets.

Proof
Suppose that $\mathcal{F}$ is a sheaf. Then the following diagram is an equalizer for any open set U and any open cover $\{U_i\mid i\in I\}$ of U:

$$\mathcal{F}(U)\xrightarrow{f}\prod_{i\in I}\mathcal{F}(U_i)\overset{g}{\underset{h}{\rightrightarrows}}\prod_{(i,j)\in I\times I}\mathcal{F}(U_i\cap U_j)$$

Since F commutes with limits, the following diagram is an equalizer:

$$F\circ\mathcal{F}(U)\xrightarrow{f}\prod_{i\in I}F\circ\mathcal{F}(U_i)\overset{g}{\underset{h}{\rightrightarrows}}\prod_{(i,j)\in I\times I}F\circ\mathcal{F}(U_i\cap U_j)$$

Hence $F\circ\mathcal{F}$ is a sheaf.

Conversely, suppose that $F\circ\mathcal{F}$ is a sheaf. Let E be the equalizer of the following diagram:

$$\prod_{i\in I}F\circ\mathcal{F}(U_i)\overset{g}{\underset{h}{\rightrightarrows}}\prod_{(i,j)\in I\times I}F\circ\mathcal{F}(U_i\cap U_j)$$

By the universal property of products and the definition of a presheaf, we obtain a morphism $\mathcal{F}(U)\to E$. Since F preserves limits, we have that $F\circ\mathcal{F}(U)$ and $F(E)$ are both equalizers of the same diagram, and so the morphism $F\circ\mathcal{F}(U)\to F(E)$ is an isomorphism. By assumption on F, we conclude that $\mathcal{F}(U)\cong E$. Hence $\mathcal{F}$ is a sheaf. ■

The following definition is inspired by the Stacks Project.

Definition 29.3.5 (Algebraic Categories)
Let $\mathcal{C}$ be a category. We say that $\mathcal{C}$ is an algebraic category if there exists a functor $F:\mathcal{C}\to\mathbf{Set}$ such that the following are true.

- $\mathcal{C}$ admits all products and filtered colimits.
- F is faithful (so that $\mathcal{C}$ is a concrete category).
- F preserves limits and filtered colimits.
- A morphism $f\in\mathcal{C}$ is an isomorphism if and only if $F(f)\in\mathbf{Set}$ is an isomorphism.

This allows us to simplify statements about sheaves with values in categories whose objects are algebraic in nature. Some examples of algebraic categories include:

$$\mathbf{Set}, \mathbf{Grp}, \mathbf{Ab}, \mathbf{Ring}, \mathbf{CRing}, \mathbf{Mod}_R, \mathbf{Alg}_R$$

They are all equipped with a forgetful functor to sets that is faithful, is the right adjoint of the free functor (take $\mathrm{id} : \mathbf{Set} \to \mathbf{Set}$ for $\mathbf{Set}$) and reflects isomorphisms.

We abuse notation and think of as if $\mathcal{F}(U)$ are sets for each open set $U \subseteq X$ (in reality only $(F \circ \mathcal{F})(U)$ are sets).

Lemma 29.3.6
Let X be a topological space. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{F}$ be a $\mathcal{C}$-presheaf on X. Let $p \in X$. Then we have

$$\mathcal{F}_p = \frac{\{(U, f) \mid U \subseteq X \text{ is an open neighborhood of } p, f \in \mathcal{F}(U)\}}{\sim}$$

where $(U, f) \sim (V, g)$ if there exists $W \subseteq U \cap V$ an open neighborhood of p such that $f|_W = g|_W$.

Proof If $\mathcal{F}$ is a $\mathbf{Set}$-presheaf, the above would be an explicit construction of the filtered colimit. Since $\mathcal{C}$ is algebraic, $F : \mathcal{C} \to \mathbf{Set}$ preserves colimits and by abuse of notation we obtain the following set-theoretic description. ∎

Think of the definition of stalks as follows: Treat f and g to be sections in $\mathcal{F}(U_1)$ and $\mathcal{F}(U_2)$ where $V \subseteq U_1 \cap U_2$ is open and contains x. Then we think of f and g to be the same function in the stalk as long as they agree on some open set that contains x.

Indeed, since we do not care about the entirety of the domain of f and g, and only care about what happens locally near x, it makes sense for us to treat them as a function when they appear to be the same locally.

Let $[(s, U)] \in \mathcal{F}_p$ be an element of the stalk. Consider the following commutative diagram:

$$\begin{tikzcd} \mathcal{F}(U) \arrow[r] \arrow[d] & \mathcal{G}(U) \arrow[d] \\ \mathcal{F}_p \arrow[r] & \mathcal{G}_p \end{tikzcd}$$

The element $s \in \mathcal{F}(U)$ maps vertically on the left to $[(s, U)] \in \mathcal{F}_p$. Since the diagram is commutative, we see that the image of the element in $\mathcal{F}_p$ from the induced map of stalks is $[(\phi(U)(s), U)]$. This is the formula for the induced map.

Proposition 29.3.7
Let X be a topological space. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{C}$-presheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism. Then φ is an isomorphism if and only if for all $p \in X$, the induced map

$$\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$$

is an isomorphism.

Proof
Suppose that ϕ is an isomorphism of sheaves. Then ϕ and ϕ^{-1} induces unique morphisms $\phi_p : \mathcal{F}_p \to \mathcal{G}_{X,p}$ and $\phi_p^{-1} : \mathcal{G}_{X,p} \to \mathcal{F}_p$ making the following diagram commute:

$$\begin{tikzcd} \mathcal{F}(U) \arrow[r, "\phi(U)"] \arrow[d] & \mathcal{G}(U) \arrow[r, "\phi^{-1}(U)"] \arrow[d] & \mathcal{F}(U) \arrow[r, "\phi^{-1}(U)"] \arrow[d] & \mathcal{G}(U) \arrow[d] \\ \mathcal{F}_p \arrow[r, dashed, "\exists!"'] & \mathcal{G}_{X,p} \arrow[r, dashed, "\exists!"'] & \mathcal{F}_p \arrow[r, dashed, "\exists!"'] & \mathcal{G}(V) \end{tikzcd}$$

On the other hand, the identity morphisms $\mathrm{id}_{\mathcal{F}}$ and $\mathrm{id}_{\mathcal{G}}$ also induces the identity map on the level of stalks. Thus we must have $\phi_p \circ \phi_p^{-1} = \mathrm{id}_{\mathcal{G}_{X,p}}$ and $\phi_p^{-1} \circ \phi_p = \mathrm{id}_{\mathcal{F}_p}$. Hence the induced morphism of stalks is an isomorphism.

Now suppose that ϕ_p is an isomorphism for each $p \in X$. We show that $\phi(U)$ is bijective for all $U \subseteq X$ open. Let $s \in \mathcal{F}(U)$ such that $\phi(U)(s) = 0$. Then for each $p \in U$, $\phi(U)(s)$ considered as an element in $\mathcal{G}_{X,p}$ is equal to 0. Since ϕ_p is injective, $s = 0$ in $\mathcal{F}_p$. This means that there exists an open neighbourhood $V_p \subseteq U$ of p such that $s \in \phi(V_p)$ is 0 for each $p \in U$. Since U is covered by the open neighbourhoods V_p for $p \in U$, by the identity axiom we conclude that $s = 0$ in $\phi(U)$. Thus ϕ is injective.

Now let $t \in \mathcal{G}(U)$. Let t_p be the corresponding element of t in $\mathcal{G}_{X,p}$ for each $p \in U$. Since ϕ_p is surjective, there exists $s_p \in \mathcal{F}_p$ such that $\phi_p(s_p) = t_p$ for each $p \in U$. Suppose that on a neighbourhood V_p of p, s_p is represented as r_p. Then $\phi(r_p) \in \mathcal{G}(V_p)$ and $t|_{V_p}$ are two elements whose germs in $\mathcal{G}_{X,p}$ are equal. Hence there exists $W_p \subseteq V_p$ containing p such that $\phi(r_p) = t|_{W_p}$ in $\mathcal{G}(W_p)$. Now U is covered by open sets of the form W_p for each $p \in U$. Let $p \in W_p$ and $q \in W_q$. Then both $r_p|_{W_p\cap W_q}$ and $r_q|_{W_p\cap W_q}$ are two sections in $\mathcal{F}(W_p \cap W_q)$ that are sent to $t|_{W_p\cap W_q} \in \mathcal{G}(W_p \cap W_q)$ by $\phi(W_p \cap W_q)$. By injectivity, they are equal. We conclude that

$$r_p|_{W_p\cap W_q} = r_q|_{W_p\cap W_q}$$

By the gluing axiom, we conclude that there exists a section $s \in \mathcal{F}(U)$ such that $s|_{W_p} = r_p$. Now $\phi(s)$ and t are two sections in $\mathcal{G}(U)$ such that for any $p \in U$, $\phi(s)|_{W_p} = t|_{W_p}$. By the identity axiom, we conclude that $\phi(s) = t$. ■

Note that the above proposition is very untrue for presheaves because as one can see, proving both injective and surjective requires to use of the sheaf axioms instead of just the presheaf datum.

Proposition 29.3.8
Let X be a topological space. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{C}$-presheaves on X. Let $\varphi, \psi : \mathcal{F} \to \mathcal{G}$ be morphisms. Then $\varphi = \phi$ if and only if for all $p \in X$, we have $\varphi_p = \phi_p$.

Proof
Clearly if $\varphi(U) = \psi(U)$ for all open sets $U \subseteq X$, we must have that the induced map of colimits are equal. So suppose instead that $\varphi_p = \psi_p$ for all $p \in X$. Consider the following commutative diagram:

$$\begin{array}{ccc} \mathcal{F}(U) & \longrightarrow & \mathcal{G}(U) \\ \downarrow & & \downarrow \\ \prod_{p\in U} \mathcal{F}_p & \longrightarrow & \prod_{p\in U} \mathcal{G}_p \end{array}$$

where the top and bottom maps are either φ or ψ and their induced product map of stalks.

I claim that the vertical maps are injective. Suppose that $s, t \in \mathcal{F}(U)$ are such that $([(s, U)])_{p\in U} = ([(t, U)])_{p\in U}$. This means that for each $p \in U$, we have that $[(s, U)] = [(t, U)] \in \mathcal{F}_p$. This means that there exists an open subset $p \in V_p \subseteq U$ such that $s|_{V_p} = t|_{V_p}$. Now the collection $\{V_p \mid p \in U\}$ forms an open cover of U. Since $\mathcal{F}$ is a sheaf, the identity axiom implies that $s = t$ on U. Hence the map is injective.

Let $s \in \mathcal{F}(U)$. Then the left vertical map sends s to $([(s, U)])_{p\in U}$ and $\prod_{p\in U} \varphi_p = \prod_{p\in U} \psi_p$ sends the element to $([(\varphi(U)(s), U)])_{p\in U} = ([(\psi(U)(s), U)])_{p\in U}$. Since the right vertical map is injective, we must have $\varphi(U)(s) = \psi(U)(s)$. Hence we have $\varphi = \psi$. ■

Definition 29.3.9 (The Category of Presheaves)
Let X be a topological space. Let $\mathcal{C}$ be a category. Define the category of presheaves on X with values in $\mathcal{C}$ to be

$$\mathbf{P}\mathcal{C}(X) = \text{Func}(\mathbf{Open}(X)^{\text{op}}, \mathcal{C})$$

Definition 29.3.10 (The Category of Sheaves over a Category) Let X be a topological space. Let $\mathcal{C}$ be a category that admits all products. Define the category $\mathcal{C}(X)$ of sheaves on X with values in $\mathcal{C}$ to be the full subcategory of $\mathbf{P}\mathcal{C}(X)$ consisting of sheaves on X with values in $\mathcal{C}$.

29.4 Defining Sheaves on a Subcategory of Open Sets

Definition 29.4.1 (Sheaves of a Space on a Subcategory of Open Sets)
Let X be a space. Let $\mathcal{S}$ be a subcategory of $\mathbf{Open}(X)$. Let $\mathcal{C}$ be an algebraic category with forgetful functor $F : \mathcal{C} \to \mathbf{Set}$. A sheaf of X on $\mathcal{S}$ is a contravariant functor $\mathcal{F} : \mathcal{S}^{\text{op}} \to \mathcal{C}$ such that $F \circ \mathcal{F}$ is a sheaf.

Definition 29.4.2 (The Category of Sheaves on a Space on a Subcategory of Open Sets)
Let X be a space. Let $\mathcal{S}$ be a subcategory of $\mathbf{Open}(X)$. Let $\mathcal{C}$ be an algebraic category. Define the category of sheaves of X on $\mathcal{S}$

$$\mathcal{C}_{\mathcal{S}}(X)$$

to be the full subcategory of $\mathrm{Hom}_{\mathbf{Cat}}(\mathcal{S}^{\text{op}}, \mathcal{C})$ consisting of sheaves of X on $\mathcal{S}$.

Proposition 29.4.3
Let X be a space. Let $\mathcal{S}$ be a subcategory of $\mathbf{Open}(X)$. Let $\mathcal{C}$ be an algebraic category. If $\mathcal{S} \subseteq \mathbf{Open}(X)$ is filteredm, then there is an equivalence of categories

$$\mathcal{C}(X) = \mathcal{C}_{\mathcal{S}}(X)$$

An example of $\mathcal{S}$ would be: any base $\mathcal{B}$ of the topology of X.

30 Properties of the Category of Sheaves

30.1 Sheafification

Definition 30.1.1 (Compatible Germs)
Let X be a space. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{F}$ be an $\mathcal{C}$-presheaf on X. Let $([(s_p,U_p)])_{p\in U}\in\prod_{p\in U}\mathcal{F}_p$. We say that the element is a compatible germ if for all $p\in U$, there exists an open subset $p\in V_p\subseteq U$ such that
$$[(s_p|_{V_p},V_p)]=[(s_q,U_q)]\in\mathcal{F}_q$$
for all $q\in V_p$.

Lemma 30.1.2
Let X be a space. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{F}$ be an $\mathcal{C}$-presheaf on X. Let $([(s_p,U_p)])_{p\in U}\in\prod_{p\in U}\mathcal{F}_p$. Then the element is a compatible germ if and only if there exists an open cover $U=\bigcup_{i\in I}U_i$ of U together with elements $f_i\in\mathcal{F}(U_i)$ such that for all $p\in U_i$ for all i, $[(f_i,U_i)]=[(s_p,U_p)]\in\mathcal{F}_p$.

Proof
Suppose that $([(s_p,U_p)])_{p\in U}\in\prod_{p\in U}\mathcal{F}_p$ is a compatible germ. Then $\{U_p\mid p\in U\}$ is a valid open cover of U together with the elements $s_p\in\mathcal{F}(U_p)$ so that the latter condition is satisfied.

Conversely, suppose that the latter condition is satisfied. Let $p\in U$. Since $\{U_i\mid i\in I\}$ is an open cover of U, there exists $i\in I$ such that $p\in U_i$. Then for all $q\in U_i$, we have $[(f_i,U_i)]=[(s_q,U_q)]\in\mathcal{F}_q$ by assumption. Hence we are done. ■

Thus not only are isomorphism of sheaves determined on the level of stalks, the morphism of sheaves themselves are determined on the level of stalks.

Definition 30.1.3 (Sheafification)
Let X be a space. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{F}$ be an $\mathcal{C}$-presheaf on X. Define the sheafification
$$\mathcal{F}^{\text{sh}}:\textbf{Open}(X)\to\mathcal{A}$$
of $\mathcal{F}$ as follows.

- For each open set $U\subseteq X$,
$$\mathcal{F}^{\text{sh}}(U)=\left\{([(s_p,U_p)])_{p\in U}\in\prod_{p\in U}\mathcal{F}_p\;\middle|\;([(s_p,U_p)])_{p\in U}\text{ is a compatible germ}\right\}$$
- For each $V\subseteq U$ an inclusion, there is a unique morphism $\mathcal{F}^{\text{sh}}(U)\to\mathcal{F}^{\text{sh}}(V)$ that sends $([(s_p,U_p)])_{p\in U}\in\in\mathcal{F}^(U)$ to
$$([(s_p,U_p)])_{p\in V}$$

Lemma 30.1.4
Let X be a space. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{F}$ be an $\mathcal{C}$-presheaf on X. Then $\mathcal{F}^{\text{sh}}$ is a $\mathcal{C}$-sheaf.

Proof
We first show that $\mathcal{F}^{\text{sh}}$ is a sheaf. Let $\{W_i\mid i\in I\}$ be an open cover for an open set $U\subseteq X$.

For the identity axiom, suppose that $([(s_p,U_p)])_{p\in U}$ and $([t_p,V_p])_{p\in U}$ in $\mathcal{F}(U)$ are such that their restriction to $\mathcal{F}^{\text{sh}}(W_i)$ are equal. This means that $([(s_p,U_p)])_{p\in W_i}=([(s_p,U_p)])_{p\in W_i}$ or in other words, that $[(s_p,U_p)]=[(t_p,V_p)]\in\mathcal{F}_p$. Then since $\{W_i\mid i\in I\}$ covers U, we must have $([(s_p,U_p)])_{p\in U}=([t_p,V_p])_{p\in U}$ in $\mathcal{F}(U)$.

For gluing, suppose that we are given $([(s_p, U_p)])_{p\in W_i} \in \mathcal{F}^{\text{sh}}(W_i)$ for each $i \in I$ such that restricting to common intersections give equal elements. This means that the element $([(s_p, U_p)])_{p\in U}$ is a well defined element of $\mathcal{F}^{\text{sh}}$. ■

Example 30.1.5
Let X be a space. Let S be a set. Then we have

$$\underline{S} = \left(\underline{S}_{\text{pre}}\right)^{\text{sh}}$$

Definition 30.1.6 (Universal Morphism of Sheafification)
Let X be a space. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{F}$ be an $\mathcal{C}$-presheaf on X. Define the universal morphism of sheafification $\theta_{\mathcal{F}}$ as follows. For each open set $U \subseteq X$, define

$$\theta_{\mathcal{F}}(U) : \mathcal{F}(U) \to \prod_{p\in U} \mathcal{F}_p$$

by $s \mapsto ([(s, U)])_{p\in U}$.

Lemma 30.1.7
Let X be a space. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{F}$ be an $\mathcal{C}$-presheaf on X. Then the following are true.
- $\text{im}(\theta_{\mathcal{F}}) = \mathcal{F}^{\text{sh}}(U)$.
- If $\mathcal{F}$ is a sheaf, then for any open subset $U \subseteq X$, $\theta_{\mathcal{F}}(U) : \mathcal{F} \to \mathcal{F}^{\text{sh}}$ is injective.

Proof

- Clearly any element in the image of the map is a compatible germ. Now conversely, suppose that $([(s_p, U_p)])_{p\in U}$ is a compatible germ. Now $\{U_p \mid p \in U\}$ is an open cover of U. Let $p, q \in U$ be arbitrary. Let $r \in U_p \cap U_q$. Then there exists an open subset $V_r \subseteq U_p$ such that $[(s_p|_{V_r}, V_r)] = [(s_q, U_q)] \in \mathcal{F}_q$. This means that there exists an open subset $W_r \subseteq V_r \cap U_q$ such that $s_p|_{W_r} = s_q|_{W_r}$. Now since $\{W_r \mid r \in U_p \cap U_q\}$ covers $U_p \cap U_q$, we must have that $s_p|_{U_i\cap U_j} = s_q|_{U_i\cap U_j}$ by the identity axiom. Then by the gluability axiom, there exists $s \in \mathcal{F}(U)$ such that $s|_{U_p} = s_p \in \mathcal{F}(U_p)$. Now clearly $([(s, U)])_{p\in U} = ([(s_p, U_p)])_{p\in U}$ since $s|_{U_p} = s_p \in \mathcal{F}(U_p)$. Hence the map is surjective.
- Suppose that $s, t \in \mathcal{F}(U)$ are such that $([(s, U)])_{p\in U} = ([(t, U)])_{p\in U}$. This means that for each $p \in U$, we have that $[(s, U)] = [(t, U)] \in \mathcal{F}_p$. This means that there exists an open subset $p \in V_p \subseteq U$ such that $s|_{V_p} = t|_{V_p}$. Now the collection $\{V_p \mid p \in U\}$ forms an open cover of U. Since $\mathcal{F}$ is a sheaf, the identity axiom implies that $s = t$ on U. Hence the map is injective.

■

Lemma 30.1.8
Let X be a space. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{F}$ be an $\mathcal{C}$-presheaf on X. Then θ induces an isomorphism

$$\theta_p : \mathcal{F}_p \xrightarrow{\cong} \mathcal{F}_p^{\text{sh}}$$

of stalks for each $p \in X$.

Proof
By the universal property of stalks as colimits, the morphism θ_p is defined by $[(s, U)] \mapsto [(([(s, U)])_{p\in U}, U)]$.

Suppose that $[(([(s, U)])_{q\in U}, U)] = [(([(t, U)])_{q\in U}, U)]$. By the definition of the equivalence relation, we have that $([(s, U)])_{q\in U}$ and $([(t, U)])_{q\in U}$ agree on U. Hence for each $q \in U$, we have

$[(s,U)] = [(t,U)]$. Again by the definition of the equivalence relation, we have $s = t$ on U. Hence θ_p is injective.

To show that θ_p is surjective, let $[(([(s_q,U_q)])_{q\in U},U)] \in \mathcal{F}_p^{\text{sh}}$. Since s is a compatible germ by definition, there exists an open cover $\{U_i \mid i \in I\}$ of U together with elements $f_i \in \mathcal{F}(U_i)$ such that $[(f_i,U_i)] = [(s_q,U_q)] \in \mathcal{F}_p$ for all $q \in U_i$ for all i. In particular, for our already fixed choice of $p \in X$, there exists $i \in I$ and $f_i \in \mathcal{F}(U_i)$ such that $[(f_i,U_i)] = [(s_p,U_p)] \in \mathcal{F}_p$. Now $[(f_i,U_i)] \in \mathcal{F}_p$ is mapped to $[(([(f_i,U_i)])_{q\in U_i},U_i)]$. I claim that

$$[(([(f_i,U_i)])_{q\in U_i},U_i)] = [(([(s_q,U_q)])_{q\in U},U)] \in \mathcal{F}_p^{\text{sh}}$$

Indeed, in the intersection $U_i = U_i \cap U$, we have $[(f_i,U_i)] = [(s_q,U_q)]$ for all $q \in U_i$. by definition. Hence we have $([(f_i,U_i)])_{q\in U_i} = ([(s_q,U_q)])_{q\in U_i}$ and hence we have surjectivity. ■

Proposition 30.1.9 (The Universal Property of Sheafification)
Let X be a space. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{F}$ be an $\mathcal{C}$-presheaf on X. Then the sheafification $\mathcal{F}^{\text{sh}}$ of $\mathcal{F}$ together with the universal morphism θ satisfies the following universal property. For any sheaf $\mathcal{G}$ on X and morphism of presheaves $\varphi : \mathcal{F} \to \mathcal{G}$, there exists a unique morphism $\varphi^{\text{sh}} : \mathcal{F}^{\text{sh}} \to \mathcal{G}$ such that the following diagram commutes:

$$\begin{tikzcd}
\mathcal{F} \arrow[r, "\theta"] \arrow[dr, "\varphi"'] & \mathcal{F}^{\text{sh}} \arrow[d, dashed, "\exists!\varphi^{\text{sh}}"] \\
& \mathcal{G}
\end{tikzcd}$$

Moreover, $\mathcal{F}^+$ is the unique (up to unique isomorphism) sheaf of X satisfying the above property.

Proof
Let $\mathcal{G}$ be a sheaf. Let $\phi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves. To define a morphism of sheaves $\mathcal{F}^{\text{sh}} \to \mathcal{G}$, we first define it on each open set U and prove that the relevant square diagrams commute. So define $\phi^{\text{sh}}(U) : \mathcal{F}^{\text{sh}}(U) \to \mathcal{G}(U)$ for each open set $U \subseteq X$ as follows. For $([(s_p,U_p)])_{p\in U}$ an element in $\prod_{p\in U} \mathcal{F}_p$, the product map $\prod_{p\in U} \phi_p$ sends the our element to $([\phi(U_p)(s_p),U_p)])_{p\in U}$. Let $([(s_p,U_p)])_{p\in U}$ be a compatible germ. I claim that $([\phi(U_p)(s_p),U_p)])_{p\in U}$ is a compatible germ. By lemma 1.2.6, there exists an open cover $\{U_i \mid i \in I\}$ of U together with elements in $f_i \in \mathcal{F}(U_i)$ such that for all $i \in I$ and all $p \in U_i$, $[(f_i,U_i)] = [(s_p,U_p)]$. Now the elements $\phi(U_i)(f_i)$ for all i proves the claim since ϕ_p maps $[(f_i,U_i)]$ to $[(\phi(U_i)(f_i),U_i)]$, $[(s_p,U_p)] \mapsto [(\phi(U_p)(s_p),U_p)]$ and $[(f_i,U_i)] = [(s_p,U_p)] \in \mathcal{F}_p$ implies that $[(\phi(U_i)(f_i,U_i)] = [(\phi(U_p)(s_p),U_p)]$. By lemma 1.2.9, we obtain a unique element in $\mathcal{G}(U)$. Hence we obtain a map $\phi^{\text{sh}}(U) : \mathcal{F}^{\text{sh}}(U) \to \mathcal{G}(U)$.

Now we show that the composite $\mathcal{F} \to \mathcal{F}^{\text{sh}} \to \mathcal{G}$ is equal to ϕ. Notice that given $s \in \mathcal{F}(U)$, it is sent to $([(s,U)]_{p\in U}$ in $\mathcal{F}^{\text{sh}}(U)$ and then to $([(\phi(U)(s),U)])_{p\in U}$ in $\mathcal{G}(U)$. Evidently, $\phi(U)(s) \in \mathcal{G}(U)$ is the unique preimage of $([(\phi(U)(s),U)])_{p\in U}$ of the map in lemma 1.2.7. Hence $s \mapsto \phi(U)(s)$ matches up with the map of presheaves $\phi : \mathcal{F} \to \mathcal{G}$.

It remains to show uniqueness of the morphism ϕ^{sh}. Suppose that $\psi : \mathcal{F}^{\text{sh}} \to \mathcal{G}$ is another morphism of sheaves such that $\phi = \psi \circ \theta$. By the above lemma, θ is an isomorphism at stalks. Hence for all $p \in X$, we have

$$\phi_p \circ \theta_p^{-1} = \psi_p$$

On the other hand, the equation $\phi = \phi^{\text{sh}} \circ \theta$ gives

$$\phi_p \circ \theta_p^{-1} = \phi_p^{\text{sh}}$$

Equating the two gives $\psi_p = \phi_p^{\text{sh}}$ for all $p \in X$. Since $\mathcal{F}^{\text{sh}}$ and $\mathcal{G}$ are sheaves, By 1.2.6 we conclude that $\psi = \phi^{\text{sh}}$ and so we are done. ■

Lemma 30.1.10
Let X be a space. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{F}$ be an $\mathcal{C}$-presheaf on X. Then the universal morphism $\theta : \mathcal{F} \to \mathcal{F}^{\text{sh}}$ is an isomorphism.

Proof
By 1.2.11, we have that θ_p is an isomorphism for all $p \in X$. Hence by 1.2.4, we have that θ is an isomorphism. ■

Thus the theory of compatible germs gives us an explicit way of understanding the elements of a sheaf by a collection of elements in the stalks.

Definition 30.1.11 (The Sheafification Functor)
Let X be a space. Let $\mathcal{C}$ be an algebraic category. Define the sheafification functor

$$(-)^{\text{sh}} : \mathbf{P}\mathcal{C}(X) \to \mathcal{C}(X)$$

as follows.
- For each presheaf $\mathcal{F}$ on X, $\mathcal{F}^{\text{sh}}$ is the sheafification of $\mathcal{F}$.
- For each morphism of presheaves $\mathcal{F} \to \mathcal{G}$, the morphism of sheaves is given by the universal property of $\mathcal{F}^{\text{sh}}$ applied to the composite $\mathcal{F} \to \mathcal{G} \to \mathcal{G}^{\text{sh}}$.

Proposition 30.1.12
Let X be a space. Let $\mathcal{C}$ be an algebraic category. Let Forget : $\mathcal{C}(X) \to \mathbf{P}\mathcal{C}(X)$ be the forgetful functor. Then there is an adjunction

$$(-)^{\text{sh}} : \mathbf{P}\mathcal{C}(X) \overset{\dashv}{\rightleftarrows} \mathcal{C}(X) : \text{Forget}$$

Proof
The universal property of sheafification gives a map from the right hom set to the left hom set. Conversely, precomposition with the universal morphism $\mathcal{F} \to \mathcal{F}^{\text{sh}}$ gives the map to the other direction. Evidently they are inverses of each other. It remains to show that this bijection is natural in both variables. In other words, we ought to show that the following two squares commute:

$$\begin{tikzcd}
\text{Hom}_{\mathbf{Set(X)}}(\mathcal{F}^{\text{sh}}, \mathcal{G}) \arrow[r, "\cong"] & \text{Hom}_{\mathbf{PSet}(X)}(\mathcal{F}, \text{Forget}(\mathcal{G})) \\
\text{Hom}_{\mathbf{Set(X)}}(\mathcal{H}^{\text{sh}}, \mathcal{G}) \arrow[r, "\cong"'] \arrow[u, "-\circ\varphi^{\text{sh}}"] & \text{Hom}_{\mathbf{PSet}(X)}(\mathcal{H}, \text{Forget}(\mathcal{G})) \arrow[u, "-\circ\varphi"']
\end{tikzcd}$$

$$\begin{tikzcd}
\text{Hom}_{\mathbf{Set(X)}}(\mathcal{F}^{\text{sh}}, \mathcal{G}) \arrow[r, "\cong"] \arrow[d, "\psi\circ-"'] & \text{Hom}_{\mathbf{PSet}(X)}(\mathcal{F}, \text{Forget}(\mathcal{G})) \arrow[d, "\text{Forget}(\psi)\circ-"] \\
\text{Hom}_{\mathbf{Set(X)}}(\mathcal{F}^{\text{sh}}, \mathcal{I}) \arrow[r, "\cong"'] & \text{Hom}_{\mathbf{PSet}(X)}(\mathcal{F}, \text{Forget}(\mathcal{I}))
\end{tikzcd}$$

for two morphisms $\varphi : \mathcal{F} \to \mathcal{H}$ and $\psi : \mathcal{G} \to \mathcal{I}$. Since $\mathbf{Sh}(X)$ is a full subcategory of $\mathbf{PSh}(X)$, clearly the square on the bottom commutes. For the top square, let $\sigma : \mathcal{H}^{\text{sh}} \to \mathcal{G}$ be a morphism. Following the left and top arrows, we obtain the morphism $\sigma \circ \varphi^{\text{sh}} \circ \theta_{\mathcal{F}}$. Following the bottom and right arrows, we obtain the morphism $\sigma \circ \theta_{\mathcal{H}} \circ \varphi$. I claim that $\varphi^{\text{sh}} \circ \theta_{\mathcal{F}} = \theta_{\mathcal{H}} \circ \varphi$. Indeed, the definition of φ^{sh} as in def 2.1.10 is given by the universal property of sheafification of the composite $\theta_{\mathcal{H}} \circ \varphi$. Hence precomposing φ^{sh} with $\theta_{\mathcal{F}}$ is precisely $\theta_{\mathcal{H}} \circ \varphi$. Thus we are done. ■

30.2 The Category of (Pre)Sheaves as an Abelian Category

Lemma 30.2.1
Let X be a topological space. Let $\mathcal{A}$ be an abelian category. Let $*$ be the zero object of $\mathcal{A}$. Then the sheaf $\underline{*}$ is the zero object of $\mathcal{A}(X)$.

Proof Let $\mathcal{F} \in \mathcal{A}(X)$ be an abelian sheaf. We first show the existence and uniqueness of a morphism $\underline{*} \to \mathcal{F}$. Suppose that $\varphi : \underline{*} \to \mathcal{F}$ is a morphism. Then for each open set $U \subseteq X$, there is a unique morphism $\varphi(U) : \underline{*}(U) = * \to \mathcal{F}(U)$. Hence φ is unique. Existence is also evident.

Similarly, we show the existence and uniqueness of a morphism $\mathcal{F} \to \underline{*}$. Suppose that $\varphi : \mathcal{F} \to \underline{*}$ is a morphism. Then for each open set $U \subseteq X$, there is a unique morphism $\varphi(U) : \mathcal{F}(U) \to * = \underline{*}(U)$. Hence φ is unique. Existence is also evident. ■

Definition 30.2.2 (The Product Sheaf)
Let X be a space. Let $\mathcal{C}$ be a category that admits all products. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{C}$-sheaves on X. Define the product $\mathcal{F} \times \mathcal{G}$ sheaf of $\mathcal{F}$ and $\mathcal{G}$ as follows.

- For each open set $U \subseteq X$, define $(\mathcal{F} \times \mathcal{G})(U) = \mathcal{F}(U) \times \mathcal{G}(U)$.
- For each inclusion of open sets $V \subseteq U$, define the morphism $(\mathcal{F} \times \mathcal{G})(U) \to (\mathcal{F} \times \mathcal{G})(V)$ by $\text{res}^U_V \times \text{res}^U_V$.

Lemma 30.2.3
Let X be a space. Let $\mathcal{C}$ be a category that admits all products. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{C}$-sheaves on X. Then $\mathcal{F} \times \mathcal{G}$ is the categorical product of $\mathcal{F}$ and $\mathcal{G}$ in $\mathcal{C}(X)$.

Proof
Firstly, the identity and gluing axiom of $\mathcal{F} \times \mathcal{G}$ can be checked coordinate wise. Clearly, there are projection morphisms $\mathcal{F} \times \mathcal{G} \to \mathcal{F}, \mathcal{G}$ open set wise. Let $\mathcal{H}$ be a sheaf with morphisms $\mathcal{H} \to \mathcal{F}, \mathcal{G}$. Then by the universal product of products of abelian groups, there is a unique morphism $\mathcal{H}(U) \to (\mathcal{F} \times \mathcal{G})(U)$ for each open set $U \subseteq X$ such that the composition $\mathcal{H}(U) \to (\mathcal{F} \times \mathcal{G})(U) \to \mathcal{F}(U), \mathcal{G}(U)$ is equal to $\mathcal{H}(U) \to \mathcal{F}(U), \mathcal{G}(U)$ respectively. Then they assemble into a morphism of sheaves $\mathcal{H} \to \mathcal{F} \times \mathcal{G}$. This morphism is unique by construction. ■

Definition 30.2.4 (The Presheaf Kernel)
Let X be a space. Let $\mathcal{A}$ be an abelian category that is an algebraic category. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{A}$-sheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism. Define the kernel $\ker(\varphi)$ of φ as follows.

- For each open set $U \subseteq X$, define $\ker(\varphi)(U) = \ker(\varphi(U))$.
- For each inclusion $V \subseteq U$ of open sets, define the restriction morphism by the universal property of kernels as follows:

$$\begin{tikzcd}
\ker(\varphi(U)) \arrow[r] \arrow[d, dashed, "\exists!"'] & \mathcal{F}(U) \arrow[r, "\varphi(U)"] \arrow[d, "\text{res}^U_V"'] & \mathcal{G}(U) \arrow[d, "\text{res}^U_V"] \\
\ker(\varphi(V)) \arrow[r] & \mathcal{F}(V) \arrow[r, "\varphi(V)"'] & \mathcal{G}(V)
\end{tikzcd}$$

Definition 30.2.5 (The Presheaf Cokernel)
Let X be a space. Let $\mathcal{A}$ be an abelian category that is an algebraic category. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{A}$-sheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism. Define the cokernel $\text{coker}(\varphi)$ of φ as follows.

- For each open set $U \subseteq X$, define $\text{coker}(\varphi)(U) = \text{coker}(\varphi(U))$.
- For each inclusion $V \subseteq U$ of open sets, define the restriction morphism by the universal property of cokernels as follows:

$$\begin{tikzcd}
\mathcal{F}(U) \arrow[r, "\varphi(U)"] \arrow[d, "\text{res}^U_V"'] & \mathcal{G}(U) \arrow[r] \arrow[d, "\text{res}^U_V"] & \text{coker}(\varphi(U)) \arrow[d, dashed, "\exists!"] \\
\mathcal{F}(V) \arrow[r, "\varphi(V)"'] & \mathcal{G}(V) \arrow[r] & \text{coker}(\varphi(V))
\end{tikzcd}$$

Proposition 30.2.6
Let X be a space. Let $\mathcal{A}$ be an abelian category that is an algebraic category. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{A}$-sheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism. Then the following are true.

- The presheaf $\ker(\varphi)$ is the categorical kernel of φ in $\mathbf{P}\mathcal{A}(X)$.
- The presheaf $\text{coker}(\varphi)$ is the categorical cokernel of φ in $\mathbf{P}\mathcal{A}(X)$.

Proof ∎

Proposition 30.2.7
Let X be a space. Let $\mathcal{A}$ be an abelian category that is an algebraic category. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{A}$-sheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism. Then the following are true.

- The presheaf $\ker(\varphi)$ is a sheaf, and is the categorical kernel of φ in $\mathcal{A}(X)$.
- The sheafification of the presheaf $\text{coker}(\varphi)$ is the categorical cokernel of φ in $\mathcal{A}(X)$.

Proof

- Let $U \subseteq X$ be an open set. Let $\{U_i \mid i \in I\}$ be an open cover of U. Let $f, g \in \ker(\varphi)(U)$ such that $\text{res}^U_{U_i}(f) = \text{res}^U_{U_i}(g)$. Consider the following commutative diagram:
$$\begin{tikzcd} \ker(\varphi(U)) \arrow[r] \arrow[d, "\text{res}^U_V"'] & \mathcal{F}(U) \arrow[d, "\text{res}^U_V"'] \\ \ker(\varphi(U_i)) \arrow[r] & \mathcal{F}(U_i) \end{tikzcd}$$
Consider the images of f and g in $\mathcal{F}(U)$. Since f and g map vertically down to the same element in $\ker(\varphi)(U_i)$, by commutativity of the diagram, the images of f and g in $\mathcal{F}(U)$ map vertically to the same element in $\mathcal{F}(U_i)$. Since this is true for all $i \in I$, the identity property of the sheaf $\mathcal{F}$ implies that the image of f and g in $\mathcal{F}(U)$ are equal. This means that $f - g \in \ker(\varphi(U))$ map to $0 \in \mathcal{F}(U)$. Since $\ker(\varphi)(U)$ is a subgroup of $\mathcal{F}(U)$, we must have $f = g$. This proves the identity property of $\ker(\varphi)$.
- We prove that $\text{coker}(\varphi)^{\text{sh}}$ satisfy the universal property of the cokernel. Let $\mathcal{H}$ be a sheaf on X with values in $\mathcal{A}$. Let $\psi : \mathcal{G} \to \mathcal{H}$ be a morphism such that the composite $\mathcal{F} \to \mathcal{G} \to \mathcal{H}$ is 0. Consider the following diagram:
$$\begin{tikzcd} \mathcal{F} \arrow[r, "\varphi"] & \mathcal{G} \arrow[r] \arrow[drr, "\psi"'] & \text{coker}(\varphi) \arrow[r, "\theta_{\text{coker}(\varphi)}"] \arrow[dr, dashed, "\exists!"] & \text{coker}(\varphi)^{\text{sh}} \arrow[d, dashed, "\exists!"] \\ & & & \mathcal{H} \end{tikzcd}$$
By the universal property of cokernels in the category of presheaves, there exists a unique morphism $\text{coker}(\varphi) \to \mathcal{H}$ such that the relevant diagram commutes. Then by the universal property of sheafification, we obtain the unique morphism required to satisfy the universal property of the cokernel.

∎

Proposition 30.2.8
Let X be a space. Let $\mathcal{A}$ be an abelian category that is an algebraic category. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{A}$-sheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism. Let $p \in X$. Then the following are true.

- $\ker(\phi)_p = \ker(\phi_p)$.
- There is a natural isomorphism $\text{coker}(\varphi)^{\text{sh}}_p \cong \text{coker}(\varphi_p)$.

Proof I claim that the induced map $\ker(\phi)_p \to \mathcal{F}_p \to \mathcal{G}_p$ is the zero map. Indeed, the map $\ker(\phi)_p \to \mathcal{F}_p$ is induced by the maps $\ker(\phi)(U) \to \mathcal{F}(U)$ for each open set $U \subseteq X$, which composes with $\mathcal{F}(U) \to \mathcal{G}(U)$ to give the zero map. Hence by the universal property of the kernel of ϕ_p, we obtain a unique morphism $\ker(\phi)_p \to \ker(\phi_p)$. Since the morphism $\ker(\phi) \to \mathcal{F}$ is the inclusion map for each open set, the morphism $\ker(\phi)_p \to \mathcal{F}_p$ is also the inclusion map and so $\ker(\phi_p) \subseteq \ker(\phi_p)$. Conversely, suppose that $[(s, U)] \in \ker(\phi_p)$. Then we have $[(\phi(U)(s), U)] = [(0, V)]$ for some $V \subseteq X$ open. This means that there exists $W \subseteq U \cap V$ open such that $\phi(U)(s)|_W = 0|_W$. But we have

$$\begin{aligned} 0|_W &= \phi(U)(s)|_W \\ &= \text{res}^U_W(\phi(U)(s)) \\ &= \phi(W)(\text{res}^U_W(s)) && (\phi \text{ is a morphism}) \\ &= \phi(W)(s|_W) \end{aligned}$$

and so $s|_W \in \ker(\phi)(W)$. By definition of the kernel of ϕ, we have $[(s|_W, W)] \in \ker(\phi)_p$. But $[(s|_W, W)] = [(s, U)]$ and so we are done.

For cokernels, we invoke the fact that colimits commutes with colimits. ■

Proposition 30.2.9
Let X be a space. Let $\mathcal{A}$ be an abelian category that is an algebraic category. Then $\mathcal{A}(X)$ is an abelian category.

Proof We showed that $\mathcal{A}(X)$ has a zero object by 2.2.1. The hom sets have the structure of an abelian group by open set wise, point wise addition. Thus addition distributes over composition. Finite products are given in def 2.2.2 and lmm 2.2.3. Kernels and cokernels are given above for arbitrary morphisms. Now for any monomorphism $\varphi : \mathcal{F} \to \mathcal{G}$ of sheaves, ■

Let $\mathcal{A}$ be an abelian category. Let f b e a morphism. Recall the following equivalent definitions.

- f is injective if and only if $\ker(\varphi) = 0$ if and only if f is a monomorphism.
- f is surjective if and only if $\text{coker}(f) = 0$ if and only if f is an epimorphism.
- f is an isomorphism if and only if f is injective and surjective.

If $\mathcal{A} = \mathbf{Set}$ recall also that injective functions and monomorphisms coincide, while surjective functions and monomorphisms coincide.

Definition 30.2.10 (The Image Presheaf)
Let X be a space. Let $\mathcal{A}$ be an abelian category that is an algebraic category. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{A}$-sheaves on X. Define the image presheaf $\text{im}(\varphi)$ by the following.

- For each open set $U \subseteq X$, define $\text{im}(\varphi)(U) = \text{im}(\varphi(U))$.
- For each inclusion of open sets $V \subseteq U$, define $\text{im}(\varphi)(U) \to \text{im}(\varphi)(V)$ by the universal property of kernels as follows:

$$\begin{tikzcd} \text{im}(\varphi)(U) = \ker(\mathcal{G}(U) \to \text{coker}(f)(U)) \arrow[r] \arrow[d, "\exists!"'] & \mathcal{G}(U) \arrow[r] \arrow[d, "\text{res}^U_V"'] & \text{coker}(\varphi)(U) \arrow[d, "\text{res}^U_V"] \\ \text{im}(\varphi)(V) = \ker(\mathcal{G}(V) \to \text{coker}(f)(V)) \arrow[r] & \mathcal{G}(V) \arrow[r] & \text{coker}(\varphi)(V) \end{tikzcd}$$

Proposition 30.2.11
Let X be a space. Let $\mathcal{A}$ be an abelian category that is an algebraic category. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{A}$-sheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism. Then the following are true.

- The image presheaf $\text{im}(\varphi)$ is the image of φ in $\mathbf{P}\mathcal{A}(X)$.
- The sheafification of the image presheaf $\text{im}(\varphi)^{\text{sh}}$ is the image of φ in $\mathcal{A}(X)$.
- There is a natural isomorphism $\text{im}(\varphi_p) = \text{im}(\varphi)_p$.

Proof

-
- Since the the kernel of the cokernel is the cokernel of the kernel in an abelian category, we have that
$$\text{im}(\varphi) = \ker(\text{coker}(\varphi)^{\text{sh}}) = \text{coker}(\ker(\varphi))^{\text{sh}} = \ker(\text{coker}(\varphi))^{\text{sh}}$$

- Since taking kernels and cokernels commute with stalk, and sheafification preserves stalks, we conclude.

■

30.3 Exact Sequences of Sheaves

From now on, we focus on the abelian category of sheaves and not presheaves. Therefore we will drop the sheafification symbol for cokernels and images since they mean the abelian categorical object that is the cokernel and the image.

Let X be a space. Let $\mathcal{A}$ be an abelian category that is an algebraic category. Since $\mathcal{A}(X)$ is an abelian category, it makes sense to talk about chain complexes and in particular, exact sequences of sheaves. Recall that a sequence of sheaves $\mathcal{F} \to \mathcal{G} \to \mathcal{H}$ is exact if $\operatorname{im}(\mathcal{F} \to \mathcal{G}) = \ker(\mathcal{G} \to \mathcal{H})$.

Proposition 30.3.1
Let X be a space. Let $\mathcal{A}$ be an abelian category that is an algebraic category. Suppose the following

$$\mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H}$$

is a sequence of $\mathcal{A}$-sheaves on X. Then the sequence is exact if and only if the following sequence of stalks

$$\mathcal{F}_p \longrightarrow \mathcal{G}_p \longrightarrow \mathcal{H}_p$$

is exact for all $p \in X$.

Proof We have that $\operatorname{im}(\mathcal{F} \to \mathcal{G}) = \ker(\mathcal{G} \to \mathcal{H})$ and taking stalks gives $\operatorname{im}(\mathcal{F}_p \to \mathcal{G}_p) = \ker(\mathcal{G}_p \to \mathcal{H}_p)$. Since taking stalks commute with images and kernels, we are done. Conversely, suppose that for all $p \in X$, we have $\operatorname{im}(\mathcal{F}_p \to \mathcal{G}_p) = \ker(\mathcal{G}_p \to \mathcal{H}_p)$. Since taking stalks commute with kernels and images, this show that $\operatorname{im}(\mathcal{F} \to \mathcal{G})_p = \ker(\mathcal{G} \to \mathcal{H})_p$. But two sheaves are isomorphic if and only if the stalks are isomorphic, we are done. ■

Thus the functor

$$(-)_p : \mathcal{A}(X) \to \mathcal{A}$$

of taking stalks is exact.

Proposition 30.3.2 Let X be a space. Let $\mathcal{A}$ be an abelian category that is an algebraic category. Let the following be a sequence of $\mathcal{A}$-sheaves on X:

$$0 \longrightarrow \mathcal{F} \xrightarrow{\varphi} \mathcal{G} \xrightarrow{\psi} \mathcal{H}$$

Then the above sequence is exact if and only if the following sequence

$$0 \longrightarrow \mathcal{F}(U) \xrightarrow{\varphi(U)} \mathcal{G}(U) \xrightarrow{\psi(U)} \mathcal{H}(U)$$

of local sections is exact for any open set $U \subseteq X$.

Proof
Suppose that the given sequence of sheaves is exact. This means that $\ker(\mathcal{F} \to \mathcal{G}) = 0$. Let $U \subseteq X$ be an open set. Then we have $0 = \ker(\varphi)(U) = \ker(\varphi(U))$. On the other hand, notice that $\operatorname{im}(\varphi) = \ker(\psi)$ implies that $\psi_p \circ \varphi_p = 0$ for all $p \in X$. Recall that the image is sheafified, meaning that an element in the image is

$$([(s_p, U_p)])_{p \in U} \in \prod_{p \in U} \operatorname{im}(\varphi)_p$$

This means that $[(s_p, U_p)] \in \operatorname{im}(\varphi)_p$. So there exists $[(t_p, V_p)] \in \mathcal{F}_p$ such that $\varphi_p([(t_p, V_p)]) = [(s_p, U_p)]$. Also, since the sheafification of a sheaf is isomorphic to the original sheaf, without loss of generality, Then we have

$$\begin{aligned}\psi(U)\left(\varphi_p([(t_p, V_p)])\right)_{p\in U} &= \psi(U)\left([(\varphi(V_p)(t_p), V_p)]\right)_{p\in U} \\ &= \left([(\psi(U)(\varphi(U)(t_p)), V_p)]\right)_{p\in U} \\ &= \left([(\psi\circ\varphi(U)(t_p), V_p)]\right)_{p\in U} \\ &= 0\end{aligned}$$

Thus $\text{im}(\varphi)(U) \subseteq \ker(\psi)(U)$. Let $s \in \ker(\psi)(U)$. ■

Thus the functor

$$(-)(U) : \mathcal{A}(X) \to \mathcal{A}$$

is left exact. It is in general not right exact.

Proposition 30.3.3
Let X be a topological space. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{C}$-sheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism. Then the following are equivalent.

- φ is injective (a monomorphism).
- For each $U \subseteq X$ open, $\varphi(U)$ is injective.
- For each $p \in X$, φ_p is injective.

Proof When $\mathcal{A}$ is not **Set**, $(1) \iff (2)$ is a special case of 2.3.2. $(1) \iff (3)$ is a special case of 2.3.1. ■

Proposition 30.3.4
Let X be a space. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{C}$-sheaves on X. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism. Then the following are equivalent.

- φ is srujective (an epimorphism).
- For each $U \subseteq X$ open and for every $g \in \mathcal{G}(U)$, there exists an open cover $\{U_i \mid i \in I\}$ of U and there exists $f_i \in \mathcal{F}(U_i)$ such that $\varphi(U_i)(f_i) = g|_{U_i}$.
- For each $p \in X$, φ_p is surjective.
- There exists a base $\mathcal{B} = \{B_i \mid i \in I\}$ of X such that $\varphi(B_i)$ is surjective for all $i \in I$.

30.4 The Direct Image and Inverse Image Functor

Definition 30.4.1 (Direct Image Sheaf)
Let X, Y be spaces. Let $f : X \to Y$ be a map. Let $\mathcal{C}$ be category. Let $\mathcal{F}$ be a $\mathcal{C}$-presheaf on X. Define the direct image of sheaf

$$f_*\mathcal{F} : \mathbf{Open}(Y) \to \mathcal{C}$$

as follows.

- For each open set $U \subseteq Y$, define $f_*\mathcal{F}(U) = \mathcal{F}(f^{-1}(U))$.
- For each inclusion $V \subseteq U$, define the induced map

$$\mathcal{F}(f^{-1}(U)) \to \mathcal{F}(f^{-1}(V))$$

  to be the map $\mathcal{F}(\text{incl.} : f^{-1}(V) \hookrightarrow f^{-1}(U))$.

Proposition 30.4.2
Let X, Y be spaces. Let $f : X \to Y$ be a map. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{F}$ be a $\mathcal{C}$-sheaf. Then $f_*\mathcal{F}$ is a sheaf on Y.

Proof
It is clear that $f_*\mathcal{F}$ is a presheaf on Y. We need to check the identity and gluing axiom. By 1.3.4, without loss of generality we can assume that $\mathcal{C} = \mathbf{Set}$.

Identity: Let $U \subseteq Y$ be an open set. Let $\{U_i \mid i \in I\}$ be an open cover of U. Denote the inclusion maps by $\iota_i : U_i \to U$. Let $s, t \in f_*\mathcal{F}(U)$. Suppose that $f_*\mathcal{F}(\iota_i)(s) = f_*\mathcal{F}(\iota_i)(t)$ for all $i \in I$. This means that $s, t \in \mathcal{F}(f^{-1}(U))$ is such that the images of s, t in $\mathcal{F}(f^{-1}(U_i))$ are equal for all $i \in I$. Since $\{f^{-1}(U_i)\}$ is an open cover of $f^{-1}(U)$ and $\mathcal{F}$ is a sheaf on X, this means that $s = t$.

Gluing: Let $U \subseteq Y$ be an open set. Let $\{U_i \mid i \in I\}$ be an open cover of U. Suppose that $s_i \in f_*\mathcal{F}(U_i) = \mathcal{F}(f^{-1}(U_i))$ is such that the images of s_i and s_j in $f_*\mathcal{F}(U_i \cap U_j) = \mathcal{F}(f^{-1}(U_i \cap U_j))$ agree for all $i, j \in I$. Since $\{f^{-1}(U_i)\}$ is an open cover of $f^{-1}(U)$ and $\mathcal{F}$ is a sheaf on X, we conclude that there exists $s \in \mathcal{F}(f^{-1}(U)) = f_*\mathcal{F}(U)$ such that the image of s in $\mathcal{F}(f^{-1}(U_i)) = f_*\mathcal{F}(U_i)$ is s_i for all $i \in I$. Thus the gluing axiom is satisfied. ■

Example 30.4.3 (Skyscraper Sheaf)
Let X be a space. Let $p \in X$. Let S be a set. Let $\underline{S}$ be the constant sheaf on $\{p\}$. Define the skyscraper sheaf of X at p to be the pushforward

$$\iota_*\underline{S}(U) = \underline{S}(\iota^{-1}(U))$$

where $\iota : \{p\} \hookrightarrow X$ is the inclusion map.

Definition 30.4.4 (The Direct Image Sheaf Functor)
Let X, Y be spaces. Let $f : X \to Y$ be a map. Let $\mathcal{C}$ be a category. Let $\mathcal{F}$ be a $\mathcal{C}$-sheaf on X. Define the direct image sheaf functor

$$f_* : \mathcal{C}(X) \to \mathcal{C}(Y)$$

by the following data.

- For each sheaf $\mathcal{F}$ of X, $f_*\mathcal{F}$ is the direct image sheaf on Y.
- For each morphism $\varphi : \mathcal{F} \to \mathcal{G}$, $f_*\varphi : f_*\mathcal{F} \to f_*\mathcal{G}$ is the morphism given by

$$\varphi(f^{-1}(U)) : \mathcal{F}(f^{-1}(U)) \to \mathcal{G}(f^{-1}(U))$$

Definition 30.4.5 (Inverse Image Sheaf)
Let X, Y be spaces. Let $f : X \to Y$ be a map. Let $\mathcal{C}$ be a sheaf that admits all filtered colimits. Let $\mathcal{G}$ be a sheaf on Y. Define the inverse image sheaf $f^{-1}\mathcal{G}$ to be the sheafification of the presheaf on X defined as follows.

- For each open set $U \subseteq X$, let $\mathcal{J}(U) \subseteq \textbf{Open}(X)$ be the full subcategory consisting of open sets of Y the contain $f(U)$. Then the presheaf sends U to $\lim_{W \in \mathcal{J}(U)} \mathcal{G}(W)$.
- For each inclusion $V \subseteq U$, define the induced map to be induced by the universal property of the limit.

Proposition 30.4.6
Let X, Y be spaces. Let $f : X \to Y$ be a map. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{G}$ be a sheaf on Y. Then the inverse image sheaf $f^{-1}\mathcal{G}$ is a sheaf.

Definition 30.4.7 (The Inverse Image Sheaf Functor)
Let X, Y be spaces. Let $f : X \to Y$ be a map. Let $\mathcal{C}$ be a sheaf that admits all filtered colimits. Let $\mathcal{F}$ be a sheaf on X. Define the inverse image sheaf functor

$$f^{-1} : \mathcal{C}(Y) \to \mathcal{C}(X)$$

by the following data.

- For each sheaf $\mathcal{F}$ of Y, $f^{-1}\mathcal{F}$ is the inverse image sheaf on X.
- For each morphism $\varphi : \mathcal{F} \to \mathcal{G}$, the morphism $f^{-1}\mathcal{F} \to f^{-1}\mathcal{G}$ is induced by the universal property of limits.

Example 30.4.8
Let Y be a space. Let $p \in Y$. Let $\iota : \{p\} \hookrightarrow Y$ be the inclusion map. Let $\mathcal{G}$ be a sheaf on Y. Then we have
$$\left(\iota^{-1}\mathcal{G}\right)(\{p\}) = \mathcal{G}_p$$

Proposition 30.4.9
Let X, Y be spaces. Let $f : X \to Y$ be a map. Let $\mathcal{C}$ be an algebraic category. Then there is an adjunction
$$f^{-1} : \mathcal{C}(Y) \overset{\dashv}{\rightleftarrows} \mathcal{C}(X) : f_*$$

Corollary 30.4.10
Let X, Y be spaces. Let $f : X \to Y$ be a map. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{G}$ be a $\mathcal{C}$-sheaf of Y. Then there is a natural isomorphism
$$(f^{-1}\mathcal{G})_p \cong \mathcal{G}_{f(p)}$$
for any $p \in X$.

Proof f^{-1} is a left adjoint and stalks are colimits. Since left adjoints preserve colimits we are done. ■

Example 30.4.11
Let X be a space. Let $U \subseteq X$ be an open subset. Let $\mathcal{F}$ be a sheaf of sets on X. Let $p \in U$. Then we have
$$(\mathcal{F}|_U)_p \cong \mathcal{F}_p$$

Proof Follows from the above corollary. ■

Lemma 30.4.12
Let Y be a space. Let $U \subseteq Y$ be an open set. Let $\iota : U \hookrightarrow Y$ be the inclusion map. Let $\mathcal{C}$ be an algebraic category. Let $\mathcal{G}$ be a $\mathcal{C}$-sheaf on Y. Then there is a natural isomorphism
$$\iota^{-1}\mathcal{G} \cong \mathcal{G}|_U$$

Proof Write $\iota^{-1}\mathcal{G}_{\text{pre}}$ the inverse image sheaf before sheafification. Recall that before sheafification, for $W \subseteq U$ an open set we have $(\iota^{-1}\mathcal{G}_{\text{pre}})(W) = \text{colim}_{V \supseteq W}\, \mathcal{G}(V)$. Since W is the initial object in the category of open sets containing W, we have that he colimit is equal to the terminal object. Hence $(\iota^{-1}\mathcal{G}_{\text{pre}})(W) = \mathcal{G}(W)$. Since $\mathcal{G}$ is already a sheaf, sheafification gives a natural isomorphism $\iota^{-1}\mathcal{G}) \cong \iota^{-1}\mathcal{G}_{\text{pre}}$ and the latter is equal to $\mathcal{G}|_U$. ■

Proposition 30.4.13
Let X, Y be spaces. Let $f : X \to Y$ be a map. Let $\mathcal{C}$ be an algebraic category. Then $\iota^{-1} : \mathcal{C}(Y) \to \mathcal{C}(X)$ is an exact functor.

Part VI

Abelian Varieties

31 Abelian Varieties

31.1

Let k be a field. Recall that an abelian variety over k is a complete connected group variety over k.

Theorem 31.1.1 (The Rigidity Lemma)

Corollary 31.1.2
Let k be a field. Let A be an abelian variety over k. Then A is commutative.

Part VII

The Theory of Operads

32 Equivalent Definitions of Operads

32.1 Operads as Objects with Compatibility Diagrams

Definition 32.1.1 (Symmetric Operads)
Let $\mathcal{C}$ be a symmetric monoidal category. An operad consists on $\mathcal{C}$ consists of the following data.

- Objects $F(n) \in \mathcal{C}$ for each $n \in \mathbb{N}$.
- Right action of the symmetric group $\rho_n : S_n \to \mathrm{Hom}_{\mathcal{C}}(F(n), F(n))$.
- A unit $e : I \to F(1)$.
- Composition operations $\gamma : F(k) \otimes F(n_1) \otimes \cdots \otimes F(n_k) \to F(n_1 + \cdots + n_k)$.

such that the following diagram commutes:

- Associativity:

$$\begin{tikzcd}
F(k) \otimes \bigotimes_{i=1}^k \left(F(n_i) \otimes \bigotimes_{j=1}^{n_i} F(m_{ij}) \right) \arrow[r, "\gamma\otimes\mathrm{id}"] \arrow[d, "\mathrm{id}\otimes\gamma"'] & F\left(\sum_{i=1}^k n_i\right) \otimes \bigotimes_{j=1}^{\sum_{i=1}^k n_i} F(m_{ij}) \arrow[d, "\gamma"] \\
F(k) \otimes \bigotimes_{i=1}^k F\left(\sum_{j=1}^{n_i} m_{ij}\right) \arrow[r, "\gamma"'] & F\left(\sum_{i=1}^k \sum_{j=1}^{n_i} m_{ij}\right)
\end{tikzcd}$$

- Unit: The following diagram commutes

$$\begin{tikzcd}
F(n) \arrow[r, "e\otimes\mathrm{id}_{F(n)}"] \arrow[d, "\mathrm{id}\otimes e^{\otimes n}"'] \arrow[dr, "\mathrm{id}_{F(n)}"] & F(1) \otimes F(n) \arrow[d, "\gamma"] \\
F(n) \otimes F(1)^{\otimes n} \arrow[r, "\gamma"'] & F(n)
\end{tikzcd}$$

- Compatibility with the symmetric group action:

Here we think of $F(n)$ as the set of all the allowed operations that has n inputs.

Definition 32.1.2 (Operad Maps)
Let $\mathcal{C}$ be a symmetric monoidal category. Let O, P be operads. A morphism of operads consists of morphism $f_n \in \mathrm{Hom}_{\mathcal{C}}(O(n), P(n))$ such that the following are true.

- Associativity:

$$\begin{tikzcd}
O(k) \otimes \bigotimes_{i=1}^k O(n_i) \arrow[r, "f"] \arrow[d, "\gamma"'] & P(k) \otimes \bigotimes_{i=1}^k P(n_i) \arrow[d, "\gamma"] \\
O\left(\sum_{i=1}^k n_i\right) \arrow[r, "f"'] & P\left(\sum_{i=1}^k n_i\right)
\end{tikzcd}$$

- Unit: The following diagram commutes

$$\begin{tikzcd}
I \arrow[r, "e"] \arrow[dr, "e"'] & O(1) \arrow[d, "f"] \\
 & P(1)
\end{tikzcd}$$

- Compatibility with the symmetric group action:

Definition 32.1.3 (Algebras over Operads)
Let $\mathcal{C}$ be a symmetric monoidal category. Let O be an operad on $\mathcal{C}$. An algebra over O consists of the following data.

- An object $X \in \mathcal{C}$.
- A collection of maps $\lambda : O(k) \otimes X^{\otimes k} \to X$

such that the following diagram commutes:

- Associativity:

$$\begin{tikzcd} O(k) \otimes \bigotimes_{i=1}^k \left(O(n_i) \otimes V^{\otimes n_i}\right) \arrow[r, "\gamma\otimes\mathrm{id}"] \arrow[d, "\mathrm{id}\otimes\lambda"'] & O\left(\sum_{i=1}^k n_i\right) \otimes V^{\otimes \sum_{i=1}^k n_i} \arrow[d, "\lambda"] \\ O(k) \otimes V^{\otimes k} \arrow[r, "\lambda"'] & V \end{tikzcd}$$

- Unit: The following diagram commutes

$$\begin{tikzcd} V \arrow[r, "e\otimes\mathrm{id}_V"] \arrow[dr, "\mathrm{id}"'] & O(1) \otimes V \arrow[d, "\lambda"] \\ & V \end{tikzcd}$$

- Compatibility with the symmetric group action:

Definition 32.1.4 (The Canonical Operad of an Object)
Let $\mathcal{C}$ be a symmetric monoidal category. Let $X \in \mathcal{C}$ be an object. Define the canonical operad $\mathrm{Op}(X)$ by the following data.
- $\mathrm{Op}(X)(n) = \mathrm{Hom}_{\mathcal{C}}(X^{\otimes n}, X)$.
- The symmetric group action is induced by shuffling the order of the copies of X in $X^{\otimes n}$.
- The unit is the unique map $I \to \mathrm{Hom}_{\mathcal{C}}(X, X)$.
- Composition is given by composition in $\mathcal{C}$.

Proposition 32.1.5
Let $\mathcal{C}$ be a symmetric monoidal category. Let O be an operad. Let $X \in \mathcal{C}$ be an object. Then the data of an algebra X over O is equivalent to a monoid map $O \to \mathrm{Op}(X)$.

32.2 $\mathbb{S}$-Modules

Definition 32.2.1 ($\mathbb{S}$-Modules)
Let R be a commutative ring. An $\mathbb{S}$-module is a collection $(M_n)_{n\in\mathbb{N}}$ where each M_n is a right $R[S_n]$-module.

Definition 32.2.2 (Morphism of $\mathbb{S}$-Modules)
Let R be a commutative ring. Let M and N be $\mathbb{S}$-modules. Let $f_n : M_n \to N_n$ be a function. We say that $(f_n)_{n\in\mathbb{N}}$ is a morphism of $\mathbb{S}$-modules if each f_n is a $k[S_n]$-module homomorphism.

Definition 32.2.3 (The Associated Schur Functor)
Let R be a commutative ring. Let M be an $\mathbb{S}$-module. Define the associated Schur functor $\widetilde{M} : \mathbf{Mod}_R \to \mathbf{Mod}_R$ by

$$\widetilde{M}(V) = \bigoplus_{n=0}^{\infty} M_n \otimes_{R[S_n]} V^{\otimes n}$$

Induced map.

Direct sum, tensor product, composition of functors.

Category of $\mathbb{S}$-modules is monoidal.

Symmetric operad is equivalent to monoid in the category of $\mathbb{S}$-modules.

Algebraic Operads are maps from Schur functor to the base module.

32.3 Multicategories

33 Higher Coherence Conditions